

Solutions

Question1

A substance 'X' (1.5 g) dissolved in 150 g of a solvent 'Y' (molar mass = 300 g mol^{-1}) led to an elevation of the boiling point by 0.5 K. The relative lowering in the vapour pressure of the solvent 'Y' is _____ $\times 10^{-2}$. (nearest integer)

[Given: K_b of the solvent = $5.0 \text{ K kg mol}^{-1}$]

Assume the solution to be dilute and no association or dissociation of X takes place in solution.

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Answer: 3

Solution:

$$\Delta T_b = i \times K_b \times m$$

$$0.5 = i \times m \times 5$$

$$i \times m = \frac{0.5}{5} = 0.1$$

$$i \times a = \frac{15}{1000}$$

(where a = moles of solute)

Now,

$$\frac{P_o - P_s}{P_o} = i X_{\text{solute}} = i \times \frac{a}{a + \frac{150}{300}}$$

$$= i \times \frac{a}{1/2} = \frac{15/1000}{1/2} = \frac{30}{1000} = 3 \times 10^{-2} = 3$$

Question2

The osmotic pressure of a living cell is 12 atm at 300 K. The strength of sodium chloride solution that is isotonic with the living cell at this temperature is _____ g L^{-1} . (Nearest integer)

Given : $R = 0.08 \text{ L atm K}^{-1} \text{ mol}^{-1}$

Assume complete dissociation of NaCl

(Given: Molar mass of Na and Cl are 23 and 35.5 g mol^{-1} respectively.)

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Answer: 15

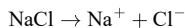
Solution:

Osmotic pressure (π) = 12 atm

Temperature (T) = 300 K

Gas constant (R) = $0.08 \text{ L atm K}^{-1} \text{ mol}^{-1}$

Sodium chloride (NaCl) is an electrolyte that breaks into ions in an aqueous solution:



Since it is given to assume complete dissociation, one unit of NaCl produces 2 ions.

Therefore, the van 't Hoff factor, $i = 2$.

According to the formula for osmotic pressure:

$$\pi = iCRT$$

Substitute the given values into the equation:

$$12 = 2 \times C \times 0.08 \times 300$$

$$12 = 2 \times 24 \times C$$

$$12 = 48 \times C$$

$$C = \frac{12}{48}$$

$$C = 0.25 \text{ mol L}^{-1}$$

Molar mass of NaCl = Molar mass of Na + Molar mass of Cl

$$\text{Molar mass of NaCl} = 23 + 35.5 = 58.5 \text{ g mol}^{-1}$$

To convert the concentration from moles per liter to grams per liter (strength), multiply the molar concentration (C) by the molar mass:

$$\text{Strength}(\text{g L}^{-1}) = \text{Molar concentration} \times \text{Molar mass}$$

$$\text{Strength} = 0.25 \text{ mol L}^{-1} \times 58.5 \text{ g mol}^{-1}$$

$$\text{Strength} = 14.625 \text{ g L}^{-1}$$

Rounding off to the nearest integer, we get 15.

The strength of the sodium chloride solution is **15** g L⁻¹.

Question3

Two liquids A and B form an ideal solution. At 320 K , the vapour pressure of the solution, containing 3 mol of A and 1 mol of B is 500 mm Hg . At the same temperature, if 1 mol of A is further added to this solution, vapour pressure of the solution increases by 20 mm Hg . Vapour pressure (in mm Hg) of B in pure state is ____ . (Nearest integer)

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Answer: 200

Solution:

For an ideal solution, by Raoult's law:

$$P_{\text{total}} = x_A P_A^0 + x_B P_B^0$$

First solution (3 mol A, 1 mol B)

$$\text{Total moles} = 3 + 1 = 4$$

$$x_A = \frac{3}{4}, \quad x_B = \frac{1}{4}$$

Given $P_{\text{total}} = 500 \text{ mm Hg}$:

$$500 = \frac{3}{4} P_A^0 + \frac{1}{4} P_B^0$$

Multiply by 4:

$$2000 = 3P_A^0 + P_B^0 \quad \dots (1)$$

After adding 1 mol A (4 mol A, 1 mol B)

$$\text{Total moles} = 4 + 1 = 5$$

$$x_A = \frac{4}{5}, \quad x_B = \frac{1}{5}$$

$$\text{New vapour pressure} = 500 + 20 = 520 \text{ mm Hg:}$$

$$520 = \frac{4}{5}P_A^0 + \frac{1}{5}P_B^0$$

Multiply by 5:

$$2600 = 4P_A^0 + P_B^0 \quad \dots (2)$$

Solve

Subtract (1) from (2):

$$2600 - 2000 = (4P_A^0 + P_B^0) - (3P_A^0 + P_B^0)$$

$$600 = P_A^0$$

Put in (1):

$$2000 = 3(600) + P_B^0 = 1800 + P_B^0$$

$$P_B^0 = 200 \text{ mm Hg}$$

Answer: 200

Question4

A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg . If the same solution boils at 320 K , then the number of moles of the solvent present in the solution is ____ . (Nearest integer)

[Given: boiling point of the pure solvent = 319.5 K,

K_b of the solvent = 0.3 K kg mol⁻¹]

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Answer: 5

Solution:

For a non-volatile, non-electrolyte solute, first use **relative lowering of vapour pressure**:

$$\frac{p^0 - p}{p^0} = x_2$$

where x_2 is the mole fraction of solute.

Given:

$$p^0 = 760 \text{ mm Hg}, \quad p = 750 \text{ mm Hg}$$

So,

$$x_2 = \frac{760 - 750}{760} = \frac{10}{760} = \frac{1}{76}$$

Now,

$$x_2 = \frac{n_2}{n_1 + n_2}$$

where n_1 = moles of solvent and n_2 = moles of solute.

Hence,

$$\frac{n_2}{n_1 + n_2} = \frac{1}{76}$$

This gives

$$76n_2 = n_1 + n_2$$

$$n_1 = 75n_2$$

Now use elevation in boiling point:

$$\Delta T_b = K_b m$$

Given:

$$\Delta T_b = 320 - 319.5 = 0.5 \text{ K}$$

and

$$K_b = 0.3 \text{ K kg mol}^{-1}$$

So molality,

$$m = \frac{\Delta T_b}{K_b} = \frac{0.5}{0.3} = \frac{5}{3}$$

But

$$m = \frac{n_2}{\text{mass of solvent in kg}}$$

$$\text{Mass of solvent} = 40 \text{ g} = 0.040 \text{ kg}$$

Thus,

$$\frac{n_2}{0.040} = \frac{5}{3}$$

$$n_2 = \frac{5}{3} \times 0.040 = 0.0667 \text{ mol}$$

Now,

$$n_1 = 75n_2 = 75 \times 0.0667 = 5.00 \text{ mol}$$

Therefore, the number of moles of solvent in the solution is

$$\boxed{5}$$

Nearest integer: $\boxed{5}$

Question5

20 g hemoglobin in a 1 L aqueous solution (A) at 300 K is separated from pure water by semi permeable membrane. At equilibrium the height of solution in a tube dipped in a solution (A) is found to be 80.0 mm higher than the tube dipped in water.

The molar mass of hemoglobin is ____ kgmol^{-1} . (Nearest integer)

(Given : $g = 10 \text{ m s}^{-2}$, $R = 8.3 \text{ kPa dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$, density of solution = 1000 kg m^{-3})

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Answer: 62

Solution:

For a solution separated from pure water by a semipermeable membrane, at equilibrium the osmotic pressure is balanced by the hydrostatic pressure.

So,

$$\pi = h\rho g$$

Given:

- $h = 80.0 \text{ mm} = 0.080 \text{ m}$
- $\rho = 1000 \text{ kg m}^{-3}$
- $g = 10 \text{ m s}^{-2}$

Therefore,

$$\pi = 0.080 \times 1000 \times 10$$

$$\pi = 800 \text{ Pa} = 0.8 \text{ kPa}$$

Now, using the osmotic pressure formula:

$$\pi = CRT$$

$$\text{where } C = \frac{n}{V}$$

So,

$$\pi = \frac{w}{MV} RT$$

Here,

- $w = 20 \text{ g}$
- $M = \text{molar mass}$
- $V = 1 \text{ L} = 1 \text{ dm}^3$
- $R = 8.3 \text{ kPa dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$
- $T = 300 \text{ K}$
- $\pi = 0.8 \text{ kPa}$

Substituting,

$$0.8 = \frac{20}{M \times 1} \times 8.3 \times 300$$

$$0.8 = \frac{20 \times 8.3 \times 300}{M}$$

$$M = \frac{20 \times 8.3 \times 300}{0.8}$$

$$M = 62250 \text{ g mol}^{-1}$$

Convert into kg mol^{-1} :

$$M = 62.25 \text{ kg mol}^{-1}$$

Nearest integer:

62 kg mol^{-1}

Question6

Elements P and Q form two types of non-volatile, non-ionizable compounds PQ and PQ₂. When 1 g of PQ is dissolved in 50 g of solvent 'A', ΔT_b was 1.176 K while when 1 g of PQ₂ is dissolved in 50 g of solvent 'A', ΔT_b was 0.689 K . (K_b of 'A' = 5 K kg mol⁻¹). The molar masses of elements P and Q (in gmol⁻¹) respectively, are :

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Options:

A.

25, 60

B.

60, 25

C.

65, 145

D.

70, 110

Answer: A

Solution:

For a non-volatile, non-ionizable solute,

$$\Delta T_b = K_b m$$

and

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

Given: mass of solvent = 50 g = 0.05 kg, mass of solute = 1 g.

So for any solute of molar mass M ,

$$m = \frac{\frac{1}{M}}{0.05} = \frac{1}{0.05M} = \frac{20}{M}$$

Hence,

$$\Delta T_b = K_b \cdot \frac{20}{M} = 5 \cdot \frac{20}{M} = \frac{100}{M}$$

So,

$$M = \frac{100}{\Delta T_b}$$

1) For PQ

$$M(PQ) = \frac{100}{1.176} \approx 85.03 \approx 85$$

Let molar masses of elements be $P = x$ and $Q = y$:

$$x + y = 85 \quad \dots (1)$$

2) For PQ_2

$$M(PQ_2) = \frac{100}{0.689} \approx 145.14 \approx 145$$

So,

$$x + 2y = 145 \quad \dots (2)$$

Subtract (1) from (2):

$$(x + 2y) - (x + y) = 145 - 85 \Rightarrow y = 60$$

Then from (1):

$$x + 60 = 85 \Rightarrow x = 25$$

Therefore, molar masses are:

$$P = 25, \quad Q = 60$$

Correct option: A (25, 60)

Question 7

Given below are two statements :

Statement I : The Henry's law constant K_H is constant with respect to variations in solution's concentration over the range for which the solution is ideally dilute.

Statement II : K_H does not differ for the same solute in different solvents.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are true

B.

Statement I is true but Statement II is false

C.

Statement I is false but Statement II is true

D.

Both Statement I and Statement II are false

Answer: B

Solution:

Statement I: True.

For an **ideally dilute solution**, Henry's law is obeyed:

$$p = K_H x$$

At a **fixed temperature**, K_H remains **constant** over the concentration range where the solution behaves ideally (i.e., dilute enough).

Statement II: False.

K_H depends on the **nature of the gas (solute)** and also the **nature of the solvent** (and temperature). So, for the **same solute**, K_H can be different in different solvents.

Correct option: B (Statement I is true but Statement II is false).

Question8

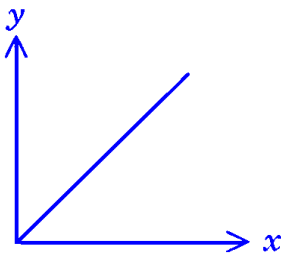
Consider a solution of CO_2 (g) dissolved in water in a closed container.

Which one of the following plots correctly represents variation of log (partial pressure of CO_2 in vapour phase above water) [y-axis] with log (mole fraction of CO_2 in water) [x-axis] at 25°C ?

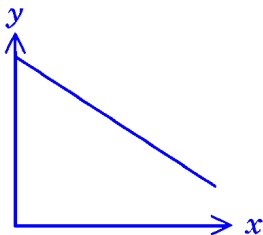
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Options:

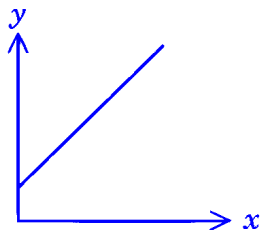
A.



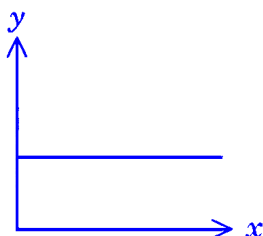
B.



C.



D.



Answer: C

Solution:

From Henry's law (as given in NCERT), the partial pressure of a gas above the solution is directly proportional to its mole fraction in the solution.

So, we write:

$$P(g) = K_H \cdot X(g)$$

$$\log P(g) = \log K_H + \log X(g)$$

Now compare this with the straight-line form $y = c + x$.

Here, $y = \log P(g)$, $x = \log X(g)$, and the constant (intercept) is $\log K_H$.

Therefore, the graph between $\log P$ (on y -axis) and $\log X$ (on x -axis) will be a straight line with slope 1 and y -intercept $\log K_H$.

Question9

At T(K), 100 g of 98% H_2SO_4 (w/w) aqueous solution is mixed with 100 g of 49% H_2SO_4 (w/w) aqueous solution. What is the mole fraction of H_2SO_4 in the resultant solution?

(Given : Atomic mass H = 1u; S = 32u; O = 16u).

(Assume that temperature after mixing remains constant)

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Options:

A.

0.9

B.

0.663

C.

0.1

D.

0.337

Answer: D

Solution:

Total mass of H_2SO_4 in the mixture:

$$= \left(100 \times \frac{98}{100}\right) + \left(100 \times \frac{49}{100}\right) = 147\text{gm}$$

Here, the first solution gives 98 g of H_2SO_4 and the second gives 49 g of H_2SO_4 , so total is 147 g.

Total mass of solution after mixing = $100 + 100 = 200$ g.

So, total mass of H_2O in the mixture:

$$\text{Total weight of } \text{H}_2\text{O} = 200 - 147 = 53\text{gm}$$

$$\text{Moles of } \text{H}_2\text{SO}_4 = \frac{147}{98} \text{ (since molar mass of } \text{H}_2\text{SO}_4 \text{ is } 98 \text{ g mol}^{-1}\text{)}.$$

$$\text{Moles of } \text{H}_2\text{O} = \frac{53}{18} \text{ (since molar mass of } \text{H}_2\text{O} \text{ is } 18 \text{ g mol}^{-1}\text{)}.$$

Now, mole fraction of H_2SO_4 is:

$$\text{Mole fraction of } \text{H}_2\text{SO}_4 = \frac{\frac{147}{98}}{\left(\frac{147}{98} + \frac{53}{18}\right)} = 0.337$$

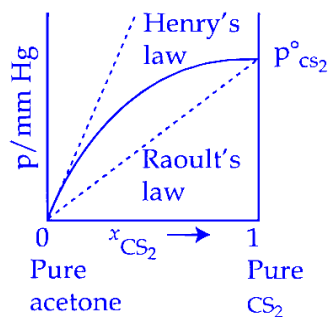
Question10

Which one of the following graphs accurately represents the plot of partial pressure of CS_2 vs its mole fraction in a mixture of acetone and CS_2 at constant temperature?

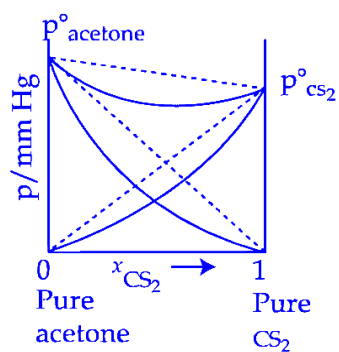
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Options:

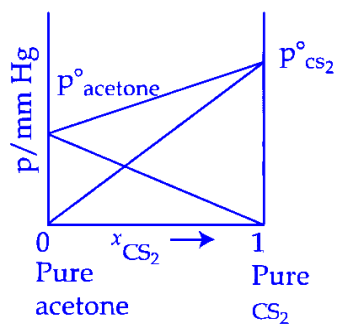
A.



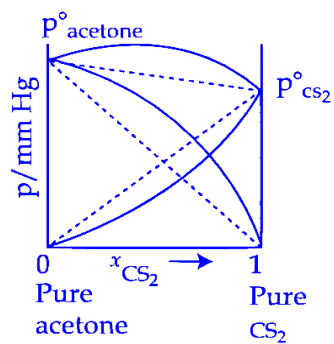
B.



C.



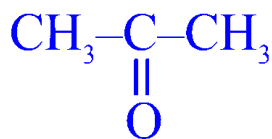
D.



Answer: A

Solution:

Mixture of CS_2 and



show positive deviation $P_{\text{CS}_2} > P_{\text{CS}_2}^\circ \cdot X_{\text{CS}_2}$

Question 11

'W' g of a non-volatile electrolyte solid solute of molar mass 'M' g mol^{-1} when dissolved in 100 mL water, decreases vapour pressure of water from 640 mm Hg to 600 mm Hg. If aqueous solution of the electrolyte boils at 375 K and K_b for water is $0.52 \text{ K kg mol}^{-1}$, then the mole fraction of the electrolyte solute (x_2) in the solution can be expressed as

(Given : density of water = 1 g/mL and boiling point of water = 373 K)

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Options:

A.

$$\frac{16}{2.6} \times \frac{W}{M}$$

B.

$$\frac{1.3}{8} \times \frac{M}{W}$$

C.

$$\frac{2.6}{16} \times \frac{M}{W}$$

D.

$$\frac{1.3}{8} \times \frac{W}{M}$$

Answer: D

Solution:

Boiling point elevation:

$$\Delta T_b = 375 - 373 = 2 \text{ K}$$

$$\text{Mass of water} = 100 \text{ mL} \times 1 \text{ g mL}^{-1} = 100 \text{ g} = 0.1 \text{ kg}$$

Molality:

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}} = \frac{(W/M)}{0.1} = 10 \frac{W}{M}$$

For electrolyte (NCERT):

$$\Delta T_b = i K_b m$$

$$2 = i(0.52) \left(10 \frac{W}{M} \right) = i \left(5.2 \frac{W}{M} \right)$$

$$i = \frac{2}{5.2} \cdot \frac{M}{W} = \frac{5}{13} \cdot \frac{M}{W}$$

Vapour pressure lowering:

$$\frac{\Delta p}{p^*} = \frac{640-600}{640} = \frac{40}{640} = \frac{1}{16}$$

For electrolyte (NCERT):

$$\frac{\Delta p}{p^*} = ix_2$$

So,

$$x_2 = \frac{1/16}{i} = \frac{1}{16} \cdot \frac{1}{\left(\frac{5}{13} \cdot \frac{M}{W}\right)} = \frac{1}{16} \cdot \frac{13}{5} \cdot \frac{W}{M} = \frac{13}{80} \cdot \frac{W}{M}$$

$$x_2 = \left(\frac{1.3}{8}\right) \frac{W}{M}$$

Correct option: D

Question12

A solution is prepared by dissolving 0.3 g of a non-volatile non-electrolyte solute 'A' of molar mass 60 g mol⁻¹ and 0.9 g of a non-volatile non-electrolyte solute ' B ' of molar mass 180 g mol⁻¹ in 100 mLH₂O at 27°C. Osmotic pressure of the solution will be

[Given: R = 0.082 L atm K⁻¹ mol⁻¹]

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Options:

A.

2.46 atm

B.

0.82 atm

C.

1.47 atm

D.

1.23 atm

Answer: A

Solution:

Moles of solute *A*:

$$n_A = \frac{0.3}{60} = 0.005 \text{ mol}$$

Moles of solute *B*:

$$n_B = \frac{0.9}{180} = 0.005 \text{ mol}$$

Total moles of solute particles (both are non-electrolytes, so no dissociation):

$$n = n_A + n_B = 0.005 + 0.005 = 0.01 \text{ mol}$$

Volume of solution (dilute solution, take volume $\approx 100 \text{ mL} = 0.1 \text{ L}$):

$$V = 0.1 \text{ L}$$

Temperature:

$$T = 27^\circ\text{C} = 300 \text{ K}$$

Osmotic pressure:

$$\pi = \frac{n}{V}RT = \frac{0.01}{0.1} \times 0.082 \times 300$$

$$\pi = 0.1 \times 24.6 = 2.46 \text{ atm}$$

Correct option: A) 2.46 atm

Question 13

Two liquids A and B form an ideal solution at temperature T K. At T K, the vapour pressures of pure A and B are 55 and 15 kNm⁻² respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

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Options:

A.

$$0.5217$$

B.

$$0.480$$

C.

$$0.340$$

D.

$$0.663$$

Answer: A

Solution:

For an ideal solution, Raoult's law applies:

$$p_A = x_A P_A^*, \quad p_B = x_B P_B^*$$

Total pressure:

$$P = p_A + p_B = 55x_A + 15(1 - x_A) = 55x_A + 15 - 15x_A = 40x_A + 15$$

Vapour-phase mole fraction (Dalton's law):

$$y_A = \frac{p_A}{P} = \frac{55x_A}{40x_A + 15} = 0.8$$

Solve:

$$55x_A = 0.8(40x_A + 15) = 32x_A + 12$$

$$23x_A = 12 \Rightarrow x_A = \frac{12}{23} = 0.5217$$

Answer: 0.5217 (Option A)

Question14

At 298 K , the mole percentage of $\text{N}_2(\text{g})$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm . What is the mole fraction of $\text{N}_2(\text{g})$ in water at 298 K ? (K_{H} for N_2 is 6.5×10^7 mmHg)

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Options:

A.

$$9.35 \times 10^{-5}$$

B.

$$9.35 \times 10^5$$

C.

$$1.23 \times 10^{-7}$$

D.

$$1.17 \times 10^{-4}$$

Answer: A

Solution:

$$P_{\text{N}_2} = K_{\text{H}} \cdot X_{\text{N}_2}$$

$$P_{\text{N}_2} = 0.8 \times 10 = 8 \text{ atm}$$

$$8 \times 760 = 6.5 \times 10^7 \times X_{\text{N}_2}$$

$$X_{\text{N}_2} = \frac{8 \times 760}{6.5 \times 10^7}$$

$$X_{\text{N}_2} = 9.35 \times 10^{-5}$$

Question15

At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg . The vapour pressure (in mm Hg) of A and B are respectively:

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Options:

A.

$$400, 300$$

B.

$$600, 400$$

C.

$$500, 200$$

D.

300, 200

Answer: C

Solution:

For an ideal solution, by Raoult's law:

$$P_{\text{total}} = x_A P_A^0 + x_B P_B^0$$

where P_A^0 , P_B^0 are vapour pressures of pure liquids A and B.

Step 1: First mixture (2 mol A + 3 mol B)

Total moles = 2 + 3 = 5

$$x_A = \frac{2}{5}, \quad x_B = \frac{3}{5}$$

Given $P_{\text{total}} = 320$ mm Hg:

$$320 = \frac{2}{5} P_A^0 + \frac{3}{5} P_B^0$$

Multiply by 5:

$$1600 = 2P_A^0 + 3P_B^0 \quad \dots (1)$$

Step 2: After adding 1 mol A and 1 mol B

New moles: A = 3, B = 4, total = 7

$$x_A = \frac{3}{7}, \quad x_B = \frac{4}{7}$$

Given new vapour pressure = 328.6 mm Hg:

$$328.6 = \frac{3}{7} P_A^0 + \frac{4}{7} P_B^0$$

Multiply by 7:

$$2300.2 = 3P_A^0 + 4P_B^0 \quad \dots (2)$$

Step 3: Solve the two equations

From (1): $2P_A^0 + 3P_B^0 = 1600$

From (2): $3P_A^0 + 4P_B^0 = 2300.2$

Multiply (1) by 3:

$$6P_A^0 + 9P_B^0 = 4800$$

Multiply (2) by 2:

$$6P_A^0 + 8P_B^0 = 4600.4$$

Subtract:

$$P_B^0 = 4800 - 4600.4 = 199.6 \approx 200$$

Put in (1):

$$2P_A^0 + 3(199.6) = 1600$$

$$2P_A^0 + 598.8 = 1600$$

$$2P_A^0 = 1001.2 \Rightarrow P_A^0 = 500.6 \approx 500$$

Final Answer

Vapour pressures of A and B are approximately:

$$P_A^0 = 500 \text{ mm Hg}, \quad P_B^0 = 200 \text{ mm Hg}$$

Correct option: C (500, 200)

Question 16

Consider the following aqueous solutions.

I. 2.2 g Glucose in 125 mL of solution.

II. 1.9 g Calcium chloride in 250 mL of solution.

III. 9.0 g Urea in 500 mL of solution.

IV. 20.5 g Aluminium sulphate in 750 mL of solution.

The correct increasing order of boiling point of these solutions will be :

[Given : Molar mass in g mol^{-1} : H = 1, C = 12, N = 14, O = 16, Cl = 35.5, Ca = 40, Al = 27 and S = 32]

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Options:

A.

III < I < II < IV

B.

II < III < IV < I

C.

II < III < I < IV

D.

I < II < III < IV

Answer: D

Solution:

$$\Delta T_b = i \cdot k_b \cdot m$$

For dilute solution ($M = m$)

We know that boiling point elevation depends on ΔT_b .

Since k_b is same for water in all cases, we compare only $i \times m$ (or $i \times M$ for dilute solutions).

Molarity	$i \times m$
(I) $M_{\text{glucose}} = \frac{2.2}{180} \times \frac{1000}{125} = 0.098$	0.098×1
(II) $M_{\text{CaCl}_2} = \frac{1.9}{111} \times \frac{1000}{250} = 0.068$	0.068×3
(III) $M_{\text{urea}} = \frac{9}{60} \times \frac{1000}{500} = 0.3$	0.3×1
(IV) $M_{\text{Al}_2(\text{SO}_4)_3} = \frac{20.5}{342} \times \frac{1000}{750} \simeq 0.08$	0.08×5

Now compare the values of $i \times m$:

(I) Glucose: $0.098 \times 1 = 0.098$

(II) CaCl_2 : $0.068 \times 3 = 0.204$

(III) Urea: $0.3 \times 1 = 0.3$

(IV) Aluminium sulphate: $0.08 \times 5 = 0.4$

Order of $\Delta T_b = \text{Al}_2(\text{SO}_4)_3 > \text{Urea} > \text{CaCl}_2 > \text{Glucose}$

So order of BP = $\text{Al}_2(\text{SO}_4)_3 > \text{Urea} > \text{CaCl}_2 > \text{Glucose}$

So Answer will be I < II < III < IV

Question 17

19.5 g of fluoro acetic acid (molar mass = 78 g mol^{-1}) is dissolved in 500 g of water at 298 K. The depression in the freezing point of water was 1°C . What is K_a of fluoro acetic acid?

(For water, $K_f = 1.86 \text{ K kg mol}^{-1}$). Assume molarity and molality to have same values.

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Options:

A.

$$10^{-6}$$

B.

$$4 \times 10^{-4}$$

C.

$$3 \times 10^{-5}$$

D.

$$3 \times 10^{-3}$$

Answer: D

Solution:

Given:

- Mass of fluoroacetic acid = 19.5 g
- Molar mass = 78 g mol^{-1}
- Mass of water = 500 g = 0.5 kg
- Depression in freezing point, $\Delta T_f = 1^\circ\text{C} = 1 \text{ K}$
- $K_f = 1.86 \text{ K kg mol}^{-1}$

We have to find K_a of fluoroacetic acid.

First, calculate the number of moles of acid:

$$n = \frac{19.5}{78} = 0.25 \text{ mol}$$

Now, molality of the solution:

$$m = \frac{0.25}{0.5} = 0.5 \text{ mol kg}^{-1}$$

For an electrolyte,

$$\Delta T_f = iK_fm$$

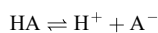
So,

$$1 = i \times 1.86 \times 0.5$$

$$1 = i \times 0.93$$

$$i = \frac{1}{0.93} \approx 1.075$$

Fluoroacetic acid is a weak monoprotic acid:



If degree of dissociation is α , then for this type of acid:

$$i = 1 + \alpha$$

Therefore,

$$\alpha = i - 1 = 1.075 - 1 = 0.075$$

Now use Ostwald's dilution law:

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

Given in the question, assume molarity and molality are same, so

$$C = 0.5$$

Thus,

$$\begin{aligned}K_a &= \frac{0.5 \times (0.075)^2}{1 - 0.075} \\&= \frac{0.5 \times 0.005625}{0.925} \\&= \frac{0.0028125}{0.925} \approx 3.04 \times 10^{-3}\end{aligned}$$

So,

$$K_a \approx 3 \times 10^{-3}$$

Hence, the correct option is:

3×10^{-3}

Option D.

Question 18

Solution A is prepared by dissolving 1 g of a protein (molar mass = 50,000 g mol⁻¹) in 0.5 L of water at 300 K. Its osmotic pressure is x bar. Solution B is made by dissolving 2 g of the same protein in 1 L of water at 300 K. Osmotic pressure of solution B is y bar. Entire solution of A is mixed with entire solution of B at the same temperature. The osmotic pressure of resultant solution is z bar. x , y and z respectively are:

($R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$)

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Options:

A.

$$9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 9.96 \times 10^{-4}$$

B.

$$9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 19.92 \times 10^{-4}$$

C.

$$4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 9.96 \times 10^{-4}$$

D.

$$4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 4.98 \times 10^{-4}$$

Answer: A

Solution:

For osmotic pressure, we use the NCERT relation:

$$\pi = CRT = \frac{n}{V} RT$$

where

- π = osmotic pressure
- n = number of moles of solute
- V = volume of solution
- $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$
- $T = 300 \text{ K}$

Protein molar mass is:

$$M = 50000 \text{ g mol}^{-1}$$

Now solve each part.

For solution A

Mass of protein = 1 g

Moles of protein:

$$n_A = \frac{1}{50000} = 2 \times 10^{-5} \text{ mol}$$

Volume:

$$V_A = 0.5 \text{ L}$$

So concentration:

$$C_A = \frac{2 \times 10^{-5}}{0.5} = 4 \times 10^{-5} \text{ mol L}^{-1}$$

Hence osmotic pressure:

$$x = \pi_A = C_A RT$$

$$x = (4 \times 10^{-5})(0.083)(300)$$

First calculate:

$$0.083 \times 300 = 24.9$$

So,

$$x = 4 \times 10^{-5} \times 24.9 = 9.96 \times 10^{-4} \text{ bar}$$

For solution B

Mass of protein = 2 g

Moles:

$$n_B = \frac{2}{50000} = 4 \times 10^{-5} \text{ mol}$$

Volume:

$$V_B = 1 \text{ L}$$

Concentration:

$$C_B = \frac{4 \times 10^{-5}}{1} = 4 \times 10^{-5} \text{ mol L}^{-1}$$

Thus,

$$y = \pi_B = C_B RT$$

$$y = (4 \times 10^{-5})(0.083)(300)$$

$$y = 9.96 \times 10^{-4} \text{ bar}$$

After mixing A and B

Total moles of protein:

$$n_{\text{total}} = 2 \times 10^{-5} + 4 \times 10^{-5} = 6 \times 10^{-5} \text{ mol}$$

Total volume:

$$V_{\text{total}} = 0.5 + 1 = 1.5 \text{ L}$$

Concentration after mixing:

$$C_{\text{mix}} = \frac{6 \times 10^{-5}}{1.5} = 4 \times 10^{-5} \text{ mol L}^{-1}$$

Again,

$$z = \pi_{\text{mix}} = C_{\text{mix}} RT$$

$$z = (4 \times 10^{-5})(0.083)(300) = 9.96 \times 10^{-4} \text{ bar}$$

So,

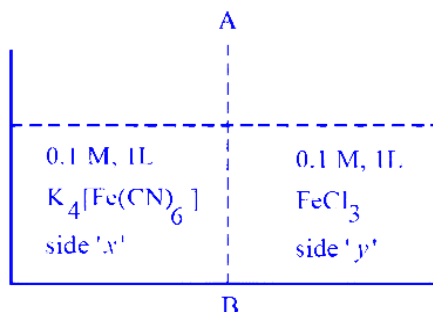
$$x = 9.96 \times 10^{-4} \text{ bar}, \quad y = 9.96 \times 10^{-4} \text{ bar}, \quad z = 9.96 \times 10^{-4} \text{ bar}$$

Therefore, the correct option is:

Option A

Question 19

At 27°C , 0.1M , 1 L $\text{K}_4[\text{Fe}(\text{CN})_6]$ aqueous solution and 0.1M , 1 L FeCl_3 aqueous solution are placed in a container separated by a semi permeable membrane AB. Assume complete dissociation of both the solutes. Which of the following statement is correct?



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Options:

A.

Blue color is formed on both sides.

B.

Ionic solutes in aqueous solution can pass through semi-permeable membrane.

C.

Solution on side 'y' is hypotonic.

D.

To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side 'x'.

Answer: C

Solution:

(A) A semi-permeable membrane (SPM) allows only solvent molecules (water) to pass through it. It does not allow ions to pass.

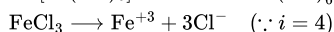
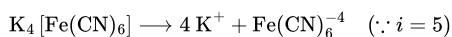
So, the solute ions from the two sides cannot come together. Therefore, no blue colour (due to mixing of ions) will appear on either side.

(B) Because ions cannot cross the SPM, only solvent molecules can move from one side to the other depending on osmotic pressure.

(C) A hypotonic solution is the one which has lower osmotic pressure (lower “effective” concentration of solute particles).

We use the NCERT relation for osmotic pressure:

As osmotic pressure $(\pi) = iCRT$



for side x : $\pi = 5CRT = 0.5RT$ ($\because c = 0.1\text{ m}$)

y : $\pi = 0.4RT$ ($\because c = 0.1\text{ m}$)

effective concentration of $x = i \times C = 0.5\text{M}$

$y = i \times C = 0.4\text{ m}$

Since, effective concentration of y is less than x . Hence, y is hypotonic.

(D) In reverse osmosis, pressure is applied on the side having higher osmotic pressure (here, side x). But this applied external pressure must be greater than the osmotic pressure difference between the two sides.

i.e. $P_{\text{applied}} > (\pi_x - \pi_y)$

Question20

When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was $x\%$ of vapor pressure of pure solvent. What is $x\%$?

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Options:

A.

50%

B.

60%

C.

70%

D.

80%

Answer: D

Solution:

Let the mole fraction of the solvent be x_1 and that of the solute be x_2 .

Given:

Moles of solute $n_2 = 0.25$

Moles of solvent $n_1 = 1$

$$x_1 = \frac{n_1}{n_1 + n_2} = \frac{1}{1 + 0.25} = \frac{1}{1.25} = 0.8$$

$$x_2 = 1 - x_1 = 1 - 0.8 = 0.2$$

For a non-volatile solute, the vapor pressure of the solution is given by:

$$P_{\text{solution}} = x_1 P^0$$

where P^0 is the vapor pressure of the pure solvent.

$$\frac{P_{\text{solution}}}{P^0} \times 100 = x_1 \times 100 = 0.8 \times 100 = 80\%$$

Final Answer:

$$x = 80\%$$

Option D: 80%

Question21

Given below are two statements :

Chamber 1	Semi-permeable membrane	Chamber 2
18 g glucose in 100 mL aqueous solution		30 g glucose in 250 mL aqueous solution

Statement I : H_2O molecules move from the chamber 1 to chamber 2 .

Statement II : The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass = 174 g/mol) in 2 L of water (at 27°C) is 0.0107 bar. (Given: $R = 0.083\text{dm}^3 \text{ bar K}^{-1} \text{ mol}^{-1}$ and assume complete dissociation of electrolyte)

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

In chamber (I), the concentration of solute is calculated as:

$$C_1 = \frac{18/180}{0.1} = 1 \text{ M}$$

In chamber (II), the concentration is given by:

$$C_2 = \frac{30/180}{250/1000} = \frac{2}{3} \text{ M}$$

Since osmotic pressure depends on concentration and $\pi_1 > \pi_2$, water will move from the side having lower osmotic pressure (chamber II) to the side having higher osmotic pressure (chamber I).

Osmotic pressure is calculated using the formula:

$$\pi = iCRT$$

For potassium sulfate (K_2SO_4), the van't Hoff factor is $i = 3$.

So,

$$\pi = 3 \times \left(\frac{50 \times 10^{-3} / 174}{2} \right) \times 0.083 \times 300$$

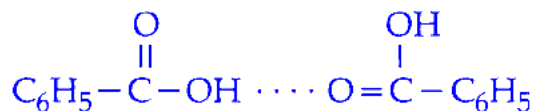
On solving,

$$\pi = 0.0107 \text{ bar}$$

Question 22

Which of the following statements are not correct?

- A. For water, magnitude of K_b is more than the magnitude of K_f .
- B. The elevation in boiling point of water when a non-volatile solute is added to it is larger in magnitude than its depression in freezing point.
- C. Osmotic pressure measurement is preferred over any other colligative property to determine molar mass of proteins and polymers.
- D. The dimerised form of benzoic acid in benzene is



Choose the correct answer from the options given below :

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Options:

- A.
- A and B only
- B.
- A and D only
- C.
- A, B and D only
- D.
- A, C and D only

Answer: C

Solution:

$$(A) K_b \text{ for water} = 0.512 \frac{^\circ\text{C-Kg}}{\text{mol}}$$

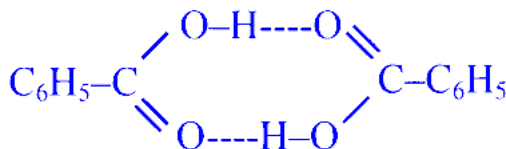
$$K_f \text{ for water} = 1.86 \frac{^\circ\text{C-Kg}}{\text{mol}}$$

So, $K_b < K_f$ for water

(B) Since, $K_b < K_f$ so $\Delta T_b < \Delta T_f$ [same equimolal solution]

(C) Osmotic pressure is used to determine molar mass of proteins & polymers

(D) dimerised form of benzoic acid in benzene



Option (A), (B) & (D) are incorrect

Question 23

Given below are two statements :

Given : Molar mass of C, H, O, Cl are 12, 1, 16 and 35.5 g mol⁻¹, respectively

Statement I : In 30% (w/w) solution of methanol in CCl₄ (at T K), the mole fraction of CCl₄ is equal to 0.33 .

Statement II : Mixture of methanol and CCl₄ shows positive deviation from Raoult's law. In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: A

Solution:

- Molar masses:
- C = 12 g mol⁻¹
- H = 1 g mol⁻¹
- O = 16 g mol⁻¹
- Cl = 35.5 g mol⁻¹

Formula masses:

- For methanol, CH₃OH:

$$M = 12 + (3 \times 1) + 16 + 1 = 32 \text{ g mol}^{-1}$$
- For carbon tetrachloride, CCl₄:

$$M = 12 + (4 \times 35.5) = 12 + 142 = 154 \text{ g mol}^{-1}$$

This means in **100 g of solution:**

- Mass of methanol = 30 g
- Mass of CCl₄ = 70 g

$$n_{\text{methanol}} = \frac{30}{32} = 0.9375 \text{ mol}$$

$$n_{\text{CCl}_4} = \frac{70}{154} = 0.4545 \text{ mol}$$

$$\text{Total moles} = 0.9375 + 0.4545 = 1.392$$

So,

$$x_{\text{methanol}} = \frac{0.9375}{1.392} = 0.674$$

$$x_{\text{CCl}_4} = \frac{0.4545}{1.392} = 0.326 \approx 0.33$$

Hence, **Statement I is correct.**

- Methanol is polar and forms hydrogen bonds.
- Carbon tetrachloride is non-polar.
- When they are mixed, the H-bonding between methanol molecules weakens (since CCl₄ cannot form hydrogen bonds with methanol).
→ The intermolecular attraction between unlike molecules decreases.

As a result, the escaping tendency of molecules increases and **vapour pressure increases**, leading to **positive deviation** from Raoult's law.

Hence, **Statement II is also correct.**

Final Answer

Both Statement I and Statement II are true.

Option A

Question 24

If A_2B is 30% ionised in an aqueous solution, then the value of van't Hoff factor (i) is _____ $\times 10^{-1}$.

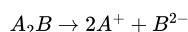
JEE Main 2025 (Online) 29th January Morning Shift

Answer: 16

Solution:

Percent ionisation of $A_2B = 30\%$

The dissociation of A_2B in aqueous solution can be represented as



1 mole of A_2B produces 2 moles of A^+ and 1 mole of B^{2-} ions.

The total number of moles of ions produced (n) from the dissociation is

$$n = 2 + 1 = 3$$

The degree of dissociation given that A_2B is 30% ionised, the degree of dissociation (λ) or degree of ionisation

$$\lambda = \frac{30}{100} = 0.3$$

Van't Hoff factor (i) can be calculated using the formula,

$$i = 1 + (n - 1)\lambda$$

Substitute the values of n and λ as,

$$i = 1 + (3 - 1) \times 0.3$$

$$= 1 + 2 \times 0.3$$

$$= 1 + 0.6$$

$$= 1.6$$

$$= 16 \times 10^{-1}$$

Question25

When 1 g each of compounds AB and AB₂ are dissolved in 15 g of water separately, they increased the boiling point of water by 2.7 K and 1.5 K respectively. The atomic mass of A (in amu) is _____ $\times 10^{-1}$ (Nearest integer)

(Given : Molal boiling point elevation constant is $0.5 \text{ K kg mol}^{-1}$)

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Answer: 25

Solution:

For AB

$$\Delta T_b = 2.7 \text{ K}$$

$$2.7 = 1 \times 0.5 \times m$$

$$m = \frac{27}{5}$$

Let molar mass of AB = x .

$$\text{So } \frac{1/x}{15} \times 1000 = \frac{27}{5}$$

$$x = 12.34$$

For AB₂

$$\Delta T_b = 1.5 \text{ K}$$

$$1.5 = 1 \times 0.5 \times m$$

$$m = 3$$

Let molar mass of AB₂ = y

$$\text{So } \frac{1/y}{15} \times 1000 = 3$$

$$y = \frac{1000}{45}$$

$$y = 22.22$$

Now let a and b be atomic masses of A and B respectively, then

$$A + b = 12.34 \quad \dots\dots (i)$$

$$A + 2b = 22.22 \quad \dots\dots (ii)$$

$$B = 22.22 - 12.34 = 9.88$$

$$\text{Now } a = 12.34 - 9.88 = 2.46$$

$$= 24.6 \times 10^{-1} = 25 \times 10^{-1}$$

Question26

Sea water, which can be considered as a 6 molar (6M) solution of NaCl , has a density of 2 g mL^{-1} . The concentration of dissolved oxygen (O₂) in sea water is 5.8 ppm . Then the concentration of dissolved oxygen (O₂) in sea water, is $x \times 10^{-4}$ m.

$x =$ _____. (Nearest integer)

Given: Molar mass of NaCl is 58.5 g mol^{-1}

Molar mass of O₂ is 32 g mol^{-1}

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Answer: 2

Solution:

Sea water is 6 Molar in NaCl , So 1000 ml of sea water contains 6 mol of NaCl .

$$\text{mass of solution} = \text{Volume} \times \text{density}$$

$$= 1000 \times 2$$

$$\text{mass of solution} = 2000 \text{ g}$$

$$\text{ppm} = \frac{\text{mass of O}_2}{2000} \times 10^6$$

$$\text{mass of O}_2 = 5.8 \times 2 \times 10^{-3}$$

$$= 1.16 \times 10^{-2} \text{ g}$$

$$\text{molality for O}_2 = \frac{1.16 \times 10^{-2} / 32}{(2000 - 6 \times 58.5)} \times 1000$$

$$= \frac{1.16 \times 10}{32 \times 1649}$$

$$= 0.000219$$

$$= 2.19 \times 10^{-4}$$

$$\text{Correct answer} \Rightarrow 2$$

Question27

Arrange the following solutions in order of their increasing boiling points.

(i) 10^{-4}MNaCl

(ii) 10^{-4}M Urea

(iii) 10^{-3}MNaCl

(iv) 10^{-2}MNaCl

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Options:

A. (i) < (ii) < (iii) < (iv)

B. (ii) < (i) < (iii) < (iv)

C. (iv) < (iii) < (i) < (ii)

D. (ii) < (i) $\equiv$ (iii) < (iv)

Answer: B

Solution:

Step 1: Identify the van't Hoff factor (i) for each solute

NaCl dissociates (ideally) into two ions:



so $i \approx 2$.

Urea ($\text{CH}_4\text{N}_2\text{O}$) is a **non-electrolyte** (does not dissociate), so $i = 1$.

Step 2: Effective molar concentration of particles

The total particle concentration for each solution is approximately ($i \times$ molarity).

(i) $10^{-4} M$ NaCl

Effective concentration = $2 \times 10^{-4} = 2 \times 10^{-4}$.

(ii) $10^{-4} M$ Urea

Effective concentration = $1 \times 10^{-4} = 1 \times 10^{-4}$.

(iii) $10^{-3} M$ NaCl

Effective concentration = $2 \times 10^{-3} = 2 \times 10^{-3}$.

(iv) $10^{-2} M$ NaCl

Effective concentration = $2 \times 10^{-2} = 2 \times 10^{-2}$.

Step 3: Compare to rank the boiling points

A larger total particle concentration (and hence larger colligative effect) corresponds to a **higher boiling point**. Arrange from **lowest** to **highest**:

Lowest: $10^{-4} M$ Urea [1×10^{-4}]

Next: $10^{-4} M$ NaCl [2×10^{-4}]

Next: $10^{-3} M$ NaCl [2×10^{-3}]

Highest: $10^{-2} M$ NaCl [2×10^{-2}]

Hence, in the format (ii) < (i) < (iii) < (iv).

Final Answer

(ii) < (i) < (iii) < (iv) (Option B)

Question 28

Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the mole fractions of component 1 in liquid and vapour phase, respectively. The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as :

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Options:

A. $\frac{P_1^0}{P_2^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

B. $\frac{P_2^0}{P_1^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

C. $\frac{P_2^0}{P_1^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

D. $\frac{P_1^0}{P_2^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

Answer: D

Solution:

For a binary solution of two volatile liquid components labeled 1 and 2, let x_1 and y_1 represent the mole fractions of component 1 in the liquid and vapor phases, respectively. The linear relationship between the inverse of these mole fractions is plotted as $\frac{1}{x_1}$ versus $\frac{1}{y_1}$.

To derive the slope and intercept of this linear plot, consider the following calculations:

Using Raoult's Law for a Liquid Solution:

For a liquid solution with volatile components 1 and 2:

$$P_1 = P_T \cdot y_1 = P_1^\circ \cdot x_1$$

Therefore:

$$\frac{P_T}{x_1} = \frac{P_1^\circ}{y_1}$$

Rearranging the Equation:

By substituting and rearranging, we have:

$$\frac{P_2^\circ + x_1(P_1^\circ - P_2^\circ)}{x_1} = \frac{P_1^\circ}{y_1}$$

Simplifying further:

$$\frac{P_2^\circ}{x_1} + (P_1^\circ - P_2^\circ) = \frac{P_1^\circ}{y_1}$$

Expressing $\frac{1}{x_1}$:

Solving for $\frac{1}{x_1}$, we obtain:

$$\frac{1}{x_1} = \left(\frac{P_1^\circ}{P_2^\circ} \right) \left(\frac{1}{y_1} \right) + \left(\frac{P_2^\circ - P_1^\circ}{P_2^\circ} \right)$$

Determining the Slope and Intercept:

The slope of the line is:

$$\text{Slope} = \frac{P_1^\circ}{P_2^\circ}$$

The intercept of the line is:

$$\text{Intercept} = \frac{P_2^\circ - P_1^\circ}{P_2^\circ}$$

In summary, for the plot of $\frac{1}{x_1}$ against $\frac{1}{y_1}$, the slope is $\frac{P_1^\circ}{P_2^\circ}$ and the intercept is $\frac{P_2^\circ - P_1^\circ}{P_2^\circ}$.

Question 29

When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg . The mole fraction of the solute in the solution is 0.2 . What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg ?

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Options:

- A. 0.2
- B. 0.4
- C. 0.8
- D. 0.6

Answer: D

Solution:

When a non-volatile solute is added to a solvent, it causes the vapour pressure of the solvent to decrease. In this scenario, when the vapour pressure decreases by 10 mm of Hg, the mole fraction of the solute in the solution is 0.2.

By understanding the relationship between vapour pressure change and mole fraction, we see that:

The change in vapour pressure ($P^\circ - P$) is directly proportional to the mole fraction of the solute (X_{solute}).

Therefore, if a 10 mm of Hg decrease corresponds to a mole fraction of 0.2, then a 20 mm of Hg decrease would correspond to a mole fraction of 0.4.

To find the mole fraction of the solvent (X_{solvent}), we use the formula:

$$X_{\text{solvent}} = 1 - X_{\text{solute}}$$

Substituting the value we found:

$$X_{\text{solvent}} = 1 - 0.4 = 0.6$$

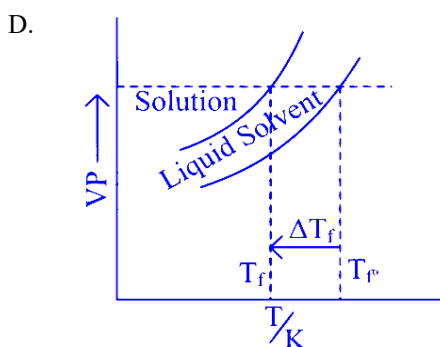
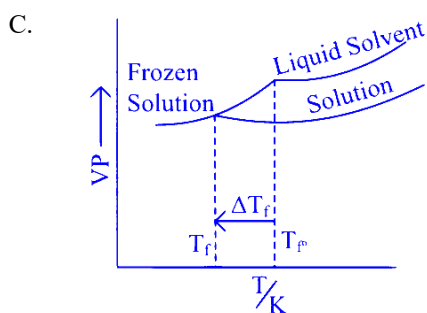
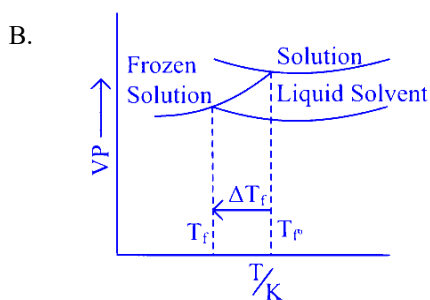
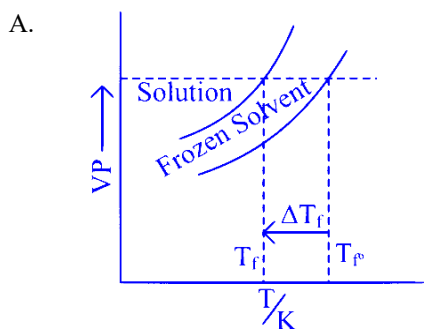
Thus, when the vapour pressure decreases by 20 mm of Hg, the mole fraction of the solvent is 0.6.

Question 30

Consider the given plots of vapour pressure (VP) vs temperature (T/K). Which amongst the following options is correct graphical representation showing ΔT_f , depression in the freezing point of a solution?

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Options:



Answer: C

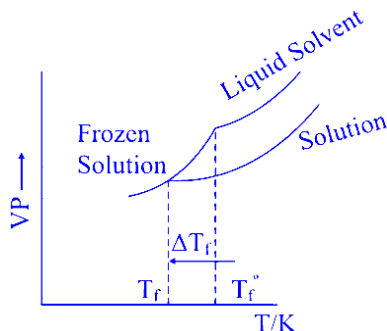
Solution:

On adding non-volatile solute in a solvent, the freezing point of solution decreases.

$$T_f < T_f^0$$

F.P. of solution < F.P. of pure solvent

Also V.P. of solution decreases on adding nonvolatile solute in a solvent.



Question31

What is the freezing point depression constant of a solvent, 50 g of which contain 1 g non volatile solute (molar mass 256 g mol^{-1}) and the decrease in freezing point is 0.40 K ?

JEE Main 2025 (Online) 28th January Morning Shift

Options:

- A. $4.43 \text{ K kg mol}^{-1}$
- B. $3.72 \text{ K kg mol}^{-1}$
- C. $5.12 \text{ K kg mol}^{-1}$
- D. $1.86 \text{ K kg mol}^{-1}$

Answer: C

Solution:

To find the freezing point depression constant (K_f) of the solvent, we use the formula for freezing point depression:

$$\Delta T_f = K_f \cdot m$$

Given:

The decrease in freezing point ΔT_f is 0.40 K .

The mass of the solute is 1 g and its molar mass is 256 g/mol .

The mass of the solvent is 50 g (or 0.050 kg).

First, calculate the molality (m):

Molality is defined as the moles of solute per kilogram of solvent.

Calculate moles of solute:

$$\text{Moles of solute} = \frac{1 \text{ g}}{256 \text{ g/mol}} = \frac{1}{256} \text{ mol}$$

Calculate molality (m):

$$m = \frac{\frac{1}{256} \text{ mol}}{0.050 \text{ kg}} = \frac{1}{256 \times 0.050} \text{ mol/kg}$$

Now, substitute into the formula to find K_f :

$$0.4 = K_f \cdot \frac{1}{256 \times 0.050}$$

Solving for K_f :

$$K_f = 0.4 \cdot 256 \times 0.050 = 5.12 \text{ K kg/mol}$$

Thus, the freezing point depression constant of the solvent is 5.12 K kg/mol.

Question32

Assume a living cell with 0.9% (w/w) of glucose solution (aqueous). This cell is immersed in another solution having equal mole fraction of glucose and water. (Consider the data upto first decimal place only)

The cell will :

JEE Main 2025 (Online) 28th January Evening Shift

Options:

- A. shrink since solution is 0.45% (ω/ω) as a result of association of glucose molecules (due to hydrogen bonding)
- B. shrink since solution is 0.5% (ω/ω)
- C. show no change in volume since solution is 0.9% (ω/ω)
- D. swell up since solution is 1% (ω/ω)

Answer: A

Solution:

Living cell = 0.9gm in 100 gm of solution %w/w = 0.9

Solution is have equal moles of glucose and water = 0.5

Weight of solution = $0.5 \times 180 + 0.5 \times 18 = 99\text{gm}$ %w/w $\simeq 90\%$

Concentrated solution = Cell will shrink

Question33

1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.015°C , while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C .

$$K_b(\text{H}_2\text{O}) = 0.52 \text{ kg mol}^{-1}$$

Which of the following is correct?

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Options:

A.

AX_2 and AY_2 (both) are completely unionised.

B.

AX_2 and AY_2 (both) are fully ionised.

C.

AX_2 is completely unionised while AY_2 is fully ionised.

D.

AX_2 is fully ionised while AY_2 is completely unionised.

Answer: D

Solution:

Mass of $\text{Ax}_2 = 1.24 \text{ g (solute)}$

Molarmass of $\text{AX}_2 = 124 \text{ g mol}^{-1}$

Mass of water = 1 kg (solvent.)

Boiling point of water = 100°C

Boiling point of water after adding solute $\text{AX}_2 = 100.0156^\circ\text{C}$

Mass of $\text{AY}_2 = 25.4 \text{ g (solute)}$

Molarmass of $\text{AY}_2 = 250 \text{ g mol}^{-1}$

Mass of water = 2 kg (Solvent)

Boiling point of water $\mp 100^\circ\text{C}$

Boiling point of water after adding solute $\text{Ay}_2 = 100.0260^\circ\text{C}$

$$K_b(\text{H}_2\text{O}) = 0.52 \text{ K kg mol}^{-1} = 0.520^\circ\text{C kg mol}^{-1}$$

The ionisation of Ax_2 and AY_2 can be determined by calculating Van't Hoff factor.

The phenomenon given in the question is elevation in boiling point. The boiling point of solvent increases when another compound (solute) is added to it.

$$\text{Relation: } \Delta T_b = K_b \cdot i \cdot m$$

$$\Delta T_b = T_b - T_b^0$$

$\Delta T_b \rightarrow$ Boiling point elevation

$T_b \rightarrow$ Boiling point of solution (solvent + solute)

$T_b^0 \rightarrow$ Boiling point of solvent

$K_b \rightarrow$ Molal elevation constant

$i \rightarrow$ Van't Hoff factor

$$m \rightarrow \text{Molality} = \frac{\text{Number of moles}}{\text{kg of solvent}}, \text{ Moles} = \frac{\text{Mass}}{\text{Molarmass}}$$

For AX_2 and AY_2 (solute), Van't Hoff factor represents how many particles a solute dissociates into when dissolved in a solvent (water).

For non-electrolytes, $i = 1$

AX_2 :

$$\text{Moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{1.249}{124 \text{ g mol}^{-1}} = 0.01 \text{ mol}$$

$$\text{Molality} = \frac{\text{Moles}}{k_g \text{ of solvent}} = \frac{0.01 \text{ mol}}{1 \text{ kg}} = 0.01 \text{ mol kg}^{-1}$$

$$\Delta T_b = K_b \times i \times m$$

$$i = \frac{\Delta T_b}{k_b \times m}$$

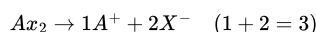
$$= \frac{T_b - T_b^0}{K_b \times m}$$

Substitute values as,

$$i = \frac{(100.0156^\circ\text{C} - 100^\circ\text{C})}{0.52^\circ\text{C kg mol}^{-1} \times 0.01 \text{ mol kg}^{-1}}$$

$$= \frac{0.0156^\circ\text{C}}{0.52 \times 0.01^\circ\text{C}} = 3$$

$i = 3$ means, there are 3 particles in solution after AX_2 dissolved in water.



Ax_2 is completely ionised.

AY_2 :

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}} = \frac{25.4 \text{ g}}{250 \text{ g mol}^{-1}} = 0.1016 \text{ mol}$$

$$\text{Molality} = \frac{\text{Moles}}{\text{kg of solvent}} = \frac{0.1016 \text{ mol}}{2 \text{ kg}} = 0.0508 \text{ mol kg}^{-1}$$

$$\Delta T_b = k_b \times i \times m$$

$$i = \frac{\Delta T_b}{k_b \times m}$$

$$= \frac{T_b - T_b^0}{k_b \times m}$$

Substitute values as,

$$i = \frac{(100.0260^\circ\text{C} - 100^\circ\text{C})}{0.52^\circ\text{C kg mol}^{-1} \times 0.0508 \text{ mol kg}^{-1}}$$

$$= \frac{0.0260^\circ\text{C}}{0.52 \times 0.0508^\circ\text{C}}$$

$$= 0.98$$

$$\approx 1$$

$i = 1$ means AY_2 is completely unionised.

AY_2 not give ionised particles when dissolved in water.

So, AX_2 is completely ionised and AY_2 is completely unionised.

Answer: Option 4) Ax_2 is fully ionised, AY_2 is completely unionised.

Question 34

Given below are two statements :

Statement (I): NaCl is added to the ice at 0°C , present in the ice cream box to prevent the melting of ice cream.

Statement (II): On addition of NaCl to ice at 0°C , there is a depression in freezing point.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

Both Statement I and Statement II are false

B.

Statement I is true but Statement II is false

C.

Both Statement I and Statement II are true

D.

Statement I is false but Statement II is true

Answer: C

Solution:

Statement I : Correct

NaCl addition to ice causes preventing the melting of ice. On adding NaCl to ice, freezing point lowers. This creates a colder mixture, preventing the ice cream from melting.

Melting point of ice is 0°C . When only ice is used to make ice cream, at 0°C ice starts melting by absorbing the energy from its environment in the form of heat. Addition of salt to ice while making the cream lowers the freezing point of the ice, allowing it to reach a colder temperature and thus the ice cream mixture freezes properly. So, the salt causes ice to melt at a lower temperature than pure ice.

Statement II : Correct

Decrease in freezing point while addition of NaCl to ice at 0°C is due to the colligative property depression in freezing point.

So, both the statements are correct.

Both statement I and statement II are true.

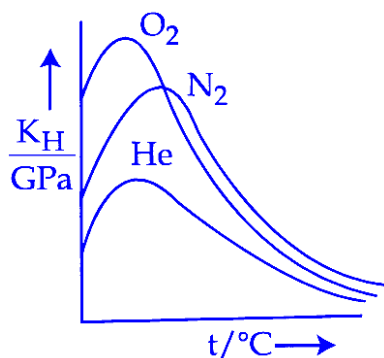
Question35

Which of the following graph correctly represents the plots of K_{H} at 1 bar for gases in water versus temperature?

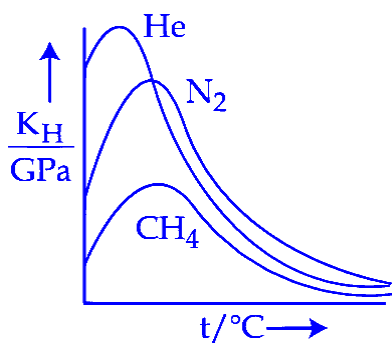
JEE Main 2025 (Online) 2nd April Morning Shift

Options:

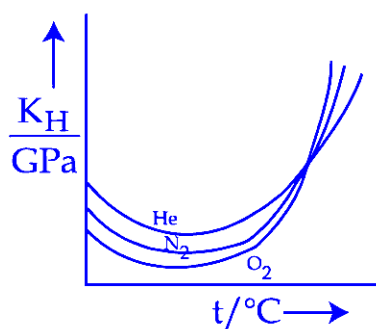
A.



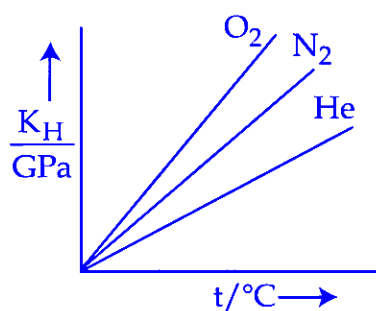
B.



C.



D.



Answer: B

Solution:

As temperature increases solubility first decrease then increase hence K_H first increase then decrease also at moderate temperature K_H value $\text{He} > N_2 > CH_4$.

Question36

A solution is made by mixing one mole of volatile liquid *A* with 3 moles of volatile liquid *B*. The vapour pressure of pure *A* is 200 mm Hg and that of the solution is 500 mm Hg. The vapour pressure of pure *B* and the least volatile component of the solution, respectively, are:

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. 1400 mm Hg, *A*

B. 1400 mm Hg, *B*

C. 600 mm Hg, A

D. 600 mm Hg, B

Answer: C

Solution:

Given:

1 mole of volatile liquid A

3 moles of volatile liquid B

Vapor pressure of pure A, $P_A^\circ = 200$ mm Hg

Vapor pressure of the solution, $P_S = 500$ mm Hg

We apply Raoult's law, which states:

$$P_S = P_A^\circ \cdot X_A + P_B^\circ \cdot X_B$$

Where:

X_A is the mole fraction of A

X_B is the mole fraction of B

P_B° is the vapor pressure of pure liquid B

Calculate the mole fractions:

$$X_A = \frac{1}{1+3} = \frac{1}{4}$$

$$X_B = \frac{3}{1+3} = \frac{3}{4}$$

Plug these into the equation:

$$500 = 200 \times \frac{1}{4} + P_B^\circ \times \frac{3}{4}$$

Simplifying:

$$500 = 50 + \frac{3}{4}P_B^\circ$$

Subtract 50 from both sides:

$$450 = \frac{3}{4}P_B^\circ$$

Multiply both sides by $\frac{4}{3}$ to solve for P_B° :

$$P_B^\circ = 600 \text{ mm Hg}$$

Since $P_A^\circ < P_B^\circ$, liquid A is the least volatile component.

In conclusion:

The vapor pressure of pure B, P_B° , is 600 mm Hg.

The least volatile component is A.

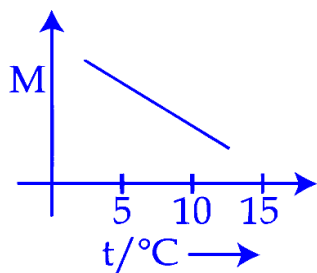
Question 37

' x ' g of NaCl is added to water in a beaker with a lid. The temperature of the system is raised from 1°C to 25°C . Which out of the following plots, is best suited for the change in the molarity (M) of the solution with respect to temperature ?

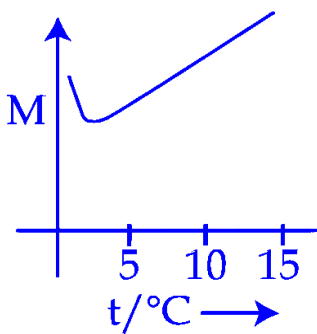
[Consider the solubility of NaCl remains unchanged over the temperature range]

Options:

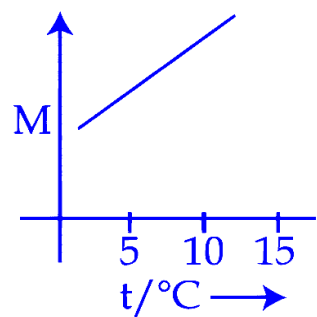
A.



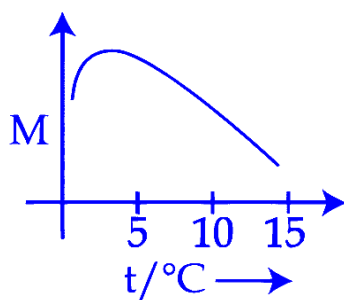
B.



C.



D.



Answer: D

Solution:

When x grams of NaCl are added to water in a beaker and the temperature is increased from 1°C to 25°C , the molarity of the solution changes due to the volumetric changes of water.

Temperature from 1°C to 4°C:

Water is densest at 4°C.

As the temperature increases from 1°C to 4°C, the water volume decreases due to increased density.

This decrease in volume results in an increase in molarity because molarity is inversely proportional to the solution's volume.

Temperature from 4°C to 25°C:

Beyond 4°C, water expands with an increase in temperature.

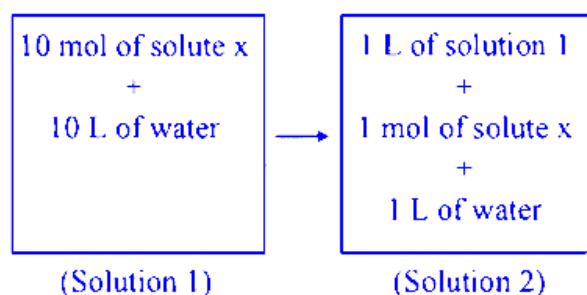
Therefore, as temperature rises to 25°C, the volume of the water increases.

The dilution leads to a decrease in molarity, since molarity is inversely proportional to volume.

Thus, the molarity first increases as temperature rises to 4°C, but then decreases as it continues to 25°C. The graphical representation of this relationship would exhibit an initial increase in molarity, followed by a decrease, correlating with changes in the volume of water due to temperature variations.

Question38

Which of the following properties will change when system containing solution 1 will become solution 2 ?

**JEE Main 2025 (Online) 3rd April Morning Shift****Options:**

- A. Molar heat capacity
- B. Concentration
- C. Gibbs free energy
- D. Density

Answer: C

Solution:

Both solutions contain the same composition, specifically 1 mole of 'x' in 1 liter of water. This means all intensive properties, such as concentration and density, will remain unchanged. However, because the total quantity of solution is greater in Solution 1 compared to Solution 2, the extensive properties will differ. Consequently, Gibbs free energy will change as it is an extensive property.

Question39

2 moles each of ethylene glycol and glucose are dissolved in 500 g of water. The boiling point of the resulting solution is:(Given : Ebullioscopic constant of water = $0.52 \text{ K kg mol}^{-1}$)

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. 377.3 K

B. 379.2 K

C. 375.3 K

D. 277.3 K

Answer: A

Solution:

Boiling Point Elevation Formula:

$$\Delta T_b = i_1 \cdot m_1 \cdot K_b + i_2 \cdot m_2 \cdot K_b$$

where:

i_1 and i_2 are the van't Hoff factors for ethylene glycol and glucose, respectively (both are 1 since they do not dissociate in solution).

m_1 and m_2 are the molalities of ethylene glycol and glucose, respectively.

K_b is the ebullioscopic constant of water ($0.52 \text{ K kg mol}^{-1}$).

Calculate Molality:

Each solute has 2 moles dissolved in 500 grams of water (0.5 kg):

$$m_1 = \frac{2 \text{ moles}}{0.5 \text{ kg}} = 4 \text{ mol kg}^{-1}$$

$$m_2 = \frac{2 \text{ moles}}{0.5 \text{ kg}} = 4 \text{ mol kg}^{-1}$$

Substitute into the Formula:

$$\Delta T_b = 1 \cdot 4 \cdot 0.52 + 1 \cdot 4 \cdot 0.52 = 4.16$$

Determine Boiling Point of Solution:

The normal boiling point of water is 373.16 K.

Add the boiling point elevation to the normal boiling point:

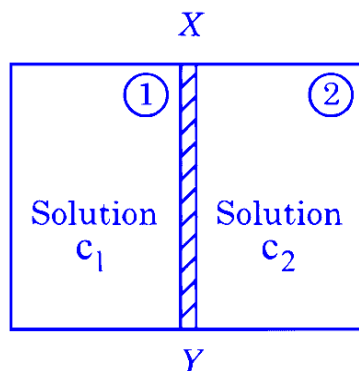
$$T_b(\text{solution}) = 373.16 + 4.16 = 377.3 \text{ K}$$

Thus, the boiling point of the resulting solution is 377.3 K.

Question40

XY is the membrane/partition between two chambers 1 and 2 containing sugar solutions of concentration c_1 and c_2 ($c_1 > c_2$) mol L^{-1} . For the reverse osmosis to take place identify the correct condition.

(Here p_1 and p_2 are pressures applied on chamber 1 and 2).



A. Membrane/Partition : Cellophane, $p_1 > \pi$

B. Membrane/Partition : Porous, $p_2 > \pi$

C. Membrane/Partition : Parchment paper, $p_1 > \pi$

D. Membrane/Partition : Cellophane, $p_2 > \pi$

Choose the correct answer from the option given below:

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. A and C Only

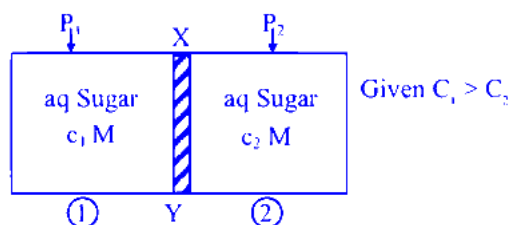
B. A and D Only

C. B and D Only

D. C Only

Answer: A

Solution:



Normal osmosis occurs from (2) to (1)

For reverse osmosis from (1) to (2)

Pressure : $P_1 > \pi$

$\therefore$ Answer [A & C] only

Question41

Given below are two statements:

Statement (I) : Molal depression constant K_f is given by $\frac{M_1 R T_f}{\Delta S_{\text{fus}}}$, where symbols have their usual meaning.

Statement (II) : K_f for benzene is less than the K_f for water.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

- A. Statement I is correct but Statement II is incorrect
- B. Both Statement I and Statement II are incorrect
- C. Statement I is incorrect but Statement II is correct
- D. Both Statement I and Statement II are correct

Answer: A

Solution:

Statement-I

$$\text{Molar depression constant } k_f = \frac{M_1 R T_f^2}{\Delta H_{\text{fus}}}$$

$$k_f = \frac{M_1 R T_f}{\left[\frac{\Delta H_{\text{fus}}}{T_f} \right]}$$

$$k_f = \frac{M_1 R T_f}{\Delta S_{\text{fus}}}$$

Hence statement-I is correct

$$\text{but } k_f \text{ for benzene} = 5.12 \frac{^\circ\text{C}}{\text{molal}}$$

$$k_f \text{ for water} = 1.86 \frac{^\circ\text{C}}{\text{molal}} \text{ Hence statement- II is incorrect}$$

Question 42

Match List - I with List - II.

List - I	List - II
(A) Solution of chloroform and acetone	(I) Minimum boiling azeotrope
(B) Solution of ethanol and water	(II) Dimerizes
(C) Solution of benzene and toluene	(III) Maximum boiling azeotrope
(D) Solution of acetic acid in benzene	(IV) $\Delta V_{\text{mix}} = 0$

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

- A.
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- B.
(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

C.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D.

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Answer: A

Solution:

To correctly match the items from List - I with those in List - II, let's analyze the characteristics of each solution:

(A) Solution of chloroform and acetone: This solution exhibits negative deviation from Raoult's law. It creates a maximum boiling azeotrope because the interactions between chloroform and acetone molecules are stronger than those in the pure components.

(B) Solution of ethanol and water: This solution shows positive deviation from Raoult's law, leading to the formation of a minimum boiling azeotrope. The interactions between ethanol and water molecules are weaker than those in their pure states.

(C) Solution of benzene and toluene: This combination forms an ideal solution where Raoult's law is obeyed across all concentrations. Thus, the volume change upon mixing, denoted as ΔV_{mix} , is zero.

(D) Solution of acetic acid in benzene: In this mixture, acetic acid tends to dimerize. The acetic acid molecules pair up, forming dimers, especially in non-polar solvents like benzene.

Based on these explanations, the correct matches are:

(A) - (III)

(B) - (I)

(C) - (IV)

(D) - (II)

Question43

Liquid A and B form an ideal solution. The vapour pressures of pure liquids A and B are 350 and 750 mm Hg respectively at the same temperature. If x_A and x_B are the mole fraction of A and B in solution while y_A and y_B are the mole fraction of A and B in vapour phase, then,

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

$$(x_A - y_A) < (x_B - y_B)$$

B.

$$\frac{x_A}{x_B} = \frac{y_A}{y_B}$$

C.

$$\frac{x_A}{x_B} < \frac{y_A}{y_B}$$

D.

$$\frac{x_A}{x_B} > \frac{y_A}{y_B}$$

Answer: D

Solution:

Liquid A and B form an ideal solution. The vapor pressures of pure liquids A and B are 350 mm Hg and 750 mm Hg, respectively, at the same temperature. Here, x_A and x_B represent the mole fractions of A and B in the solution, and y_A and y_B are their mole fractions in the vapor phase.

Let's begin by comparing the vapor pressures:

$$P_A^\circ < P_B^\circ$$

$$\frac{P_A^o}{P_B^o} < 1$$

The relationship between the mole fractions in the vapor phase and the solution can be expressed as:

$$\frac{y_A}{y_B} = \frac{P_A^o}{P_B^o} \cdot \frac{x_A}{x_B}$$

Since $\frac{P_A^o}{P_B^o} < 1$, it follows that:

$$\frac{\frac{y_A}{y_B}}{\frac{x_A}{x_B}} < 1$$

Which implies:

$$\frac{y_A}{y_B} < \frac{x_A}{x_B}$$

This indicates that the mole fraction ratio of A to B in the vapor phase is less than that in the solution.

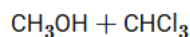
Question44

Which of the following binary mixture does not show the behaviour of minimum boiling azeotropes?

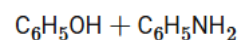
JEE Main 2025 (Online) 8th April Evening Shift

Options:

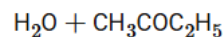
A.



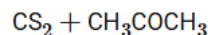
B.



C.



D.

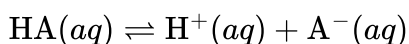


Answer: B

Solution:

A binary mixture of $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_5\text{NH}_2$ exhibits negative deviation from Raoult's law. This means that the vapor pressure of the solution is lower than the vapor pressures of the pure components, $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_5\text{NH}_2$. Consequently, the boiling point of the solution is higher than the boiling points of the pure substances. Therefore, this mixture forms a maximum boiling azeotrope.

Question45



The freezing point depression of a 0.1 m aqueous solution of a monobasic weak acid HA is 0.20 °C. The dissociation constant for the acid is

Given: $K_f(\text{H}_2\text{O}) = 1.8 \text{ K kg mol}^{-1}$, molality $\equiv$ molarity

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$$1.90 \times 10^{-3}$$

B.

$$1.38 \times 10^{-3}$$

C.

$$1.1 \times 10^{-2}$$

D.

$$1.89 \times 10^{-1}$$

Answer: B

Solution:

Freezing Point Depression:

$$\Delta T_f = i \times K_f \times m$$

Given:

$$\Delta T_f = 0.2^\circ\text{C}, \quad K_f = 1.8 \text{ K kg mol}^{-1}, \quad m = 0.1 \text{ m}$$

Substituting the given values:

$$0.2 = i \times 1.8 \times 0.1$$

Solving for i :

$$i = \frac{0.2}{1.8 \times 0.1} = \frac{20}{18} = \frac{10}{9}$$

Degree of Dissociation (α):

For the reaction $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$:

$$i = 1 + \alpha$$

Given $i = \frac{10}{9}$:

$$\frac{10}{9} = 1 + \alpha$$

$$\alpha = \frac{1}{9}$$

Dissociation Constant (K_{eq}):

$$K_{eq} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

At equilibrium:

$$[\text{H}^+] = [\text{A}^-] = \alpha \times C = \frac{1}{9} \times 0.1$$

$$[\text{HA}] = 0.1 \times (1 - \alpha) = 0.1 \times \left(1 - \frac{1}{9}\right)$$

Substituting these into K_{eq} :

$$K_{eq} = \frac{(0.1 \times \frac{1}{9})^2}{0.1 \times (1 - \frac{1}{9})}$$

Simplifying:

$$K_{eq} = \frac{0.1 \times (\frac{1}{81})}{0.1 \times \frac{8}{9}} = \frac{1}{720}$$

Therefore:

$$K_{eq} = 1.38 \times 10^{-3}$$

Question 46

A solution of two miscible liquids showing negative deviation from Raoult's law will have :

[27-Jan-2024 Shift 1]

Options:

A.

increased vapour pressure, increased boiling point

B.

increased vapour pressure, decreased boiling point

C.

decreased vapour pressure, decreased boiling point

D.

decreased vapour pressure, increased boiling point

Answer: D**Solution:**

Solution with negative deviation has

$$P_T < P_A^0 X_A + P_B^0 X_B$$

$$P_A < P_A^0 X_A$$

$$P_B < P_B^0 X_B$$

If vapour pressure decreases so boiling point increases.

Question47

The quantity which changes with temperature is:**[27-Jan-2024 Shift 2]****Options:**

A.

Molarity

B.

Mass percentage

C.

Molality

D.

Mole fraction

Answer: A**Solution:**

$$1. \text{ Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution}}$$

Since volume depends on temperature, molarity will change upon change in temperature.

Question48

A solution of H_2SO_4 is 31.4% H_2SO_4 by mass and has a density of 1.25g/mL. The molarity of the H_2SO_4 solution is M (nearest integer) [Given molar mass of $\text{H}_2\text{SO}_4 = 98\text{gmol}^{-1}$]

[29-Jan-2024 Shift 1]

Answer: 4

Solution:

$$\begin{aligned} M &= \frac{n_{\text{solute}}}{V} \times 1000 \\ &= \frac{\left(\frac{31.4}{98}\right)}{\left(\frac{100}{1.25}\right)} \times 1000 \\ &= 4.005 \approx 4 \end{aligned}$$

Question49

The osmotic pressure of a dilute solution is $7 \times 10^5 \text{Pa}$ at 273K. Osmotic pressure of the same solution at 283K is $\times 10^4 \text{Nm}^{-2}$.

[29-Jan-2024 Shift 1]

Answer: 72.56

Solution:

$$\begin{aligned} \pi &= CRT \\ \Rightarrow \frac{\pi_1}{\pi_2} &= \frac{T_1}{T_2} \\ \Rightarrow \pi_2 &= \frac{\pi_1 T_2}{T_1} = \frac{7 \times 10^5 \times 283}{273} \\ &= 72.56 \times 10^4 \text{Nm}^{-2} \end{aligned}$$

Question50

Molality of 0.8M H_2SO_4 solution (density 1.06gcm^{-3}) is $\times 10^{-3}\text{m}$.

[29-Jan-2024 Shift 2]

Answer: 815

Solution:

$$m = \frac{M \times 1000}{d_{\text{sol}} \times 1000 - M \times \text{Molar mass}_{\text{solute}}}$$

$$815 \times 10^{-3} \text{m}$$

Question51

What happens to freezing point of benzene when small quantity of naphthalene is added to benzene?

[30-Jan-2024 Shift 1]

Options:

A.

Increases

B.

Remains unchanged

C.

First decreases and then increases

D.

Decreases

Answer: D

Solution:

On addition of naphthalene to benzene there is depression in freezing point of benzene.

Question52

The mass of sodium acetate (CH_3COONa) required to prepare 250mL of 0.35M aqueous solution is _____g. (Molar mass of CH_3COONa is 82.02g mol^{-1})

[30-Jan-2024 Shift 1]

Answer: 7

Solution:

$$\text{Moles} = \text{Molarity} \times \text{Volume in litres}$$

$$= 0.35 \times 0.25$$

$$\text{Mass} = \text{moles} \times \text{molar mass}$$

$$= 0.35 \times 0.25 \times 82.02 = 7.18\text{g}$$

Ans. 7

Question53

The solution from the following with highest depression in freezing point/lowest freezing point is

[30-Jan-2024 Shift 2]

Options:

A.

180g of acetic acid dissolved in water

B.

180g of acetic acid dissolved in benzene

C.

180g of benzoic acid dissolved in benzene

D.

180g of glucose dissolved in water

Answer: A

Solution:

ΔT_f is maximum when $i \times m$ is maximum.

$$1) m_1 = \frac{180}{60} = 3, i = 1 + \alpha$$

Hence

$$\Delta T_f = (1 + \alpha) \cdot k_f = 3 \times 1.86 = 5.58^\circ\text{C} (\alpha < 1)$$

$$2) m_2 = \frac{180}{60} = 3, i = 0.5, \Delta T_f = \frac{3}{2} \times k_f' = 7.68^\circ\text{C}$$

$$3) m_3 = \frac{180}{122} = 1.48, i = 0.5, \Delta T_f = \frac{1.48}{2} \times k_f' = 3.8^\circ\text{C}$$

$$4) m_4 = \frac{180}{180} = 1, i = 1, \Delta T_f = 1 \cdot k_f' = 1.86^\circ\text{C}$$

As per NCERT, $k_f'(\text{H}_2\text{O}) = 1.86\text{K} \cdot \text{kg mol}^{-1}$

$$k_f'(\text{Benzene}) = 5.12\text{K} \cdot \text{kg mol}^{-1}$$

Question 54

If a substance 'A' dissolves in solution of a mixture of 'B' and 'C' with their respective number of moles as n_A , n_B and n_C , mole fraction of C in the solution is:

[30-Jan-2024 Shift 2]

Options:

A.

$$\frac{n_C}{n_A \times n_B \times n_C}$$

B.

$$\frac{n_C}{n_A + n_B + n_C}$$

C.

$$\frac{n_C}{n_A + n_B + n_C}$$

D.

$$\frac{n_B}{n_A + n_B}$$

Answer: B

Solution:

$$\text{Mole fraction of C} = \frac{n_C}{n_A + n_B + n_C}$$

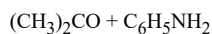
Question55

Identify the mixture that shows positive deviations from Raoult's Law

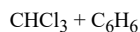
[31-Jan-2024 Shift 1]

Options:

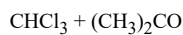
A.



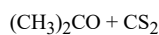
B.



C.



D.



Answer: D

Solution:

$(\text{CH}_3)_2\text{CO} + \text{CS}_2$ Exhibits positive deviations from Raoult's Law

Question56

We have three aqueous solutions of NaCl labelled as ' A ', ' B ' and ' C ' with concentration 0.1M, 0.01M & 0.001M, respectively. The value of van t' Haft factor (i) for these solutions will be in the order.

[1-Feb-2024 Shift 1]

Options:

A.

$$i_A < i_B < i_C$$

B.

$$i_A < i_C < i_B$$

C.

$$i_A = i_B = i_C$$

D.

$$i_A > i_B > i_C$$

Answer: A

Solution:

Salt	Values of i (for different conc. of a Salt)		
	0.1 M	0.01M	0.001M
NaCl	1.87	1.94	1.94

i approach 2 as the solution become very dilute.

Question57

Mass of ethylene glycol (antifreeze) to be added to 18.6kg of water to protect the freezing point at -24°C is _____ kg (Molar mass in gmol^{-1} for ethylene glycol 62, K_f of water = 1.86K kgmol^{-1})

[1-Feb-2024 Shift 2]

Answer: 15

Solution:

$$\Delta T_f = iK_f \times \text{molality}$$

$$24 = (1) \times 1.86 \times \frac{W}{62 \times 18.6}$$

$$W = 14880 \text{ gm}$$

$$= 14.880 \text{ kg}$$

Question58

In the depression of freezing point experiment

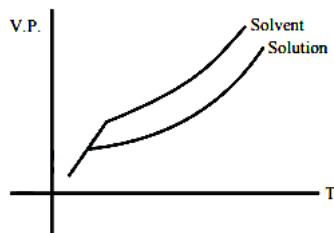
- A. Vapour pressure of the solution is less than that of pure solvent
 - B. Vapour pressure of the solution is more than that of pure solvent
 - C. Only solute molecules solidify at the freezing point
 - D. Only solvent molecules solidify at the freezing point
- [24-Jan-2023 Shift 1]

Options:

- A. A and D only
- B. B and C only
- C. A and C only
- D. A only

Answer: A

Solution:



Vapour pressure (V.P.) of solvent is greater than vapour pressure (V.P.) of solution.
Only solvent freezes.

Question59

The Total pressure observed by mixing two liquid A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively.

The Total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is _____ mm Hg. (Nearest integer)

Consider the liquids and solutions behave ideally.

[24-Jan-2023 Shift 2]

Answer: 314

Solution:

Let V.P. of pure A be P_A^0

Let V.P of pure B be P_B^0

When $X_A = 0.7$ & $X_B = 0.3$

$P_s = 350$

$\Rightarrow P_A^0 \times 0.7 + P_B^0 \times 0.3 = 350 \dots (i)$

When $X_A = 0.2$ & $X_B = 0.8$

$P_s = 410$

$\Rightarrow P_A^0 \times 0.2 + P_B^0 \times 0.8 = 410 \dots (ii)$

Solving (i) and (ii)

$P_A^0 = 314 \text{ mm Hg}$

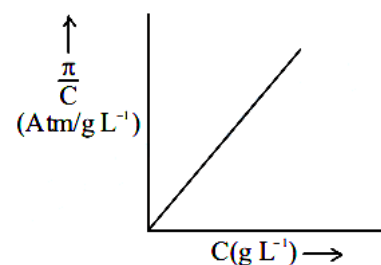
$P_B^0 = 434 \text{ mm Hg}$

$= (314)$

Question60

The osmotic pressure of solutions of PVC in cyclohexanone at 300K are plotted on the graph.

The molar mass of PVC is _____ g mol^{-1}
(Nearest integer)



(Given : $R = 0.083 \text{ L atm K}^{-1} \text{ mol}^{-1}$)
 [25-Jan-2023 Shift 1]

Answer: 41500

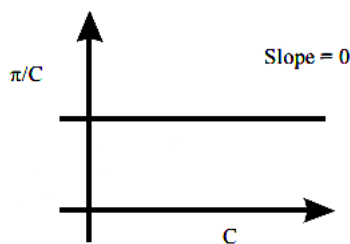
Solution:

$$\pi = M'RT = \left(\frac{W/M}{V} \right) RT$$

$$\Rightarrow \pi = \left(\frac{W}{V} \right) \left(\frac{1}{M} \right) RT = C \left(\frac{RT}{M} \right)$$

$$\Rightarrow \frac{\pi}{C} = \frac{RT}{M} \neq f(c)$$

If we assume graph between $\frac{\pi}{C}$ and C



Assuming π vs C graph

$$\text{Slope} = \frac{RT}{M} = \frac{0.083 \times 300}{M} = 6 \times 10^{-4}$$

$$\therefore M = \frac{0.083 \times 300}{6 \times 10^{-4}} = \frac{830 \times 300}{6}$$

$$= 41,500 \text{ gm/mole}$$

Question61

The number of pairs of the solution having the same value of the osmotic pressure from the following is

(Assume 100% ionization)

- A. $0.500 \text{ M C}_2\text{H}_5\text{OH(aq)}$ and 0.25 M KBr(aq)
- B. $0.100 \text{ M K}_4[\text{Fe}(\text{CN})_6] \text{ (aq)}$ and $0.100 \text{ M FeSO}_4(\text{NH}_4)_2\text{SO}_4\text{(aq)}$
- C. $0.05 \text{ M K}_4[\text{Fe}(\text{CN})_6]\text{(aq)}$ and 0.25 M NaCl (aq)
- D. 0.15 M NaCl(aq) and $0.1 \text{ M BaCl}_2 \text{ (aq)}$
- E. $0.02 \text{ M KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O(aq)}$ and 0.05 M KCl(aq)

[25-Jan-2023 Shift 2]

Answer: 4

Solution:

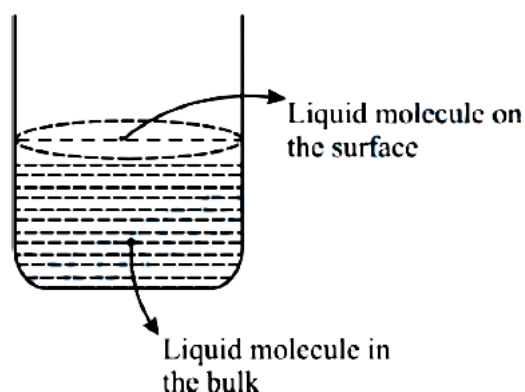
$$\pi = iCRT$$

$$\pi \propto iC$$

A, B, D and E have same value of osmotic pressure.

Question62

Based on the given figure, the number of correct statement/s is/are _____



- A. Surface tension is the outcome of equal attractive and repulsion forces acting on the liquid molecule in bulk.
 - B. Surface tension is due to uneven forces acting on the molecules present on the surface.
 - C. The molecule in the bulk can never come to the liquid surface.
 - D. The molecules on the surface are responsible for vapour pressure if the system is a closed system.
- [25-Jan-2023 Shift 2]

Answer: 2

Solution:

B and D options are correct.

Question63

Solid Lead nitrate is dissolved in 1 litre of water. The solution was found to boil at 100.15°C . When 0.2 mol of NaCl is added to the resulting solution, it was observed that the solution froze at -0.8°C . The solubility product of PbCl_2 formed is _____ $\times 10^{-6}$ at 298K. (Nearest integer)

Given : $K_b = 0.5\text{K kg mol}^{-1}$ and $K_f = 1.8\text{K kg mol}^{-1}$.

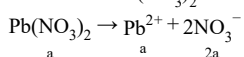
Assume molality to be equal to molarity in all cases.

[29-Jan-2023 Shift 1]

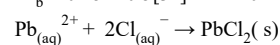
Answer: 13

Solution:

Let a mole $\text{Pb}(\text{NO}_3)_2$ be added



$$\Delta T_b = 0.15 = 0.5[3a] \Rightarrow a = 0.1$$



$$t = 0.1 \quad 0.2$$

$$t = \infty \quad (0.1 - x) \quad (0.2 - 2x)$$

In final solution

$$\Delta T_f = 0.8 = 1.8 \left[\frac{0.3 - 3x + 0.2 + 0.2}{1} \right]$$

$$\Rightarrow x = \frac{2.3}{27}$$

$$\Rightarrow K_{\text{sp}} = \left(0.1 - \frac{2.3}{27} \right) \left(0.2 - \frac{4.6}{27} \right)^2 = 13 \times 10^{-6}$$

Question64

Match List I with List II.

List-I	List-II
A. van't Hoff factor, i	I. Cryoscopic constant
B. k_f	II. Isotonic solutions
C. Solutions with same osmotic pressure	III. $\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$
D. Azeotropes	IV. Solutions with same composition of vapour above it

Choose the correct answer from the options given below :
[29-Jan-2023 Shift 2]

Options:

- A. A-III, B-I, C-II, D-IV
- B. A-III, B-II, C-I, D-IV
- C. A-III, B-I, C-IV, D-II
- D. A-I, B-III, C-II, D-IV

Answer: A

Solution:

(A) van't Hoff factor, i

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

(B) k_f = Cryoscopic constant

(C) Solutions with same osmotic pressure are known as isotonic solutions.

(D) Solutions with same composition of vapour over them are called Azeotrope.

Question65

Match List-I and List-II.

List-I	List-II
A. Osmosis	I. Solvent molecules pass through semi permeable membrane towards solvent side.
B. Reverse osmosis	II. Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes.
C. Electro osmosis	III. Solvent molecules pass through semi permeable membrane towards solution side.
D. Electrophoresis	IV. Dispersion medium moves in an electric field.

Choose the correct answer from the options given below:
[29-Jan-2023 Shift 2]

Options:

- A. A-I, B-III, C-IV, D-II
- B. A-III, B-I, C-IV, D-II

C. A-III, B-I, C-II, D-IV

D. A-I, B-III, C-II, D-IV

Answer: B

Solution:

- A. Osmosis III
 - B. Reverse osmosis I
 - C. Electro osmosis IV
 - D. Electrophoresis II
-

Question66

A solution containing 2g of a non-volatile solute in 20g of water boils at 373.52K. The molecular mass of the solute is _____ g mol^{-1} . (Nearest integer)

Given, water boils at 373K, K_b for water = 0.52K kg mol^{-1}

[30-Jan-2023 Shift 1]

Answer: 100

Solution:

$$\Delta T_b = 373.52 - 373$$

$$= 0.52$$

$$\Delta T_b = K_b \cdot m$$

$$0.52 = 0.52 \times \frac{2}{\text{Molar Mass}} \times \frac{1}{20 \times 10^{-3}}$$

$$\text{Molar Mass} = 100\text{g/mol}$$

Question67

A 300 mL bottle of soft drink has 0.2M CO_2 dissolved in it. Assuming CO_2 behaves as an ideal gas, the volume of the dissolved CO_2 at STP is _____ mL. (Nearest integer)

Given: At STP, molar volume of an ideal gas is

$$22.7\text{L mol}^{-1}$$

[30-Jan-2023 Shift 1]

Answer: 1362

Solution:

$$\text{Mole of } \text{CO}_2 = 0.2\text{M} \times (300 \times 10^{-3})\text{L}$$

$$= 0.06 \text{ Mole}$$

$$\text{Volume of } 0.06 \text{ mole } \text{CO}_2 \text{ at S.T.P}$$

$$= 0.06 \times 22.7$$

$$= 1.362\text{L}$$

Question68

Lead storage battery contains 38% by weight solution of H_2SO_4 . The van't Hoff factor is 2.67 at this concentration. The temperature in Kelvin at which the solution in the battery will freeze is _____ (Nearest integer).

Given $K_f = 1.8 \text{ K kg mol}^{-1}$

[30-Jan-2023 Shift 2]

Answer: 243

Solution:

$$\Delta T_f = i \cdot K_f \cdot m$$

$$\Rightarrow \Delta T_f = 2.67 \times 1.8 \times \frac{38}{98} \times \frac{1000}{62}$$

$$\Rightarrow \Delta T_f = 30.05$$

$$\therefore \text{F.P.} = 243 \text{ K}$$

Question69

The total pressure of a mixture of non-reacting gases X (0.6g) and Y (0.45g) in a vessel is 740 mm of Hg. The partial pressure of the gas X is mm of Hg. (Nearest Integer)

(Given : molar mass X = 20 and Y = 45 g mol^{-1})

[31-Jan-2023 Shift 1]

Answer: 555

Solution:

$$P_X = \chi_X \cdot P_r$$
$$= \frac{\frac{0.6}{20}}{\frac{0.6}{20} + \frac{0.45}{45}} \times 740$$
$$P_X = 555 \text{ mm Hg}$$

Question70

At 27°C , a solution containing 2.5g of solute in 250.0 mL of solution exerts an osmotic pressure of 400 Pa. The molar mass of the solute is _____ g mol^{-1} (Nearest integer)

(Given : $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)

[31-Jan-2023 Shift 1]

Answer: 62250

Solution:

$$\pi = CRT$$

$$\frac{400 \text{ Pa}}{10^5} = \frac{\frac{2.5 \text{ g}}{M_0}}{250/1000 \text{ L}} \times 0.83 \frac{\text{L} \cdot \text{bar}}{\text{K} \cdot \text{mol}} \times 300 \text{ K}$$

$$M_0 = 62250$$

Question 71

Evaluate the following statements for their correctness.

(A) The elevation in boiling point temperature of water will be same for 0.1M NaCl and 0.1M urea.

(B) Azeotropic mixtures boil without change in their composition

(C) Osmosis always takes place from hypertonic to hypotonic solution

(D) The density of 32% H_2SO_4 solution having molarity 4.09M is approximately 1.26 g mL^{-1}

(E) A negatively charged sol is obtained when KI solution is added to silver nitrate solution.

Choose the correct answer from the options given below :

[31-Jan-2023 Shift 2]

Options:

A. B, D, and E only

B. A, B, and D only

C. A and C only

D. B and D only

Answer: D

Solution:

(A) $\Delta T_b \propto i \times c$

(B) Azeotropic mixtures have same composition in both liquid and vapour phase.

(C) Osmosis always takes place from hypotonic to hypertonic solution.

(D) $M = \frac{30 \times 10 \times 1.26}{98} \approx 4.09 \text{ M}$

(E) When KI solution is added to AgNO_3 solution, positively charged solution results due to adsorption of Ag^+ ions in dispersion medium

AgI/Ag^+

Positively charged

Question 72

20% of acetic acid is dissociated when its 5g is added to 500 mL of water. The depression in freezing point of such water is $\text{_____} \times 10^{-3} \text{ }^\circ \text{C}$. Atomic mass of C, H and O are 12, 1 and 16 a.m.u. respectively.

[Given : Molal depression constant and density of respectively.

[1-Feb-2023 Shift 2]

Answer: 372

Solution:

$$i = 1 + (n - 1)\alpha$$

$$(i = 1 + 0.2(2 - 1) = 1.2$$

$$\Delta T_f = i K_f m$$

$$\Delta T_f = 1.2 \times 1.86 \times \frac{5 \times 1000}{60 \times 500}$$

$$\Delta t_f = 3.72$$

$$\Delta T_f = 372 \times 10^{-2}$$

Question73

Mass of Urea (NH_2CONH_2) required to be dissolved in 1000g of water to reduce the vapour pressure of water by 25% is.....g. (Nearest integer)

Given: Molar mass of N, C, O and H are 14, 12, 16 and 1g mol^{-1} respectively.

[6-Apr-2023 shift 1]

Answer: 1111

Solution:

$$\frac{P^0 - P_s}{P_s} = \frac{n_{\text{solute}}}{n_{\text{solvent}}} = \frac{\frac{x}{60}}{\frac{1000}{18}} = \frac{P^0 - 0.75P^0}{0.75P^0}$$

$$\Rightarrow x = \frac{10000}{9} = 1111 \text{ gm}$$

Question74

Consider the following pairs of solution which will be isotonic at the same temperature. The number of pairs of solutions is/are.....

A. 1M aq. NaCl and 2M aq. Urea

B. 1M aq. CaCl_2 and 1.5M aq. KCl

C. 1.5M aq. AlCl_3 and 2M aq. Na_2SO_4

D. 2.5M aq. KCl and 1M aq. $\text{Al}_2(\text{SO}_4)_3$

[6-Apr-2023 shift 2]

Answer: 4

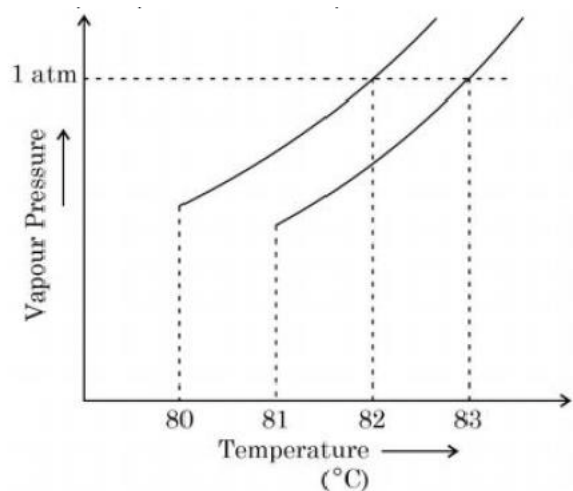
Solution:

$$\pi = icRT$$

A, B, C and D are isotonic pairs.

Question75

The vapour pressure vs. temperature curve for a solution solvent system is shown below:



The boiling point of the solvent is _____ °C
[8-Apr-2023 shift 1]

Answer: 82

Solution:

Boiling point of solvent is 82°C

Boiling point of solvent is 83°C

Question76

If the boiling points of two solvents X and Y (having same molecular weights) are in the ratio 2 : 1 and their enthalpy of vaporizations are in the ratio 1 : 2, then the boiling point elevation constant of X is m times the boiling point elevation constant of Y . The value of m is _____ (nearest integer)
[8-Apr-2023 shift 2]

Answer: 8

Solution:

$$K_b = \frac{RT_b^2 m}{1000 \Delta H_{\text{vap}}}$$

$$\frac{(K_b)_x}{(K_b)_y} = \frac{(T_b^2 M)_x}{(T_b^2 M)_y} \times \frac{(\Delta H)_y}{(\Delta H)_x} = \left(\frac{2}{1}\right)^2 \times \left(\frac{2}{1}\right) = \frac{8}{1}$$

Question77

If the degree of dissociation of aqueous solution of weak monobasic acid is determined to be 0.3 , then the observed freezing point will be _____ % higher than the expected/theoretical freezing point.
(Nearest integer)
[10-Apr-2023 shift 1]

Answer: 30

Solution:

For mono basic acid $\rightarrow n = 2$

$$i = 1 + (n - 1)\alpha = 1 + (2 - 1)0.3$$

$$i = 1.3$$

$$\% \text{ increase} = \frac{(\Delta T_f)_{\text{obs}} - (\Delta T_f)_{\text{cal}}}{(\Delta T_f)_{\text{cal}}} \times 100$$

$$= \frac{K_f \times i \times m - K_f \times m}{K_f \times m} \times 100$$

$$= \frac{i - 1}{1} \times 100 = 30\%$$

Question78

An aqueous solution of volume 300cm^3 contains 0.63g of protein. The osmotic pressure of the solution at 300 K is 1.29 mbar. The molar mass of the protein is _____ gmol^{-1} Given : $R = 0.083\text{L bar K}^{-1}\text{mol}^{-1}$ [10-Apr-2023 shift 2]

Answer: 40535

Solution:

$$\therefore \pi = CRT$$

$$\pi = \frac{n}{V}RT$$

$$\pi = \frac{w}{V} \frac{RT}{M}$$

$$M = \frac{wRT}{\pi \times V}$$

$$M = \frac{0.63 \times 0.083 \times 300}{1.29 \times 10^{-3} \times 300 \times 10^{-3}}$$

$$M = 40535 \text{ gm/mol}$$

Question79

0.004 MK_2SO_4 solution is isotonic with 0.01M glucose solution. Percentage dissociation of K_2SO_4 is (Nearest integer) [11-Apr-2023 shift 1]

Answer: 75

Solution:

For isotonic solution

$$(ic)_{\text{glucose}} = (ic)_{\text{K}_2\text{SO}_4}$$

$$0.01 = i(0.004)$$

$$i = \frac{0.01}{0.004} = \frac{10}{4} = \frac{5}{2}$$

$$1 + (n - 1)\alpha = \frac{5}{2}$$

$$1 + (3 - 1)\alpha = \frac{5}{2} \quad (\because n = 3 \text{ for } \text{K}_2\text{SO}_4)$$

$$2\alpha = \frac{3}{2}$$

$$\alpha = \frac{3}{4} \rightarrow 75\%$$

Question80

What weight of glucose must be dissolved in 100g of water to lower the vapour pressure by 0.20 mmHg ?

(Assume dilute solution is being formed)

Given : Vapour pressure of pure water is 54.2 mmHg at room temperature. Molar mass of glucose is 180gmol^{-1}

[11-Apr-2023 shift 2]

Options:

A. 2.59g

B. 3.59g

C. 3.69g

D. 4.69g

Answer: C

Solution:

$$\frac{p^0 - p_s}{p^0} = \frac{n}{N} \text{ (for dilute solution)}$$

$$\frac{0.2}{54.2} = \frac{n \times 18}{100}$$

$$n = \frac{100}{271 \times 18}$$

$$w = \frac{100 \times 180}{271 \times 18}; w = 3.69\text{g}$$

Question81

80 mole percent of MgCl_2 is dissociated in aqueous solution. The vapour pressure of 1.0 molal aqueous solution of MgCl_2 at 38°C is _____ mmHg. (Nearest integer)

Given : Vapour pressure of water at 38°C is 50 mm Hg.

[12-Apr-2023 shift 1]

Answer: 48

Solution:

$$\text{MgCl}_2 \rightarrow \text{Mg}^{+2} + 2\text{Cl}^-$$

$$1 - \alpha \quad \alpha \quad 2\alpha$$

$$i = 1 + 2\alpha (\alpha = 0.8)$$

$$i = 2.6$$

$$\frac{\Delta p}{p^0} = \frac{i \times n_2}{n_1}$$

$$\Delta p = 2.34$$

$$p_s = 47.66$$

$$p_s \cong 48$$

Question82

Solution of 12g of non-electrolyte (A) prepared by dissolving it in 1000 mL of water exerts the same osmotic pressure as that of 0.05M glucose solution at the same temperature. The empirical formula of A is CH_2O . The molecular mass of A is _____ g. (Nearest integer)

[13-Apr-2023 shift 1]

Answer: 240

Solution:

$$\pi_A = \pi_{\text{glucose}}$$

$$C_A RT = CRT$$

$$\frac{12/M_A}{1} = 0.05$$

$$M_A (\text{Molar mass of A}) = \frac{12}{0.05} = \frac{1200}{5} = 240 \text{ gm}$$

Question83

Sea water contains 29.25%NaCl and 19%MgCl₂ by weight of solution. The normal boiling point of the sea water is _____ °C (Nearest integer)

Assume 100% ionization for both NaCl and MgCl₂

Given : $K_b(\text{H}_2\text{O}) = 0.52 \text{ K kg mol}^{-1}$

Molar mass of NaCl and MgCl₂ is 58.5 and 95gmol⁻¹ respectively.

[13-Apr-2023 shift 2]

Answer: 116

Solution:

$$\text{Amount of solvent} = 100 - (29.25 + 19) = 51.75 \text{ g}$$

$$\Delta T_b = \left[\frac{2 \times 29.25 \times 1000}{58.5 \times 51.75} + \frac{3 \times 19 \times 1000}{95 \times 51.75} \right] \times 0.52$$

$$\Delta T_b = 16.075$$

$$\Delta T_b = (T_b)_{\text{solution}} - (T_b)_{\text{solvent}}$$

$$(t)_{\text{solution}} = 100 + 16.07$$

$$= 116.07^\circ\text{C}$$

Question84

The vapour pressure of 30%(w/v) aqueous solution of glucose is _____ mm Hg at 25°C.

[Given : The density of 30%(w/v), aqueous solutions of glucose is 1.2gcm⁻³ and vapour pressure of pure water is 24 mm Hg.]

(Molar mass of glucose is 180gmol⁻¹)

[15-Apr-2023 shift 1]

Answer: 23

Solution:

$$\begin{aligned}\frac{24 - P_s}{P_s} &= \frac{m \times 18}{1000} \\ \text{wt of solute} &= 30 \text{ gm} \\ \text{Volume of solution} &= 100 \text{ mL} \\ \text{wt. of solution} &= 1.2 \times 100 = 120 \text{ gm} \\ \text{wt. of solvent} &= 120 - 30 = 90 \text{ gm} \\ m \frac{30 \times 1000}{180 \times 90} &= 185 \\ \frac{24 - P_s}{P_s} &= \frac{1.85 \times 18}{1000} \\ 24 - P_s &= 0.0333 P_s \\ P_s(1.033) &= 24 \\ P_s &= 23.22\end{aligned}$$

Question85

The osmotic pressure of blood is 7.47 bar at 300K. To inject glucose to a patient intravenously, it has to be isotonic with blood. The concentration of glucose solution in g L^{-1} is
(Molar mass of glucose = 180 gmol^{-1} , $R = 0.083 \text{ L bar}^{-1} \text{ mol}^{-1}$) _____ (Nearest integer)
[24-Jun-2022-Shift-1]

Answer: 54

Solution:

$$\begin{aligned}7.47 &= C \times 0.083 \times 300 \\ (\pi &= CRT) \\ (\text{Where } C &\text{ represents the concentration of glucose solution and } \pi \text{ represents osmotic pressure}) \\ C &= \frac{7.47}{0.083 \times 300} (\text{mol L}^{-1}) \\ \text{which in gm/L} &= \frac{7.47}{0.083 \times 300} \times 180 \\ &= 54 \text{ gm/L}\end{aligned}$$

Question86

A company dissolves 'x' amount of CO_2 at 298K in 1 litre of water to prepare soda water. $X =$ _____
 $\times 10^{-3} \text{ g}$. (nearest integer)
(Given : partial pressure of CO_2 at 298K = 0.835 bar.
Henry's law constant for CO_2 at 298K = 1.67 kbar.
Atomic mass of H, C and O is 1, 12, and 16 gmol^{-1} , respectively)
[24-Jun-2022-Shift-2]

Answer: 1221

Solution:

According to Henry's law, partial pressure of a gas is given by

$$P_g = (K_H)X_g$$

where X_g is mole fraction of gas in solution

$$0.835 = 1.67 \times 10^3 (X_{\text{CO}_2})$$

$$X_{\text{CO}_2} = 5 \times 10^{-4}$$

$$\text{Mass of CO}_2 \text{ in 1L water} = 1221 \times 10^{-3} \text{ g}$$

Question87

1L aqueous solution of H_2SO_4 contains 0.02m mol H_2SO_4 . 50% of this solution is diluted with deionized water to give 1L solution (A). In solution (A), 0.01m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is $\underline{\hspace{1cm}} \times 10^3$ m mols
[25-Jun-2022-Shift-1]

Answer: 0

Solution:

$$n_{\text{H}_2\text{SO}_4} \text{ in Sol}^n\text{A} = 50\% \text{ of original solution} = 0.01 \text{ m mol}$$

$$n_{\text{H}_2\text{SO}_4} \text{ in Final solution} = 0.01 + 0.01$$

$$= 0.02 \text{ m mol}$$

$$= 0.00002 \times 10^3 \text{ m mol}$$

The answer 0

Question88

Solute A associates in water. When 0.7g of solute A is dissolved in 42.0g of water, it depresses the freezing point by 0.2°C . The percentage association of solute A in water, is :
[Given : Molar mass of A = 93 g mol^{-1} . Molal depression constant of water is $1.86 \text{ K kg mol}^{-1}$.]
[25-Jun-2022-Shift-2]

Options:

A. 50%

B. 60%

C. 70%

D. 80%

Answer: D

Solution:

$$\Delta T = i k_f \times m$$

$$0.2 = i \times 1.86 \times \frac{0.7}{93} \times \frac{1000}{42}$$

$$i = \frac{0.2 \times 93 \times 6}{1.86 \times 100}$$

$$i = 0.60$$



$$1 - \alpha = \frac{\alpha}{2}$$

$$i = 1 - \alpha + \frac{\alpha}{2}$$

$$i = 1 - \frac{\alpha}{2}$$

$$1 - \frac{\alpha}{2} = 0.60$$

$$1 - 0.60 = \frac{\alpha}{2}$$

$$\alpha = 0.80$$

Question89

A 0.5 percent solution of potassium chloride was found to freeze at -0.24°C . The percentage dissociation of potassium chloride is ____ (Nearest integer)
(Molal depression constant for water is 1.80K kg mol^{-1} and molar mass of KCl is 74.6gmol^{-1})
[26-Jun-2022-Shift-1]

Answer: 98

Solution:

$$\Delta T_f = iK_b m$$

$$i = \frac{0.24 \times 99.5 \times 74.6}{1.80 \times 0.5 \times 1000}$$

$$= 1.98$$

$$\alpha = \frac{i-1}{n-1} = \frac{0.98}{1} = 0.98 = 98\%$$

Question90

The osmotic pressure exerted by a solution prepared by dissolving 2.0g of protein of molar mass 60 kg mol^{-1} in 200mL^{m} of water at 27°C is _____ Pa. [integer value] (use $R = 0.083\text{L}_{\text{bar}}\text{mol}^{-1}\text{K}^{-1}$)
[26-Jun-2022-Shift-2]

Answer: 415

Solution:

$$\pi = iCRT$$

$$= \frac{1 \times 2}{60000 \times 0.2} \times 0.083 \times 300$$

$$= 0.00415\text{ bar } (\because 1.\text{ bar } = 10^5\text{ Pa})$$

So, $0.00415 \times 10^5\text{ Pa} = 415\text{ Pa}$

Question91

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : At 10°C , the density of a 5M solution of KCl [atomic masses of K& Cl are $39\&35.5\text{gmol}^{-1}$ respectively], is ' x ' gml^{-1} . The solution is cooled to -21°C . The molality of the solution will remain unchanged.
Reason (R): The molality of a solution does not change with temperature as mass remains unaffected

with temperature.

In the light of the above statements, choose the correct answer from the options given below :
[27-Jun-2022-Shift-1]

Options:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A).
- B. Both (A) and (R) are true but (R) is not the correct explanation of (A).
- C. (A) is true but (R) is false.
- D. (A) is false but (R) is true.

Answer: A

Solution:

Molality and Mass are temperature Independent so on changing temp., molality and mass remain unchanged.

Question92

2g of a non-volatile non-electrolyte solute is dissolved in 200g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 8. The elevation in boiling points of A and B are in the ratio $\frac{x}{y}$ (x : y). The value of y is _____ (Nearest integer)
[27-Jun-2022-Shift-1]

Answer: 8

Solution:

$$\begin{aligned}\Delta T_b &= k_b m \\ \frac{(\Delta T_b)_A}{(\Delta T_b)_B} &= \frac{(k_b)_A}{(k_b)_B} \\ &= \frac{1}{8} = \frac{x}{y} \\ \therefore y &= 8\end{aligned}$$

Question93

A solution containing 2.5×10^{-3} kg of a solute dissolved in 75×10^{-3} kg of water boils at 373.535K. The molar mass of the solute is _____ g mol⁻¹. [nearest integer] (Given : $K_b(\text{H}_2\text{O}) = 0.52\text{K kg mol}^{-1}$ and boiling point of water = 373.15K)
[27-Jun-2022-Shift-2]

Answer: 45

Solution:

$$\begin{aligned}W_{\text{solute}} &= 2.5 \times 10^{-3} \text{ kg} \\ W_{\text{solvent}} &= 75 \times 10^{-3} \text{ kg} \\ \Delta T_b &= 373.535 - 373.15\end{aligned}$$

$$= 0.385\text{K}$$

$$K_b(\text{H}_2\text{O}) = 0.52\text{K kg mol}^{-1}$$

$$\Delta T_b = \frac{K_b \times 10^3 \times W_{\text{solute}}}{M_{\text{solute}} \times W_{\text{solvent}}}$$

$$M_{\text{solute}} = \frac{0.52 \times 10^3 \times 2.5 \times 10^{-3}}{75 \times 10^{-3} \times 0.385}$$

$$= 45.02$$

$$\approx 45$$

Question94

The vapour pressures of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is ____

[28-Jun-2022-Shift-1]

Answer: 14

Solution:

$$\frac{y_B}{1-y_B} = \frac{P_B^\circ}{P_A^\circ} \left[\frac{X_B}{1-X_B} \right]$$

$$\Rightarrow \frac{y_B}{1-y_B} = \frac{100}{50} \left[\frac{0.7}{0.3} \right] = \frac{14}{3}$$

$$\Rightarrow y_B = \frac{14}{17}$$

Question95

1.2 mL of acetic acid is dissolved in water to make 2.0L of solution. The depression in freezing point observed for this strength of acid is 0.0198°C . The percentage of dissociation of the acid is ____ (Nearest integer)

[Given : Density of acetic acid is 1.02gmL^{-1} , Molar mass of acetic acid is 60gmol^{-1} , $K_f(\text{H}_2\text{O}) = 1.85\text{K kg mol}^{-1}$]

[29-Jun-2022-Shift-1]

Answer: 5

Solution:

$$M = d \times V = 1.02 \times 1.2 = 1.224 \text{ gm}$$

$$\text{Moles of acetic acid} = 0.0204 \text{ moles in 2L}$$

$$\text{So molality} = 0.0102 \text{ mol/kg}$$

$$\text{Now } \Delta T_f = i \times K_f \times M$$

$$i = 1 + \alpha \text{ for acetic acid}$$

$$0.0198 = (1 + \alpha) \times 1.85 \times 0.0102$$

$$\alpha = 0.04928$$

$$\cong 5\%$$

Question96

Elevation in boiling point for 1.5 molal solution of glucose in water is 4K. The depression in freezing point for 4.5 molal solution of glucose in water is 4K. The ratio of molal elevation constant to molal depression constant (K_b/K_f) is _____

[29-Jun-2022-Shift-2]

Answer: 3

Solution:

$$\Delta T_b = iK_b m$$

$$\Delta T_f = iK_f m$$

$$\frac{4}{4} = \frac{K_b \cdot 1.5}{K_f \cdot 4.5}$$

$$\frac{K_b}{K_f} = 3$$

Question97

The depression in freezing point observed for a formic acid solution of concentration 0.5 mL^{-1} is 0.0405°C . Density of formic acid is 1.05 g mL^{-1} . The Van't Hoff factor of the formic acid solution is nearly : (Given for water $k_f = 1.86 \text{ k kg mol}^{-1}$)

[25-Jul-2022-Shift-1]

Options:

A. 0.8

B. 1.1

C. 1.9

D. 2.4

Answer: C

Solution:

$$\Delta T_f \text{ of formic acid} = 0.0405^\circ\text{C}$$

$$\text{Concentration} = 0.5 \text{ mL/L}$$

$$\text{and density} = 1.05 \text{ g/mL}$$

$$\therefore \text{Mass of formic acid in solution} = 1.05 \times 0.5 \text{ g} = 0.525 \text{ g}$$

$$\therefore \text{According to Van't Hoff equation,}$$

$$\Delta T_f = i k_f \cdot m$$

$$0.0405 = i \times 1.86 \times \frac{0.525}{46 \times 1}$$

$$(\text{Assuming mass of 1L water} = \text{kg})$$

$$i = \frac{0.0405 \times 46}{1.86 \times 0.525} = 1.89 \approx 1.9$$

Question98

Two solutions A and B are prepared by dissolving 1g of non-volatile solutes X and Y, respectively in 1 kg of water. The ratio of depression in freezing points for A and B is found to be 1 : 4. The ratio of molar masses of X and Y is

[25-Jul-2022-Shift-2]

Options:

- A. 1 : 4
- B. 1 : 0.25
- C. 1 : 0.20
- D. 1 : 5

Answer: B

Solution:

$$\Delta T_f = iK_f \times m$$

$$\frac{\Delta T_{f(A)}}{\Delta T_{f(B)}} = \frac{1}{4}$$

$$\frac{i \times K_f \times \frac{1}{M_A} \times 1}{i \times K_f \times \frac{1}{M_B} \times 1} = \frac{1}{4}$$

$$\frac{M_B}{M_A} = \frac{1}{4}$$

$$M_A : M_B = 4 : 1$$

Question99

The elevation in boiling point for 1 molal solution of non-volatile solute A is 3K. The depression in freezing point for 2 molal solution of A in the same solvent is 6K. The ratio of K_b and K_f i.e., K_b/K_f is 1 : X. The value of X is [nearest integer]
[26-Jul-2022-Shift-2]

Answer: 1

Solution:

Molality of a solution of non volatile solute (A) = 1

Elevation in boiling point is given by

$$\Delta T_b = K_b m$$

$$3 = K_b \times 1$$

Molality of (A) in the same solvent = 2

Depression in freezing point is given by

$$\Delta T_f = K_f m$$

$$6 = K_f \times 2 \quad \dots (2)$$

Dividing (1) by (2)

$$\frac{K_b}{K_f} = \frac{1}{X} = \frac{1}{1}$$

$$\therefore X = 1$$

Question100

Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous solution of a non-volatile solute B. The relation between molecular weights of A and B is
[27-Jul-2022-Shift-1]

Options:

A. $M_A = 4M_B$

$$\text{B. } M_B = 4M_A$$

$$\text{C. } M_A = 8M_B$$

$$\text{D. } M_B = 8M_A$$

Answer: B

Solution:

For A : 100 gm solution $\rightarrow$ 2 gm solute A

$$\therefore \text{Molality} = \frac{2/M_A}{0.098}$$

For B : 100 gm solution $\rightarrow$ 8 gm solute B

$$\therefore \text{Molality} = \frac{8/M_B}{0.092}$$

$$\because (\Delta T_B)_A = (\Delta T_B)_B$$

$$\therefore \text{Molality of A} = \text{Molality of B}$$

$$\therefore \frac{2}{0.098M_A} = \frac{8}{0.092M_B}$$

$$\frac{2}{98} \times \frac{92}{8} = \frac{M_A}{M_B}$$

$$\frac{1}{4.261} = \frac{M_A}{M_B}$$

$$\therefore M_B = 4.261 \times M_A$$

Question101

When a certain amount of solid A is dissolved in 100g of water at 25°C to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is 23.76 mmHg. The number of moles of solute A added is _____. (Nearest Integer)
[27-Jul-2022-Shift-2]

Answer: 5.55

Solution:

$\therefore$ Dilute solution given:

$$\frac{P^0 - P_s}{P^0} \sim \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

$$\frac{P^0 - P^0/2}{P^0} = \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

$$n_{\text{Solute}} \sim \frac{n_{\text{solvent}}}{2} = \frac{100}{18 \times 2} = 2.78 \text{ mol}$$

More accurate approach:

$$\frac{P^0 - P_s}{P_s} = \frac{n_{\text{Solute}}}{n_{\text{solvent}}}$$

$$\frac{P^0 - P^0/2}{P^0/2} = \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

$$n_{\text{Solute}} = n_{\text{solvent}} = \frac{100}{18} = 5.55 \text{ mol}$$

Question102

150g of acetic acid was contaminated with 10.2g ascorbic acid ($C_6H_8O_6$) to lower down its freezing point by $(x \times 10^{-1})^\circ\text{C}$. The value of x is _____. (Nearest integer)

[Given $K_f = 3.9 \text{ K kg mol}^{-1}$; molar mass of ascorbic acid $= 176 \text{ g mol}^{-1}$]

[28-Jul-2022-Shift-1]

Answer: 15

Solution:

M.wt. of Acetic acid $= 60 \text{ g}$

M.wt. of Ascorbic acid $= 176 \text{ g}$

$$\Delta T_f = K_f m$$

$$\Delta T_f = \frac{3.9 \times 10.2 \times 1000}{176 \times 150}$$

$$\Delta T_f = 1.506$$

$$= 15.06 \times 10^{-1}$$

$$= 15$$

Question 103

A gaseous mixture of two substances A and B, under a total pressure of 0.8 atm is in equilibrium with an ideal liquid solution. The mole fraction of substance A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is atm. (Nearest integer)

[28-Jul-2022-Shift-2]

Answer: 2

Solution:

Given that $X_A = 0.2$, $Y_A = 0.5$, $P_T = 0.8 \text{ atm}$

We know that $P_A = Y_A \times P_T$

$$P_A = 0.5 \times 0.8 = 0.4$$

$$\text{Now } P_A = X_A \times P_A^\circ \Rightarrow P_A^\circ = \frac{0.4}{0.2} = 2 \text{ atm}$$

Question 104

If O_2 gas is bubbled through water at 303K, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____. (Nearest Integer)

(Given : Henry's Law constant for O_2 at 303K is 46.82k bar and partial pressure of $\text{O}_2 = 0.920 \text{ bar}$)

(Assume solubility of O_2 in water is too small, nearly negligible)

[29-Jul-2022-Shift-1]

Answer: 1

Solution:

$$P = K_H \times X$$

$$0.920 \text{ bar} = 46.82 \times 10^3 \text{ bar} \times \frac{\text{mol of}_2}{\text{mol of}_2 \text{ O}}$$

$$0.920 = 46.82 \times 10^3 \times \frac{\text{mol of}_2}{1000/18}$$

$$0.920 = 46.82 \times n_{\text{O}_2}$$

$$P = \frac{0.920}{46.82 \times 18} = n_{\text{O}_2}$$

$$\Rightarrow 1.09 \times 10^{-3} n_{\text{O}_2}$$

$$\Rightarrow m \text{ mol of O}_2 = 1$$

Question105

'x' g of molecular oxygen (O₂) is mixed with 200g of neon (Ne). The total pressure of the nonreactive mixture of O₂ and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is. _____.

[Given: Molar mass of O₂ – 32g^{mol}mol⁻¹.

Molar mass of .Ne = 20g^{mol}mol⁻¹]

[29-Jul-2022-Shift-2]

Answer: 80

Solution:

O₂ + N e

Xgm 200gm

P_{total} = 25 bar ; P_{Ne} = 20

P_{O₂} + P_{Ne} = 25

P_{O₂} = 25 – 20 = 5 bar

$$5 = \frac{\frac{x}{32}}{\frac{x}{32} + \frac{200}{20}} \times 25$$

$$\frac{1}{5} = \frac{\frac{x}{32}}{\frac{x}{32} + 10}$$

$$\frac{1}{5} = \frac{x \times 32}{32(x + 320)}$$

$$5x = x + 320$$

$$4x = 320$$

$$x = \frac{320}{4} = 80 \text{ gm}$$

Question106

1.80g of solute A was dissolved in 62.5cm³ of ethanol and freezing point of the solution was found to be 155.1K. The molar mass of solute A is g mol – 1 _____.

[Given : Freezing point of ethanol is 156.0K.

Density of ethanol is 0.80gcm⁻³.

Freezing point depression constant of ethanol is 2.00K kg mol⁻¹]

[29-Jul-2022-Shift-2]

Answer: 80

Solution:

Mass of solvent = $d \times v = 0.8 \times 62.5 = 50$ gram

$$\Delta T_f = k_f \times m$$

$$0.9 = 2 \left[\frac{1.8 \times 1000}{M_{\text{Solute}} \times 50} \right]$$

$$M_{\text{Solute}} = \left(\frac{2 \times 1.8 \times 1000}{0.9 \times 50} \right) = 80$$

Question107

C_6H_6 freezes at $5.5^\circ C$. The temperature at which a solution 10g of C_4H_{10} in 200g of C_6H_6 freeze is
 $^\circ C$. (The molal freezing point depression constant of C_6H_6 is $5.12^\circ C/m$.)

[24 Feb 2021 Shift 2]

Answer: 1

Solution:

Pure solvent : $C_6H_6(l)$

Given, $T_f^\circ = 5.5^\circ C$

$K_f = 5.12^\circ C/m \Rightarrow m = 200g$

$m_{\text{solute}} = 10g$

Molar mass of solute $C_4H_{10} = 12 \times 4 + 10 = 58$

Solute (C_4H_{10}) is non-dissociative;

$$\therefore i = 1$$

$$\therefore \Delta T_f = iK_f m$$

$$\Rightarrow (T_f^\circ - T_f^1) = 1 \times 5.12 \times \frac{(10/58)}{(200/1000)}$$

$$5.5 - T_f^1 = 5.12 \times 5 \times 10$$

$$58$$

$$\text{or } T_f^1 \approx 1^\circ C$$

Question108

When 12.2g of benzoic acid is dissolved in 100g of water, the freezing point of solution was found to be $-0.93^\circ C$ ($K_f(H_2O) = 1.86K \text{ kg mol}^{-1}$). The number (n) of benzoic acid molecules associated (assuming 100% association) is

[26 Feb 2021 Shift 2]

Answer: 2

Solution:



Assuming 100% association ($\alpha = 1$),

$$\Rightarrow i = 1 - \alpha \left(1 - \frac{1}{n} \right) = \frac{1}{n} [\because \alpha = 1]$$

$$\text{Now, } \Delta T_f = K_f \times m \times i$$

$$0 - (0.93) = 1.86 \times \frac{w_B \times 1000}{w_A \times M_B} \times \frac{1}{n}$$

$$[\because w_B = \text{mass of PhCOOH} = 12.2\text{g}]$$

$$w_A = \text{mass of H}_2\text{O} = 100\text{g}$$

$$M_B = \text{molar mass of PhCOOH}]$$

$$= 122\text{g mol}^{-1}$$

$$= 1.86 \times \frac{12.2 \times 1000}{100 \times 122} \times \frac{1}{n}$$

$$\Rightarrow n = \frac{1.86 \times 12.2 \times 1000}{0.93 \times 100 \times 122} = 2$$

$$\therefore \text{Number of benzoic acid molecules associated, } n = 2$$

Question 109

If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 2.5K rise in the boiling point of the solution is molal. (Rounded off to the nearest integer) [$K_b = 0.52\text{K kg mol}^{-1}$]

[25 Feb 2021 Shift 2]

Answer: 3

Solution:

As AB is a binary electrolyte,

$$\therefore AB \rightleftharpoons A^+ + B^-, n = 2$$

$$i = 1 + \alpha(n - 1) = 1 + \frac{75}{100}(2 - 1) = 1.75$$

$$\text{Given, } \Delta T_b = 2.5\text{K}$$

$$K_b = 0.52\text{K kg mol}^{-1}$$

$$\Delta T_b = K_b \times m \times i$$

$$m = \frac{\Delta T_b}{K_b \times i} = \frac{2.5}{0.52 \times 1.75}$$

$$\therefore = 2.74 \sim 3\text{mol/kg}$$

Question 110

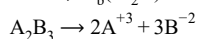
1 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1atm is K (Rounded off to the nearest integer). [Given, K_b for $(H_2O) = 0.52\text{K kg mol}^{-1}$]

[25 Feb 2021 Shift 1]

Answer: 375

Solution:

$$\text{Given, } K_b(H_2O) = 0.52\text{K kg mol}^{-1}$$



No. of ions, $n = 5$, concentration, $m = 1$ molal (Given)

$$\alpha = \frac{60}{100} = 0.6$$

$$\Delta T_b = i \cdot K_f \cdot m$$

$$= [(1 + (n - 1)\alpha) \times K_f \times m]$$

$$\Delta T_b = [(1 + (5 - 1)0.6) \times 0.52 \times 1]$$

$$\begin{aligned}
 &= (1 + 2.4) \times 0.52 \\
 \Delta T_b &= 1.768 \\
 T_b &= 1.768 + 373.15 \\
 &= 374.91 = 375\text{K}
 \end{aligned}$$

Question 111

When 9.45g of ClCH_2COOH is added to 500 mL of water, its freezing point drops by 0.5°C . The dissociation constant of ClCH_2COOH is $x \times 10^{-3}$. The value of x is _____ (Rounded off to the nearest integer) $[\text{K}_{\text{f}(\text{H}_2\text{O})} = 1.86\text{K kg mol}^{-1}]$
[24 Feb 2021 Shift 1]

Answer: 35

Solution:

$$\begin{aligned}
 \text{ClCH}_2\text{COOH} &\rightleftharpoons \text{ClCH}_2\text{COO}^- + \text{H}^+ \\
 \text{Total dissociated} &= 1 + \alpha \\
 \Rightarrow i &= 1 + \alpha \\
 \Delta T_f &= i k_f m \\
 \Rightarrow 0.5 &= (1 + \alpha)(1.86) \times \frac{9.45}{94.5} \times \frac{1000}{500} \\
 \Rightarrow 0.5 &= (1 + \alpha)(1.86)(0.2) \\
 \Rightarrow 1 + \alpha &= 1.34 \Rightarrow \alpha = 0.34 \\
 \text{ClCH}_2\text{COOH} &\rightleftharpoons \text{ClCH}_2\text{COO}^- + \text{H}^+ \\
 \begin{array}{ccccc}
 \text{At } t &= & 0 & & \alpha & & 0 & & 0 \\
 \text{At time } t & & C - C\alpha & & C\alpha & & C\alpha & &
 \end{array} \\
 K_a &= \frac{(C\alpha)^2}{C - C\alpha} = \frac{C\alpha^2}{1 - \alpha} \\
 \Rightarrow K_a &= \frac{0.2 \times (0.34)^2}{1 - 0.34} = 0.035 \\
 \therefore K_a &= 35 \times 10^{-3}.
 \end{aligned}$$

Question 112

A solute a dimerises in water. The boiling point of a 2 molar solution of A is 100.52°C . The percentage association of A is (Round off to the nearest integer) [Use : K_b for water = 0.52K kg mol^{-1} , boiling point of water = 100°C]
[18 Mar 2021 Shift 2]

Answer: 100

Solution:

$$\begin{aligned}
 \text{Given : } T_b &= 100.52^\circ\text{C} \\
 T_b^\circ &= 100^\circ\text{C} \\
 \Rightarrow K_b &= 0.52\text{K kg mol}^{-1} \\
 \text{Elevation in boiling point,} \\
 \Delta T_b &= T_b - T_b^\circ \\
 \Delta T_b &= 100.52 - 100
 \end{aligned}$$

$$\Delta T_b = 0.52^\circ\text{C}$$

For dimerisation (van't Hoff factor) $i = \left(1 - \frac{\alpha}{2}\right)$

$$\therefore \Delta T_b = i \times K_b \times m$$

$$0.52 = \left(1 - \frac{\alpha}{2}\right) \times 0.52 \times 2$$

$$\alpha = 1$$

So, percentage association = 100%

Question 113

2 molal solution of a weak acid HA has a freezing point of 3.885°C . The degree of dissociation of this acid is $\times 10^{-3}$.

(Round off to the nearest integer).

**[Given : Molal depression constant of water = 1.85K kgmol^{-1} , freezing point of pure water = 0°C]
[18 Mar 2021 Shift 1]**

Answer: 50

Solution:

Given

Depression in freezing point,

$$K_f = 1.85\text{K kgmol mol}^{-1}$$

$$T_f^\circ = 0^\circ\text{C}$$

$$T_f = 3.885^\circ\text{C}$$

$$\Delta T_f = T_f - T_f^\circ$$

$$= 3.885 - 0 = 3.885^\circ\text{C}$$

$$\Delta T_f = 3.885^\circ\text{C}$$

$$\Delta T_f = i \times K_f \times m$$

$$3.885 = i \times 1.85 \times 2$$

$$\text{but, } i = (1 + \alpha)$$

$$3.885 = (1 + \alpha) \times 1.85 \times 2$$

$$\Rightarrow \alpha = \frac{0.185}{3.7} \Rightarrow \alpha = 0.05$$

$$\alpha = 50 \times 10^{-3}$$

Question 114

A 1 molal $\text{K}_4\text{Fe}(\text{CN})_6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another solution which contains 18.1 weight per cent of a non-electrolytic solute A. The molar mass of A is u. (Round off to the nearest integer).

[Density of water = 1.0gcm^{-3}]

[17 Mar 2021 Shift 2]

Answer: 85

Solution:

	$\text{K}_4\text{Fe}(\text{CN})_6 \rightleftharpoons 4\text{K}^+ + [\text{Fe}(\text{CN})_6]^{4-}$	
Initial conc.	1m	0
Final conc.	$(1 - 0.4)\text{m}$	4×0.4
	$= 0.6\text{m}$	$= 1.6\text{m}$

Effective molality = $0.6 + 1.6 + 0.4 = 2.6\text{m}$

As elevation in boiling point is a colligative property which depends on the amount of solute. So, to have same boiling point, the molality of two solutions should be same.

Molality of non-electrolyte solution = molality of $\text{K}_4[\text{Fe}(\text{CN})_6] = 2.6\text{m}$

Now, 18.1 weight per cent solution means 18.1g solute is present in 100g solution and hence, $(100 - 18.1) = 81.9\text{g}$ water.

Molality = $\frac{(\text{Mass of solute} / \text{Molar mass of solute})}{\text{Mass of solvent (g)}} \times 1000$

Now, $2.6 = \frac{18.1/M}{81.9/1000}$

where, M is the molar mass of non-electrolyte solute

Molar mass of solute, M = 85

Question115

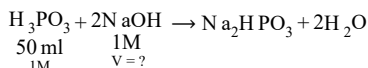
The exact volumes of 1M NaOH solution required to neutralise 50mL of 1M H_3PO_3 solution and 100mL of 2M H_3PO_2 solution, respectively, are
[16 Mar 2021 Shift 2]

Options:

- A. 100mL and 100mL
- B. 100mL and 50mL
- C. 100mL and 200mL
- D. 50mL and 50mL

Answer: C

Solution:



Millimoles of $\text{H}_3\text{PO}_3 = M \times V = 1 \times 50 = 50$

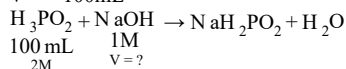
For 1 millimole of H_3PO_3 , we require 2 millimoles of NaOH.

For 50 millimole of H_3PO_3 , we require $(2 \times 50) = 100$ millimoles of NaOH.

Millimoles of NaOH = $M \times V = 100$

$1 \times V = 100$

$V = 100\text{mL}$



Millimoles of $\text{H}_3\text{PO}_2 = M \times V = 2 \times 100 = 200$

For 1 millimole of H_3PO_2 , we require 1 millimoles of NaOH.

For 200 millimole of H_3PO_2 , we require 200 millimoles of NaOH.

So, volume of NaOH = 200mL

Question116

A 6.50 molal solution of KOH (aq) has a density of 1.89gcm^{-3} . The molarity of the solution is
mol dm⁻³ (Round off to the nearest integer).

[Atomic masses: K : 39.0u, O : 16.0u, H : 1.0u]

[16 Mar 2021 Shift 1]

Answer: 9

Solution:

$$\text{Molality (m)} = \frac{M \times 1000}{\{(1000d) - M \times M_{\text{solute}}\}}$$

where, M = molarity

d = density of solution

M_{solute} = molar mass of solute

Putting the values,

$$6.50 = \frac{M \times 1000}{(1000 \times 1.89) - (M \times 56)}$$

$$\text{On solving, } M = \frac{12285}{1364}$$

$$M = 9 \text{ molar}$$

Question 117

At 20°C, the vapour pressure of benzene is 70 torr and that of methyl benzene is 20 torr. The mole fraction of benzene in the vapour phase at 20°C above an equimolar mixture of benzene and methyl benzene is _____ $\times 10^{-2}$. (Nearest integer)
[20 Jul 2021 Shift 1]

Answer: 78

Solution:

$$P_B^\circ = 40$$

$$P_T^\circ = 20$$

$$K_B = 0.5 = K_T$$

$$\begin{aligned} \text{Now } y_B &= \frac{K_B P_B^\circ}{K_B P_B^\circ + K_T P_T^\circ} \\ &= \frac{70 \times 0.5}{70 \times 0.5 + 20 \times 0.5} \end{aligned}$$

Question 118

1.46g of a biopolymer dissolved in a 100mL water at 300K exerted an osmotic pressure of 2.42×10^{-3} bar

The molar mass of the biopolymer is _____ $\times 10^4 \text{ gmol}^{-1}$. (Round off to the Nearest Integer)

[Use : R = 0.083L bar mol⁻¹K⁻¹]

[27 Jul 2021 Shift 1]

Answer: 15

Solution:

$$\pi = CRT$$

π = osmotic pressure

C = molarity

T = Temperature of solution

let the molar mass be M gm / mol

$$2.42 \times 10^{-3} \text{ bar} =$$

$$\left(\frac{1.46\text{g}}{\frac{M \text{ gm/mol}}{0.11}} \right) \times \left(\frac{0.0831 - \text{bar}}{\text{mol} - \text{K}} \right) \times (300\text{K})$$

$$\Rightarrow M = 15.02 \times 10^4 \text{ g/mol}$$

Question 119

When 3.00g of a substance 'X' is dissolved in 100g of CCl_4 , it raises the boiling point by 0.60K. The molar mass of the substance 'X' is _____ g mol⁻¹. (Nearest integer).

[Given K_b for CCl_4 is 5.0K kgmol⁻¹]

[25 Jul 2021 Shift 2]

Answer: 250

Solution:

$$\Delta T_b = K_b \times \text{molality}$$

$$0.60 = 5 \times \left(\frac{3/M}{100/100} \right)$$

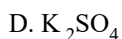
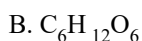
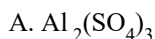
$$M = 250$$

Question 120

Which one of the following 0.06M aqueous solutions has lowest freezing point?

[22 Jul 2021 Shift 2]

Options:



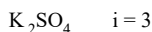
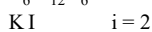
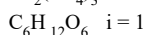
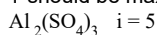
Answer: A

Solution:

$$T_f - T_f' = iK_f m$$

For minimum T_f'

'i' should be maximum.



Question 121

The water having more dissolved O_2 is :

[22 Jul 2021 Shift 2]

Options:

- A. boiling water
- B. water at 80°C
- C. polluted water
- D. water at 4°C

Answer: D

Solution:

On heating concentration of O_2 in water decreases.

So boiling water and water at 80°C having less O_2 concentration. Polluted water also having less O_2 concentration. So water at 4°C having maximum O_2 concentration.

Question122

The density of NaOH solution is 1.2gcm^{-3} . The molality of this solution is _____ m.
(Round off to the Nearest Integer)

[Use : Atomic masses : N a : 23.0u O : 16.0u H : 1.0u

Density of H_2O : 1.0gcm^{-3}]

[27 Jul 2021 Shift 1]

Answer: 5

Solution:

Consider 1l solution

mass of solution = $(1.2 \times 1000)\text{g}$
= 1200gm

Neglecting volume of NaOH

Mass of water = 1000gm

⇒ Mass of NaOH = $(1200 - 1000)\text{gm}$
= 200gm

⇒ Moles of NaOH = $\frac{200\text{g}}{50\text{g/mol}} = 5\text{mol}$

⇒ molality = $\frac{5\text{mol}}{1\text{kg}} = 5\text{m}$

Question123

CO_2 gas is bubbled through water during a soft drink manufacturing process at 298K . If CO_2 exerts a partial pressure of 0.835 bar then xm mol of CO_2 would dissolve in 0.9L of water. The value of x is _____. (Nearest integer)

(Henry's law constant for CO_2 at 298K is 1.67×10^3 bar)

[25 Jul 2021 Shift 1]

Answer: 25

Solution:

From Henry's law

$$P_{\text{gas}} = K_H \cdot X_{\text{gas}}$$

$$0.835 = 1.67 \times 10^3 \times \frac{n(\text{CO}_2)}{\frac{0.9 \times 1000}{18}}$$

$$n(\text{CO}_2) = 0.025$$

$$\text{Millimoles of CO}_2 = 0.025 \times 1000 = 25$$

Question 124

The vapour pressures of A and B at 25°C are 90 mmHg and 15 mmHg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is _____. (Nearest integer)
[20 Jul 2021 Shift 2]

Answer: 1

Solution:

Given $P_A^\circ = 90 \text{ mmHg}$

, at 25°C

$P_B^\circ = 15 \text{ mmHg}$

$$\text{and } \left. \begin{array}{l} X_A = 0.6 \\ X_B = 0.4 \end{array} \right\} P_T = X_A P_A^\circ + X_B P_B^\circ$$

$$= (0.6 \times 90) + (0.4 \times 15)$$

$$= 54 + 6 = 60 \text{ mm}$$

Now mol fraction of B in the vapour phase

$$\text{i.e. } Y_B = \frac{P_B}{P_T} = \frac{X_B P_B^\circ}{60} = 0.1 = 1 \times 10^{-1}$$

therefore: $x = 1$

Question 125

40 g of glucose (Molar mass = 180) is mixed with 200 mL of water. The freezing point of solution is K. (Nearest integer)
[Given, $K_f = 1.86 \text{ K kg mol}^{-1}$, density of water = 1.00 g cm^{-3} , freezing point of water = 273.15 K]
[27 Aug 2021 Shift 2]

Answer: 271

Solution:

40 g of glucose mixed with 200 mL of water

180 g of glucose = 1 moles of glucose

40 g of glucose mol = 0.22 mol

1 mL of water = 1 g of water [$d = 1 \text{ g/cm}^3$, 1 mL = 1 cm³]

200 mL of water = 200 g of water

$$\Delta T_f = K_f m$$

where, ΔT_f = depression in freezing point,

K_f = molal elevation constant = $1.86 \text{ K kg mol}^{-1}$ and

m = molality of solution

$$\Delta T_f = \frac{1.86 \times 0.22}{200} \times 1000, (\because 1000 \text{ g} = 1 \text{ kg})$$

$$\Delta T_f = 2 \text{ K}$$

$$\therefore (T_f - T_f') = 2 \left[\begin{array}{l} T_f = \text{freezing point of water}(273\text{K}) \\ T_f' = \text{freezing point of solution} \end{array} \right]$$

$$273\text{K} - T_f' = 2\text{K}$$

$$T_f' = (273 - 2)\text{K} = 271\text{K}$$

Question126

1 kg of 0.75 molar aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is (Nearest integer)

[Given, $K_f(\text{H}_2\text{O}) = 1.86 \text{ Kkg mol}^{-1}$]

[27 Aug 2021 Shift 1]

Answer: 518

Solution:

Let mass of water (initially present) = x g

Mass of sucrose = $(1000 - x)\text{g}$

$$\text{Moles of sucrose} = \frac{1000 - x}{342}$$

$$\text{Molality} = \frac{\text{moles of sucrose}}{\text{mass of water(initially)}}$$

$$0.75 = \frac{\left(\frac{1000 - x}{342}\right)}{\left(\frac{x}{1000}\right)}$$

$$\frac{x}{1000} = \frac{1000 - x}{342 \times 0.75}$$

$$256.5x = 10^6 - 1000x$$

$$\Rightarrow x = 795.86 \text{ g}$$

Moles of sucrose = 0.5969

New mass of H_2O = a kg

Depression in freezing point $\Delta T_f = K_f \times m$

$$4 = \frac{0.5969}{a} \times 1.86$$

$$\Rightarrow a = 0.2775 \text{ kg}$$

$$\text{Ice separated} = 795.86 - 277.5$$

$$= 518.3 \text{ g} \approx 518 \text{ g.}$$

Question127

If the degree of dissociation of aqueous solution of weak monobasic acid is determined to be 0.3 , then the [27 Aug 2021 Shift 1]

Answer: 30

Solution:

For mono basic acid $\rightarrow n = 2$

$$i = 1 + (n-1)\alpha = 1 + (2-1)0.3$$

$$i = 1.3$$

$$\% \text{ increase} = \frac{(\Delta T_f)_{\text{obs}} - (\Delta T_f)_{\text{cal}}}{(\Delta T_f)_{\text{cal}}} \times 100$$

$$= \frac{K_f \times i \times m - K_f \times m}{K_f \times m} \times 100$$

$$= \frac{i-1}{1} \times 100 = 30\%$$

Question 128

1.22g of an organic acid is separately dissolved in 100g of benzene ($K_b = 2.6 \text{ K kg mol}^{-1}$) and 100g of acetone ($K_b = 1.7 \text{ K kg mol}^{-1}$). The acid is known to dimerise in benzene but remain as a monomer in acetone. The boiling point of the solution in acetone increases by 0.17°C . The increase in boiling point of solution in benzene in $^\circ\text{C}$ is $x \times 10^{-2}$. The value of x is..... (Nearest integer)

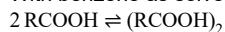
[Atomic mass: C = 12.0, H = 1.0, O = 16.0]

[31 Aug 2021 Shift 2]

Answer: 13

Solution:

With benzene as solvent, RCOOH dimerises.



$$\therefore i = \frac{1}{2}$$

$$\Delta T_b = i \times K_b \times m$$

where, ΔT_b = boiling point elevation,

K_b = ebulliscopic constant

m = molality

i = van't Hoff factor

$$\Delta T_b = \frac{1}{2} \times 2.6 \times \frac{1.22/M_w}{100/1000} \dots (i)$$

With acetone as solvent, no dimerisation.

$$\therefore i = 1$$

$$\Delta T_b = i \times K_b \times m$$

$$0.17 = 1 \times 1.7 \times \frac{1.22/M_w}{100/1000} \dots (ii)$$

Eq. (i) divide by Eq. (ii),

$$\frac{\Delta T_b}{0.17} = \frac{\frac{1}{2} \times 2.6 \times \frac{1.22/M_w}{100/1000}}{1 \times 1.7 \times \frac{1.22/M_w}{100/1000}}$$

$$\Rightarrow \Delta T_b = \frac{0.26}{2} = 13 \times 10^{-2}$$

$$x = 13$$

Question 129

Which one of the following 0.10M aqueous solutions will exhibit the largest freezing point depression?

[31 Aug 2021 Shift 1]

Options:

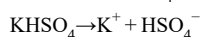
A. Hydrazine

- B. Glucose
- C. Glycine
- D. KHSO_4

Answer: D

Solution:

The van't Hoff factor (i) is highest for KHSO_4 as it dissociate into its ions. van't Hoff factor is the ratio of total number of ions produced after complete dissociation to before dissociation. Hydrazine, glucose and glycine will not dissociate to produce ions. Whereas, KHSO_4 dissociates into ions.



Hence, i will highest for KHSO_4 .

$\therefore$ Colligative property i.e. freezing point depression will be highest for KHSO_4 as $\Delta T_f \propto i$ [$\because m$ is constant i.e. 0.1]

Hence, correct option is (d).

Question130

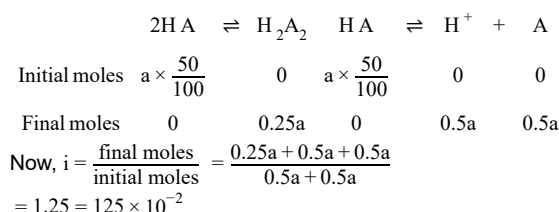
In a solvent 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is $\underline{\hspace{2cm}} \times 10^{-2}$.

(Round off to the nearest integer)

[27 Jul 2021 Shift 2]

Answer: 125

Solution:



Question131

83g of ethylene glycol dissolved in 625g of water. The freezing point of the solution isK. (Nearest integer)

[Use, molal freezing point depression constant of water = 1.86K kg mol^{-1} , Freezing point of water = 273K and Atomic masses : C = 12.0u, O = 16.0u, H = 1.0u]

[26 Aug 2021 Shift 2]

Answer: 269

Solution:

$$\Delta T_f = iK_f \times m$$

For ethylene glycol, $i = 1$

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{Weight of solvent(kg)}}$$

$$\Delta T_f = 1 \times 1.86 \times \frac{83 \times 1000}{62 \times 625} \approx 3.98$$

$$\begin{aligned} \text{Freezing point of solution} &= \text{Freezing point} - \Delta T_f \text{ of water} \\ &= 273 - 3.98 \approx 269. \end{aligned}$$

Question132

Of the following four aqueous solutions, total number of those solutions whose freezing point is lower than that of 0.10M

$\text{C}_2\text{H}_5\text{OH}$ is (Integer answer)

(i) 0.10M $\text{Ba}_3(\text{PO}_4)_2$

(ii) 0.10M Na_2SO_4

(iii) 0.10M KCl

(iv) 0.10M Li_3PO_4

[26 Aug 2021 Shift 1]

Answer: 4

Solution:

As 0.10M $\text{C}_2\text{H}_5\text{OH}$ is non-dissociative in nature and all salts

[0.10M $\text{Ba}_3(\text{PO}_4)_2$, 0.10M Na_2SO_4 ,

0.10M KCl, 0.10M Li_3PO_4] are electrolyte and dissociative in nature.

$$\Delta T_f = iK_f m$$

Where i is van't Hoff factor.

$i > 1$ for all other electrolytes where dissociation takes place.

So, for all these electrolytes effective molarity is more than 0.10. So, all the electrolytes given will have low freezing point.

Question133

The molarity of the solution prepared by dissolving 6.3g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is (Nearest integer)

[Atomic mass H = 1.0, C = 12.0, O = 16.0]

[31 Aug 2021 Shift 1]

Answer: 20

Solution:

$$\begin{aligned} \text{Molarity} &= \frac{\text{number of moles of solute}}{\text{volume of solution(in litre)}} \\ &= \frac{\text{weight of solute/molecular mass of solute}}{\text{volume of solution(in litre)}} \end{aligned}$$

Molecular mass of solute [$\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$] = 126g

$$x \times 10^{-2} = \frac{6.3\text{g}/126\text{g}}{250\text{ mL}} \times 1000$$

$$x \times 10^{-2} = 0.2$$

$$x \times 10^{-2} = 20 \times 10^{-2}$$

$$\therefore x = 20$$

Question134

100 mL of Na_3PO_4 solution contains 3.45g of sodium. The molarity of the solution is $\times 10^{-2} \text{ mol L}^{-1}$.
(Nearest integer)
[Atomic masses – Na = 23.0u, O = 16.0u,, P = 31.0u]
[26 Aug 2021 Shift 2]

Answer: 50

Solution:

$$\begin{aligned}\text{Number of mole of Na}^+ \text{ ions} &= \frac{\text{mass of sodium}}{\text{molar mass of sodium}} = \frac{3.45}{23} \\ \text{So, number of mole of Na}_3\text{PO}_4 &= \frac{\text{moles of sodium}}{3} \\ \{\text{as 1 molecule of Na}_3\text{PO}_4 \text{ contains 3 atoms of Na}\} \\ &= \frac{3.45}{3 \times 23} = 0.05 \\ \text{Molarity} &= \frac{\text{Moles of Na}_3\text{PO}_4}{\text{Volume of solution(in L)}} = \frac{0.05}{100} \times 1000 \\ \therefore \text{Answer is 50}\end{aligned}$$

Question135

If 80g of copper sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is dissolved in deionised water to make 5L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3} \text{ mol L}^{-1}$. The value of x is
[Atomic masses Cu = 63.54u, S = 32u, O = 16u, H = 1u]
[1 Sep 2021 Shift 2]

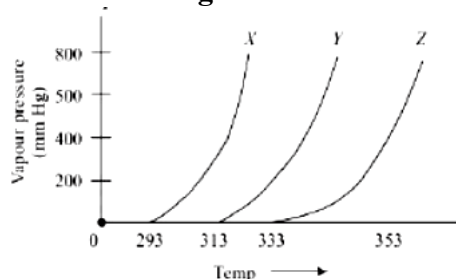
Answer: 64

Solution:

$$\begin{aligned}\text{Given, mass of CuSO}_4 \cdot 5\text{H}_2\text{O} &= 80\text{g} \\ \text{The concentration of copper sulphate solution is } &x \times 10^{-3} \text{ mol/L.} \\ \text{Molarity} &= \frac{\text{Number of moles of solute}}{\text{Volume of solution(L)}} \dots(i) \\ \text{Molar mass of CuSO}_4 \cdot 5\text{H}_2\text{O} &= 63.54 + 32 + 16 \times 4 \\ &= 5 \times 18 = 249.54 \text{ g/mol} \\ \text{Number of moles of solute} &= \frac{\text{Weight of solute}}{\text{Molecular mass of solute}} \\ &= \frac{80\text{g}}{249.54\text{g/mol}} = 0.32 \text{ mol} \\ \text{Volume of solution} &= 5 \text{ L} \\ \text{From Eq. (i),} \\ \text{Molarity} &= \frac{0.3205}{5} = 64.11 \times 10^{-3} \text{ mol/L} \\ \therefore x &= 64.11 \\ \text{or } x &\approx 64 \\ \text{Hence, answer is 64.}\end{aligned}$$

Question136

The vapour pressure of solution will be less than the vapour pressure of pure solvent, so some vapour molecules will get condensed to maintain new equilibrium.



The following inferences are made:

(A) X has higher intermolecular interactions compared to Y .

(B) X has lower intermolecular interactions compared to Y .

(C) Z has lower intermolecular interactions compared to Y .

The correct inference(s) is/are:

[Jan. 08, 2020(I)]

Options:

A. (A) and (C)

B. (A)

C. (B)

D. (C)

Answer: C

Solution:

At a particular temperature as intermolecular force of attraction increases vapour pressure decreases. Thus, intermolecular forces are inversely proportional to vapour pressure and directly proportional to temperature. Therefore X has lower intermolecular interactions compared to Y.

Question 137

At 35°C, the vapour pressure of CS₂ is 512 mmHg and that of acetone is 344 mmHg. A solution of CS₂ in acetone has a total vapour pressure of 600 mmHg. The false statement amongst the following is:

[Jan. 07, 2020(I)]

Options:

A. Raoult's law is not obeyed by this system

B. a mixture of 100 mL CS₂ and 100 mL acetone has a volume < 200 mL

C. CS₂ and acetone are less attracted to each other than to themselves

D. heat must be absorbed in order to produce the solution at 35°C

Answer: B

Solution:

Mixture of carbon disulphide and acetone will show positive deviation from Raoult's Law. The dipolar interaction between solute (CS₂) solvent (acetone) molecules in solution are weaker. So the vapour pressure of solution will be greater than the individual vapour pressure of pure components.

Question 138

Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non volatile solute are together sealed in a container. Over time:
[Jan. 07,2020 (II)]

Options:

- A. the volume of the solution increases and the volume of the solvent decreases
- B. the volume of the solution decreases and the volume of the solvent increases
- C. the volume of the solution and the solvent does not change
- D. the volume of the solution does not change and the volume of the solvent decreases

Answer: A

Solution:

There will be lowering in vapour pressure in second beaker.

Question139

How much amount of NaCl should be added to 600g of water ($\rho = 1.00\text{g/mL}$) to decrease the freezing point of water to -0.2°C ? _____. (The freezing point depression constant for water = 2K kgmol^{-1})
[NV, Jan. 09,2020 (I)]

Answer: 1.75

Solution:

$$\Delta T_f = iK_f m$$

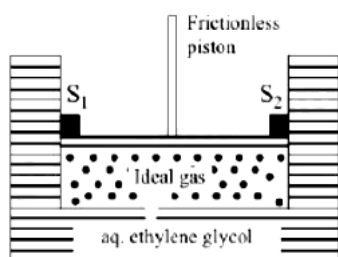
$$0.2 = 2 \times 2 \times \frac{w}{58.5} \times \frac{1000}{600}$$

$$w = \frac{0.2 \times 58.5 \times 600}{1000 \times 4} = \frac{1.2 \times 58.5}{40} = 1.75\text{g}$$

Question140

A cylinder containing an ideal gas (0.1mol of 1.0dm^3) is in thermal equilibrium with a large volume of 0.5mol of aqueous solution of ethylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be _____.

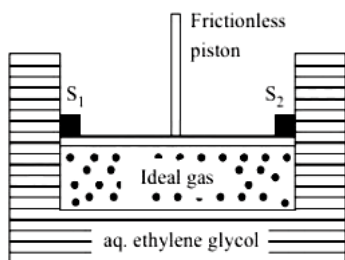
(Given, $K_f(\text{water}) = 2.0\text{K kgmol}^{-1}$, $R = 0.08\text{dm}^3\text{atmK}^{-1}\text{mol}^{-1}$)



[NV, Jan.09,2020 (II)]

Answer: 2.18

Solution:



$$K_f = 2.0 \text{ K kg mol}^{-1}; m = 0.5$$

$$\Delta T_f = K_f m = 0.5 \times 2$$

$$T_{\text{initial}} = 272 \text{ K}$$

$$n = 0.1 \text{ mol } V = 1 \text{ dm}^3$$

$$P_{\text{gs}} = \frac{nRT}{V} = \frac{0.1 \times 0.08 \times 272}{1} = 2.176 \text{ atm}$$

After releasing piston

$$P_1 V_1 = P_2 V_2$$

$$2.176 \times 1 = 1 \times V_2$$

$$V_2 = 2.18 \text{ dm}^3$$

Question 141

A set of solutions is prepared using 180g of water as a solvent and 10g of different non-volatile solutes A, B and C. The relative lowering of vapour pressure in the presence of these solutes are in the order

[Given, molar mass of

A = 100 g mol^{-1} ; B = 200 g mol^{-1} ; C = $10,000 \text{ g mol}^{-1}$]

[Sep. 06, 2020(II)]

Options:

- A. $B > C > A$
- B. $C > B > A$
- C. $A > B > C$
- D. $A > C > B$

Answer: C

Solution:

Relative lowering in vapour pressure (RLVP)

$$= \frac{P - P_s}{P} = \frac{n}{n + N}$$

$n \rightarrow$ moles of solute

$N \rightarrow$ moles of solvent

$$n_A = \frac{10}{100}, n_B = \frac{10}{200}, n_C = \frac{10}{10000}$$

$$\text{Moles of solvent (H}_2\text{O)} = \frac{180}{18} = 10 \text{ mol}$$

$$\text{RLV } P_A = \frac{0.1}{10.1} = \frac{1}{101}$$

$$\text{RLV } P_B = \frac{0.05}{10.05} = \frac{1}{201}$$

$$\text{RLV } P_C = \frac{10^{-3}}{10} = 10^{-4}$$

From the above relation

$$\text{RLV } P(A) > \text{RLV } P(B) > \text{RLV } P(C)$$

Question142

At 300K, the vapour pressure of a solution containing 1 mole of n -hexane and 3 moles of n -heptane is 550mm of Hg. At the same temperature, if one more mole of n heptane is added to this solution, the vapour pressure of the solution increases by 10mm of H g. What is the vapour pressure in mmH g of n -heptane in its pure state _____ ?
[NV, Sep. 04, 2020(I)]

Answer: 600

Solution:

$$P_{\text{total}} = P_{\text{hexane}}^0 \cdot X_{\text{hexane}} + P_{\text{heptane}}^0 \cdot X_{\text{heptane}}$$

$$550 = [P_{\text{hexane}}^0] \times \frac{1}{4} + [P_{\text{heptane}}^0] \times \frac{3}{4}$$

$$\Rightarrow 2200 = P_{\text{hexane}}^0 + 3P_{\text{heptane}}^0 \dots\dots(i)$$

$$560 = [P_{\text{hexane}}^0] \times \frac{1}{5} + [P_{\text{heptane}}^0] \times \frac{4}{5}$$

$$\Rightarrow 2800 = P_{\text{hexane}}^0 + 4P_{\text{heptane}}^0 \dots\dots(ii)$$

From equation (i) and (ii),
 $P_{\text{heptane}}^0 = 600\text{mm of H g}$

Question143

Henry's constant (in k bar) for four gases α , β , γ and δ in water at 298K is given below:

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3kgm^{-3} at 298K)

This table implies that:

[Sep. 03, 2020(I)]

Options:

- A. α has the highest solubility in water at a given pressure
- B. solubility of γ at 308K is lower than at 298K
- C. The pressure of a 55.5 molal solution of γ is 1 bar
- D. The pressure of 55.5 molal solution of δ is 250 bar

Answer: D

Solution:

(a) From Henry's law $p = K_H(x)$

Higher the value of K_H smaller will be the solubility of the gas, so γ is more soluble.

(b) Though solubility of gases will decrease with increase in temperature but this conclusion can not be drawn from the given table.

(c) For γ

$$(p)_\gamma = (K_H)_\gamma \cdot (x)_\gamma$$

$$= 2 \times 10^{-5} \left[\frac{55.5}{55.5 + \frac{1000}{18}} \right] = 10^{-5} \text{ k bar} = 10^{-2} \text{ bar}$$

(d) For $\delta \Rightarrow$

$$p_\delta = (K_H)_\delta \cdot (x)_\delta$$

$$= 0.5 \left[\frac{55.5}{55.5 + \frac{1000}{18}} \right]$$

$$= 0.5 \times 0.5 = 0.25 \text{ k bar} = 250 \text{ bar}.$$

Question 144

An open beaker of water in equilibrium with water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules: [Sep. 02, 2020 I]

Options:

- A. leaves the vapour increases
- B. leaves the solution increases
- C. leaves the solution decreases
- D. leaves the vapour decreases

Answer: A

Solution:

The vapour pressure of solution will be less than the vapour pressure of pure solvent, so some vapour molecules will get condensed to maintain new equilibrium.

Question 145

The elevation of boiling point of 0.10m aqueous $\text{CrCl}_3 \cdot x\text{NH}_3$ solution is two times that of 0.05m aqueous CaCl_2 solution. The value of x is _____.

[Assume 100% ionisation of the complex and CaCl_2 , coordination number of Cr as 6, and that all NH_3 molecules are present inside the coordination sphere]

[NV, Sep. 06, 2020 (I)]

Answer: 5

Solution:

Molality of CaCl_2 solution = 0.05m

$$\Delta T_b = iK_b m = 3 \times K_b \times 0.05 = 0.15K_b$$

Molality of $\text{CrCl}_3 \cdot x\text{NH}_3$ = 0.10m

$$\Delta T_b' = iK_b \times 0.10 \Rightarrow \Delta T_b' = 2 \Delta T_b$$

$$iK_b \times 0.10 = 2 \times 0.15K_b \Rightarrow i = 3$$

Since, co-ordination number of Cr is 6.

$\therefore$ The complex is $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$.

$\therefore x = 5$.

Question 146

The osmotic pressure of a solution of NaCl is 0.10atm and that of a glucose solution is 0.20atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2L of the

glucose solution is $x \times 10^{-3} \text{ atm}$ · x is _____. (nearest integer)
 [NV, Sep.04, 2020 (II)]

Answer: 167

Solution:

$$\text{For NaCl : } \pi_1 = iC_1RT \Rightarrow C_1 = \frac{0.10}{2RT}$$

$$\text{For Glucose: } \pi_2 = C_2RT \Rightarrow C_2 = \frac{0.20}{RT}$$

When 1L of NaCl solution and 2L glucose solution are mixed.

$$\therefore C_1' = \frac{0.10}{6RT} \text{ and } C_2' = \frac{0.20 \times 2}{3RT} = \frac{0.40}{3RT}$$

$$\therefore \pi_{\text{Total}} = iC_1'RT + C_2'RT = \frac{0.10}{3} + \frac{0.40}{3} = \frac{0.50}{3}$$

$$\pi_{\text{Total}} = 167 \times 10^{-3} \text{ atm}$$

Question147

If 250cm^3 of an aqueous solution containing 0.73g of a protein A is isotonic with one litre of another aqueous solution containing 1.65g of a protein B, at 298K , the ratio of the molecular masses of A and B is _____ $\times 10^{-2}$ (to the nearest integer).
 [NV, Sep. 03, 2020 (II)]

Answer: 177

Solution:

$$\pi_A = iC_A RT, \pi_B = iC_B RT$$

For isotonic solution,

$$\pi_A = \pi_B \quad i_1 C_1 = i_2 C_2 \quad (\text{For protein } i = 1)$$

$$\Rightarrow C_1 = C_2$$

$$\Rightarrow \frac{0.73 \times 1000}{M_A \times 250} = \frac{1.65}{M_B \times 1}$$

$$\therefore \frac{M_A}{M_B} = \frac{0.73 \times 4}{1.65} = 1.77 = 177 \times 10^{-2}$$

Question148

A solution of sodium sulfate contains 92g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is:
 [Jan. 9, 2019(I)]

Options:

A. 12

B. 4

C. 8

D. 16

Answer: B

Solution:

$$\text{Number of moles in 92g of Na}^+ = \frac{92}{23} = 4\text{mol}$$

$$\text{Molality (m)} = \frac{\text{Number of moles}}{\text{Mass of solvent (in kg)}}$$

$$\therefore m = \frac{4}{1} = 4\text{mol kg}^{-1}$$

Question149

Liquids A and B form an ideal solution in the entire composition range. At 350K, the vapor pressures of pure A and pure B are $7 \times 10^3 \text{Pa}$ and $12 \times 10^3 \text{Pa}$, respectively. The composition of the vapour is in equilibrium with a solution containing 40 mole percent of A at this temperature is:
[Jan. 10,2019(I)]

Options:

A. $x_A = 0.37$; $x_B = 0.63$

B. $x_A = 0.28$; $x_B = 0.72$

C. $x_A = 0.4$; $x_B = 0.6$

D. $x_A = 0.76$; $x_B = 0.24$

Answer: B

Solution:

$$P_A^\circ = 7 \times 10^3$$

$$P_B^\circ = 12 \times 10^3$$

$$x_A' = 0.4; x_B' = 1 - 0.4$$

$$x_B' = 0.6$$

$$P_{\text{total}} = P_A^\circ x_A' + P_B^\circ x_B'$$

$$= 7 \times 10^3 \times 0.4 + 12 \times 10^3 \times 0.6$$

$$= (7 \times 0.4 + 12 \times 0.6) \times 10^3 = 10^4$$

$$\therefore x_A = \frac{P_A^\circ x_A'}{P_{\text{total}}} = \frac{7 \times 10^3 \times 0.4}{10^4}$$

$$\therefore x_A = 0.28, x_B = 1 - 0.28 = 0.72$$

Question150

Which one of the following statements regarding Henry's law is not correct?
[Jan. 9,2019(I)]

Options:

A. Higher the value of K_H at a given pressure, higher is the solubility of the gas in liquids.

B. Different gases have different K_H (Henry's law constant) values at the same temperature.

C. The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.

D. The value of K_H increases with increase of temperature and K_H is function of the nature of the gas.

Answer: A

Solution:

The solubility of the gas in liquids decreases with the increase in value of K_H at a given pressure.

Question151

Freezing point of a 4% aqueous solution of X is equal to freezing point of 12% aqueous solution of Y .

If molecular weight of X is A, then molecular weight of Y is:

[Jan. 12,2019 (I)]

Options:

A. 3A

B. 2A

C. A

D. 4A

Answer: A

Solution:

$$(\Delta T_f)_x = (\Delta T_f)_y$$

$$K_f m_x = K_f m_y$$

$$m_x = m_y$$

$$\frac{\text{No. of moles of } x}{\text{wt. of solvent in kg}} = \frac{\text{No. of moles of } y}{\text{wt. of solvent in kg}}$$

Given, 4% aqueous solution of x = 4g of solute, x present in 100g of H_2O

12% aqueous solution of y = 12g of solute, y present in 100g of H_2O

$$\text{Now, } \frac{4g}{m_1} \times \frac{1000}{100} = \frac{12g}{m_2} \times \frac{1000}{100}$$

$$\frac{4g}{A} = \frac{12g}{m_2} \quad [m_1 = A]$$

$$m_2 = 3A$$

Question152

Molecules of benzoic acid (C_6H_5COOH) dimerise in benzene. 'w' g of the acid dissolved in 30g of benzene shows a depression in freezing point equal to 2K . If the percentage association of the acid to form dimer in the solution is 80 , then w is :

(Given that $K_f = 5K \text{ kgmol}^{-1}$, Molar mass of benzoic acid = 122gmol^{-1})

[Jan. 12, 2019 (II)]

Options:

A. 2.4g

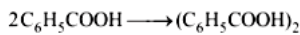
B. 1.0g

C. 1.5g

D. 1.8g

Answer: A

Solution:



w g

Dimer

$$t = 0 \quad 1 \quad 0$$

$$t = t_{\text{eq}} \quad 1 - \alpha \quad \frac{\alpha}{2}$$

$$\text{Moles at equilibrium} = 1 - \alpha + \frac{\alpha}{2} = 1 - \frac{\alpha}{2}$$

$$i = \frac{\text{Total moles at equilibrium}}{\text{Initial moles}} = \frac{1 - \frac{\alpha}{2}}{1}$$

$$i = \frac{1 - \frac{0.8}{2}}{1} = 0.6$$

$$\Delta T_f = i K_f m$$

$$2 = 0.6 \times 5 \times \left(\frac{\frac{w}{122}}{30} \right) \times 1000$$

$$w = 2.4 \text{ g}$$

Question 153

The freezing point of a diluted milk sample is found to be -0.2°C , while it should have been -0.5°C for pure milk. How much water has been added to pure milk to make the diluted sample?
[Jan. 11, 2019 (I)]

Options:

- A. 1 cup of water to 2 cups of pure milk
- B. 3 cups of water to 2 cups of pure milk
- C. 1 cup of water to 3 cups of pure milk
- D. 2 cups of water to 3 cups of pure milk

Answer: B

Solution:

Freezing point of diluted milk = -0.2°C

$$\Delta T_f' = 0.2^\circ\text{C}$$

Freezing point of pure milk = -0.5°C

$$\Delta T_f = 0.5^\circ\text{C}$$

$$\frac{\Delta T_f'}{\Delta T_f} = \frac{K_f \times m}{K_f \times m}; m = \frac{\text{mole of solute}}{\text{mass of solvent (kg)}}$$

Moles of solute are same in both samples.

$$\therefore \frac{0.5}{0.2} = \frac{W'}{W}$$

$$\frac{W'}{W} = \frac{5}{2}; W' = \frac{5}{2}W$$

2 cups of pure milk is mixed with 3 cups of water to make 5 cups of diluted milk.

Question 154

K_2HgI_4 is 40% ionised in aqueous solution. The value of its van't Hoff factor (i) is :
[Jan. 11, 2019 (II)]

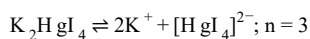
Options:

- A. 1.6
- B. 1.8
- C. 2.0

D. 2.2

Answer: B

Solution:



$$\therefore \alpha = \frac{i-1}{n-1}; 0.4 = \frac{i-1}{3-1}$$

$$i = 1.8$$

Question155

Elevation in the boiling point for 1 molal solution of glucose is 2K . The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2K . The relation between K_b and K_f is:

[Jan. 10, 2019 (II)]

Options:

A. $K_b = 1.5K_f$

B. $K_b = K_f$

C. $K_b = 0.5K_f$

D. $K_b = 2K_f$

Answer: D

Solution:

According to the question we can write

$$\Delta T_b = K_b m \Rightarrow K_b(1) = 2 \Rightarrow K_b = 2\text{K m}^{-1}$$

$$\Delta T_f = K_f m \Rightarrow K_f(2) = 2 \Rightarrow K_f = 1\text{K m}^{-1}$$

$$\text{So, } \frac{K_b}{K_f} = \frac{2}{1}$$

$$\therefore K_b = 2K_f$$

Question156

A solution containing 62g ethylene glycol in 250g water is cooled to -10°C . If K_f for water is 1.86K kg mol^{-1} , the amount of water (in g) separated as ice is:

[Jan. 9, 2019 (II)]

Options:

A. 48

B. 32

C. 64

D. 16

Answer: C

Solution:

As we know,

$$\Delta T_f = K_f \cdot m$$

$$\Rightarrow 10 = 1.86 \times \frac{62 \times 1000}{62 \times w}$$

$$\Rightarrow w = \frac{1.86 \times 1000}{10} = 186\text{g}$$

Total amount of water = 250g

$\therefore$ The amount of water separated as ice

$$= 250 - 186 = 64\text{g}$$

Question157

The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is :
[April 12, 2019 (I)]

Options:

A. 13.88×10^{-2}

B. 13.88×10^{-1}

C. 13.88

D. 13.88×10^{-3}

Answer: C

Solution:

Let, 1 mole be present in the solution,

$$n_{\text{solute}} = 0.2\text{mol}$$

$$n_{\text{solvent}} = 0.8\text{mol}$$

$$= 0.8 \times 18\text{g molality}$$

$$= \frac{\text{Moles of solute}}{\text{Kilogram of solvent}}$$

$$\therefore m = \frac{0.2 \times 1000}{0.8 \times 18} = \frac{1000}{4 \times 18} \approx 13.88$$

Question158

What would be the molality of 20% (mass/mass) aqueous solution of KI ? (molar mass of KI = 166g mol^{-1})
[April 9, 2019 (II)]

Options:

A. 1.08

B. 1.35

C. 1.48

D. 1.51

Answer: D

Solution:

20% W/W KI solution (Given)

i.e. 100g solution contains 20g KI

$$\therefore \text{Mass of solvent} = 100 - 20 = 80\text{g}$$

No. of moles of KI is $\frac{20}{166} = 0.121 \text{ mol}$

$\therefore \text{Molality} = \frac{0.121 \times 1000}{80} = 1.51 \text{ m}$

Question 159

1g of a non-volatile non-electrolyte solute is dissolved in 100g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$, is:

[April 10, 2019 (II)]

Options:

A. 5 : 1

B. 10 : 1

C. 1 : 5

D. 1 : 0.2

Answer: C

Solution:

Ebullioscopic constant (molal deviation const.) is given by,

$$k_b = \frac{\Delta T_b}{m} \Rightarrow \frac{(k_b)_A}{(k_b)_B} = \frac{1}{5}$$

$$\therefore \frac{(\Delta T_b)_A}{(\Delta T_b)_B} = \frac{(k_b)_A}{(k_b)_B} = \frac{1}{5}$$

Question 160

Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is:

(x_M = Mole fraction of 'M' in solution;

x_N = Mole fraction of 'N' in solution;

y_M = Mole fraction of 'M' in vapour phase;

y_N = Mole fraction of 'N' in vapour phase)

[April 9, 2019 (I)]

Options:

A. $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

B. $(x_M - y_M) < (x_N - y_N)$

C. $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

D. $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

Answer: D

Solution:

$$P_M^0 = 450 \text{ mmHg}, P_N^0 = 700 \text{ mmHg}$$

$$P_M = P_M^0 x_M = y_M P_T$$

$$\Rightarrow P_M^0 = \frac{y_M}{x_M} (P_T) \text{ Similarly,}$$

$$P_N^0 = \frac{y_N}{x_N} (P_T)$$

$$\text{Given, } P_M^0 < P_N^0$$

$$\Rightarrow \frac{y_M}{x_M} < \frac{y_N}{x_N}$$

$$\Rightarrow \frac{y_M}{y_N} < \frac{x_M}{x_N}$$

Question 161

The vapour pressures of pure liquids A and B are 400 and 600 mmHg, respectively at 298K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are:
[April 8, 2019(I)]

Options:

- A. 450 mmHg, 0.4, 0.6
- B. 500 mmHg, 0.5, 0.5
- C. 450 mmHg, 0.5, 0.5
- D. 500 mmHg, 0.4, 0.6

Answer: D

Solution:

$$P = x_B P_B^0 + x_A P_A^0$$

$$= 0.5 \times 600 + 0.5 \times 400 = 300 + 200 = 500$$

Using the relation $p_i = y_i P_{\text{Total}}$, we can calculate the mole fractions of the components in vapour phase.

$$P_B = y_B P_{\text{total}}$$

$$y_B = \frac{P_B}{P_{\text{Total}}} = \frac{300}{500} = \frac{3}{5} = 0.6$$

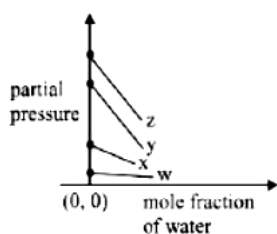
$$y_A = \frac{P_A}{P_{\text{Total}}} = \frac{200}{500} = \frac{2}{5} = 0.4$$

Question 162

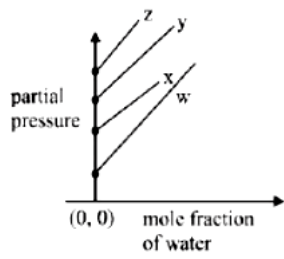
For the solution of the gases w, x, y and z in water at 298K, the Henry's law constants (K_H) are 0.5, 2, 35 and 40 kbar, respectively. The correct plot for the given data is:
[April 8, 2019 (II)]

Options:

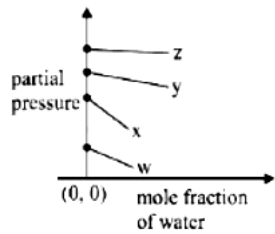
A.



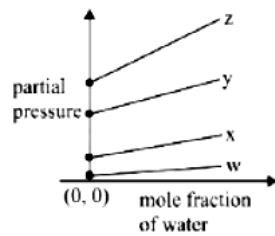
B.



C.



D.



Answer: A

Solution:

According to Henry's law

$$P = K_H \cdot X_{\text{gas}}$$

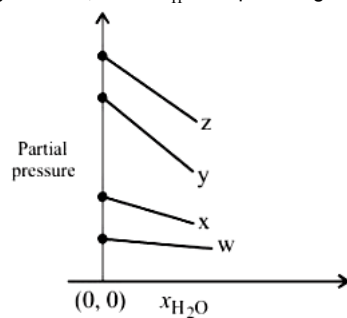
$$\because X_{\text{gas}} + X_{\text{H}_2\text{O}} = 1$$

$$\therefore X_{\text{gas}} = 1 - X_{\text{H}_2\text{O}}$$

$$\therefore P = K_H (1 - X_{\text{H}_2\text{O}})$$

$$\therefore P = K_H - K_H \cdot X_{\text{H}_2\text{O}}$$

$$y = c + mx; m = -K_H \Rightarrow \text{slope is negative.}$$



Gas	K_H
w	0.5
x	2
y	35
z	50

Question163

A solution is prepared by dissolving 0.6g of urea (molar mass = 60g mol^{-1}) and 1.8g of glucose (molar mass = 180g mol^{-1}) in 100mL, of water at 27°C . The osmotic pressure of the solution is:

($R = 0.08206\text{LatmK}^{-1}\text{mol}^{-1}$)

[April 12, 2019 (II)]

Options:

A. 8.2atm

B. 2.46atm

C. 4.92atm

D. 1.64atm

Answer: C

Solution:

Osmotic pressure (π) = CRT

Since, there are two solutes i.e. urea and glucose.

$$\therefore \pi = (C_1 + C_2)RT$$

$$= \left(\frac{0.6}{60 \times 0.1} + \frac{1.8}{180 \times 0.1} \right) \times 0.0821 \times 300$$

$$= 0.2 \times 0.0821 \times 300 = 4.926\text{atm}$$

Question164

At room temperature, a dilute solution of urea is prepared by dissolving 0.60g of urea in 360g of water. If the vapour pressure of pure water at this temperature is 35mmHg, lowering of vapour pressure will be :

(molar mass of urea = 60g mol^{-1})

[April 10, 2019 (I)]

Options:

A. 0.027mmHg

B. 0.028mmHg

C. 0.017mmHg

D. 0.031mmHg

Answer: C

Solution:

Relative lowering of vapour pressure, is given by,

$$\frac{p^\circ - p}{p^\circ} = x_A = \frac{n_A}{n_A + n_B} = \frac{n_A}{n_B}$$

$$\text{Given, } p^\circ = 35\text{mmHg, } n_{\text{urea}} = \frac{0.60}{60}, n_{\text{water}} = \frac{360}{18}$$

$$\frac{p^\circ - p}{35} = \frac{0.6 \times 18}{60 \times 360} = \frac{1}{2000}$$

$$\Delta p = p^\circ - p = 0.017$$

Question165

The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01M BaCl₂ in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol L⁻¹) in solution is:
[April 9, 2019 (I)]

Options:

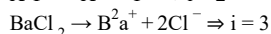
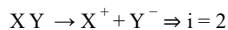
- A. 4×10^{-2}
- B. 6×10^{-2}
- C. 4×10^{-4}
- D. 16×10^{-4}

Answer: B

Solution:

We know, $\pi = iCRT$; $\pi_{xy} = 4\pi_{BaCl_2}$

Since both are ionic solute, i.e.,



Now,

$$2[XY] = 4 \times 3 \times [0.01]$$

$$[XY] = 0.06$$

$$= 6 \times 10^{-2} \text{ mol/L}$$

Question166

Molar depression constant for a solvent is 4.0K kgmol⁻¹. The depression in the freezing point of the solvent for 0.03mol kg⁻¹ solution K₂SO₄ is:

(Assume complete dissociation of the electrolyte)

[April 9, 2019 (II)]

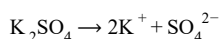
Options:

- A. 0.18K
- B. 0.24K
- C. 0.12K
- D. 0.36K

Answer: D

Solution:

Dissociation of Potassium Sulphate (K₂SO₄),



i (Van't Hoff factor) = 3

We know that, $\Delta T_f = iK_f m$

where, K_f is molal depression constant and m is molality.

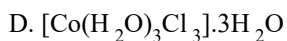
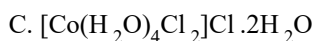
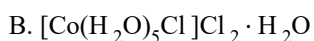
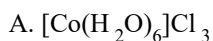
$$= 3 \times 4 \times 0.03 = 0.36K$$

Question167

For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point?

[2018]

Options:



Answer: D

Solution:

Number of particles (i)	
(a) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$	4
(b) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$	3
(c) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$	2
(d) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$	1

z; where $\Delta T_f = (T_f - T_f')$

Remember, greater the no. of particles, lower will be the freezing point. Compound (d) will have the highest freezing point due to least number of particles.

Question 168

The mass of non-volatile, non-electrolyte solute (molar mass = 50g mol^{-1}) needed to be dissolved in 114g octane to reduce its vapour pressure to 75%, is
[Online April 16, 2018]

Options:

A. 37.5g

B. 75g

C. 150g

D. 50g

Answer: C

Solution:

Molar mass of octane = 114g/mol

From the lowering of vapour pressure we have,

$$\frac{\Delta P}{P} = \frac{\frac{W_2}{M_2}}{\frac{W_2}{M_2} + \frac{W_1}{M_1}}$$
$$\frac{75}{100} = \frac{\frac{W_2}{50\text{g/mol}}}{\frac{W_2}{50\text{g/mol}} + \frac{114\text{g}}{114\text{g/mol}}}$$
$$0.75 = \frac{\frac{W_2}{50}}{\frac{W_2}{50} + 1}$$

$$\frac{W_2}{50} + 1 = \frac{W_2}{50 \times 0.75}$$

$$W_2 = 150\text{g}$$

Note : W_2 and M_2 are mass and molar mass of solute whereas W_1 and M_1 are mass and molar mass of octane.

Question 169

Two 5 molal solutions are prepared by dissolving a non-electrolyte, non-volatile solute separately in the solvents X and Y. The molecular weights of the solvents are M_X and M_Y , respectively where

$M_X = \frac{3}{4}M_Y$. The relative lowering of vapour pressure of the solution in X is "m" times that of the solution in Y. Given that the number of moles of solute is very small in comparison to that of solvent, the value of "m" is:

[Online April 15, 2018 (II)]

Options:

A. $\frac{3}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{4}{3}$

Answer: A

Solution:

The relationship between molar masses of the two solvents is

$$M_X = \frac{3}{4}M_Y \dots (i)$$

The relative lowering of vapour pressure of the two solutions is

$$\left(\frac{\Delta P}{P}\right)_X = m \left(\frac{\Delta P}{P}\right)_Y$$

But, the relative lowering of vapour pressure of solutions is directly proportional to the mole fraction of solute. Given 5 molal solution, means 5 moles of solute are dissolved in 1kg (or 1000g) of solvent.

$$\text{The number of moles of solvent} = \frac{1000\text{g}}{M}$$

$$\text{The mole fraction of solute} = \frac{5}{1000/M}$$

$$= M \times \frac{5}{1000}$$

$$\text{Hence, } M_X \times \frac{5}{1000} = m \times M_Y \times \frac{5}{1000} \dots (ii)$$

Substitute equation (i) in equation (ii)

$$\frac{3}{4} \times M_Y \times \frac{5}{1000} = m \times M_Y \times \frac{5}{1000}$$

$$m = \frac{3}{4}$$

Question 170

A solution is prepared by mixing 8.5g of CH_2Cl_2 and 11.95 g of CHCl_3 . If vapour pressure of CH_2Cl_2 and CHCl_3 at 298 K are 415 and 200mmHg respectively, the mole fraction of CHCl_3 in vapour form is:

(Molar mass of Cl = 35.5gmol^{-1})

[Online April 9, 2017]

Options:

A. 0.162

B. 0.675

C. 0.325

D. 0.486

Answer: C

Solution:

Molar mass of $\text{CHCl}_3 = 119.5 \text{ g/mol}$

Molar mass of $\text{CH}_2\text{Cl}_2 = 85 \text{ g/mol}$

Moles of $\text{CHCl}_3 = \frac{11.95}{119.5} = 0.1 \text{ mol}$

Moles of $\text{CH}_2\text{Cl}_2 = \frac{8.5}{85} = 0.1 \text{ mol}$

Mole fraction of $\text{CHCl}_3 = \frac{0.1}{0.2} = 0.5 \text{ mol}$

Mole fraction of $\text{CH}_2\text{Cl}_2 = \frac{0.1}{0.2} = 0.5 \text{ mol}$

(Given -

Vapour pressure of $\text{CHCl}_3 = 200 \text{ mmHg} = 0.263 \text{ atm}$.

Vapour pressure of $\text{CH}_2\text{Cl}_2 = 415 \text{ mmHg} = 0.546 \text{ atm}$.) (1 atm = 760 mmHg)

$\therefore P_{(\text{total})} = \text{Mole fraction of } \text{CHCl}_3 \times$

(Vapour pressure of CHCl_3) + Mole fraction of CH_2Cl_2

$\times$ (Vapour pressure of CH_2Cl_2)

$= 0.5 \times 0.263 + 0.5 \times 0.546 = 0.4045$

Mole fraction of CHCl_3 in vapour form

$= \frac{\text{vapour pressure}}{P_{\text{total}}} = \frac{0.1315}{0.4045} = 0.325$.

Question 171

The freezing point of benzene decreases by 0.45°C when 0.2g of acetic acid is added to 20g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be:

(K_f for benzene = $5.12 \text{ K kg mol}^{-1}$)

[2017]

Options:

A. 64.6%

B. 80.4%

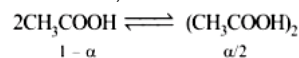
C. 74.6%

D. 94.6%

Answer: D

Solution:

In benzene,



$$i = 1 - \alpha + \alpha/2 = 1 - \alpha/2$$

Here α is degree of association

$\Delta T_f = i \times K_f \times m$

$$0.45 = \left(1 - \frac{\alpha}{2}\right) (5.12) \frac{\left(\frac{0.2}{60}\right)}{\frac{20}{1000}}$$

$$1 - \frac{\alpha}{2} = 0.527$$

$\alpha = 0.945$
 % degree of association = 94.6%

Question 172

5g of Na_2SO_4 was dissolved in xg of H_2O . The change in freezing point was found to be 3.82°C . If Na_2SO_4 is 81.5% ionised, the value of x

(K_f for water = $1.86^\circ\text{C kg mol}^{-1}$) is approximately:

(molar mass of S = 32 g mol^{-1} and that of Na = 23 g mol^{-1})

[Online April 8, 2017]

Options:

A. 15g

B. 25g

C. 45g

D. 65g

Answer: C

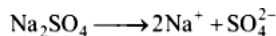
Solution:

Molality (experimental)

$$= \frac{\Delta T_f}{K_f} = \frac{3.82}{1.86} = 2.054 \text{ mol} / 1000 \text{ g solvent}$$

$$\text{Molality (theoretical)} = \frac{\text{moles of solute}}{\text{wt. of solvent (g)}} \times 1000$$

$$= \frac{5 \text{ g} / 142 \text{ g/mol}}{x} \times 1000$$



Moles before

dissociation	1	0	0
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Moles after

dissociation	$1 - \alpha$	2α	α
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$$\text{Von't Hoff factor (i)} = \frac{\text{Moles after dissociation}}{\text{Moles before dissociation}}$$

$$= \frac{(1 - \alpha) + 2\alpha + \alpha}{1}$$

Na_2SO_4 is ionised 81.5%

means $\alpha = 0.815$

$$= \frac{(1 - 0.815) + 2 \times 0.815 + 0.815}{1}$$

$$= 2.63$$

$$i = \frac{\text{Observed molality}}{\text{Calculated molality}}$$

$$2.63 = \frac{2.054}{\frac{0.0352}{x} \times 1000}$$

$$x = 45.07 \text{ g}$$

Question 173

The solubility of N_2 in water at 300K and 500 torr partial pressure is 0.01 g L^{-1} . The solubility (in g L^{-1}) at 750 torr partial pressure is:

[Online April 9, 2016]

Options:

A. 0.0075

B. 0.005

C. 0.02

D. 0.015

Answer: D

Solution:

According to Henry's law

$$\frac{P_1}{P_2} = \frac{S_1}{S_2}$$
$$\frac{500}{750} = \frac{0.01}{S_2}$$
$$\therefore S_2 = \frac{750 \times 0.01}{500} = 0.015 \text{ g/L}$$

Question174

An aqueous solution of a salt MX_2 at certain temperature has a van't Hoff factor of 2 . The degree of dissociation for this solution of the salt is:
[Online April 10,2016]

Options:

A. 0.50

B. 0.33

C. 0.67

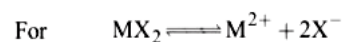
D. 0.80

Answer: A

Solution:

Van't Hoff factor (i)

$$= \frac{\text{No. of particles after association or dissociation}}{\text{No. of particles before association or dissociation}}$$



$$t = 0 \quad 1 \quad 0 \quad 0$$

$$\text{at eq.} \quad 1 - \alpha \quad \alpha \quad 2\alpha$$

$$\text{Total no. of particles} = 1 - \alpha + \alpha + 2\alpha = 1 + 2\alpha$$

$$\therefore i = \frac{1 + 2\alpha}{1} = 2$$

$$\therefore \alpha = 0.50 \text{ or } 50\%$$

Question175

18 g glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is added to 178.2g water. The vapour pressure of water (in torr) for this aqueous solution is:
[2016, 2006]

Options:

A. 752.4

B. 759.0

C. 7.6

D. 76.0

Answer: A

Solution:

According to Raoult's Law

$$\frac{P^\circ - P_s}{P_s^\circ} = \frac{W_B \times M_A}{M_B \times W_A} \dots (i)$$

Here P° = Vapour pressure of pure solvent,

P_s = Vapour pressure of solution

W_B = Mass of solute, W_A = Mass of solvent

M_B = Molar mass of solute, M_A = Molar Mass of solvent

Vapour pressure of pure water at 100°C (by assumption = 760 torr)

By substituting values in equation (i) we get,

$$\frac{760 - P_s}{P_s^\circ} = \frac{18 \times 18}{180 \times 178.2} \dots (ii)$$

On solving (ii) we get

$P_s = 752.4$ torr

Question176

A solution at 20°C is composed of 1.5mol of benzene and 3.5mol of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively:

[Online April 10,2015]

Options:

A. 35.8 torr and 0.280

B. 38.0 torr and 0.589

C. 30.5 torr and 0.389

D. 30.5 torr and 0.480

Answer: B

Solution:

Total V.P. of solution = $P_A^\circ X_A + P_B^\circ X_B$

Given, $P_A^\circ = 74.7$ torr, $P_B^\circ = 22.3$ torr

$n_{\text{benzene}} = 1.5$ mol, $n_{\text{toluene}} = 3.5$ mol

$n_{\text{solution}} = 1.5 + 3.5 = 5$ mol

$$x_A = \frac{n_{\text{benzene}}}{n_{\text{solution}}} = \frac{1.5}{5} = 0.3$$

$$x_B = \frac{n_{\text{toluene}}}{n_{\text{solution}}} = \frac{3.5}{5} = 0.7$$

Total V.P. of solution = $(0.3 \times 74.7 + 0.7 \times 22.3)$ torr

= $(22.4 + 15.6)$ torr = 38 torr

Mole fraction of benzene in vapour form = $\frac{22.4}{38} = 0.589$

Question177

The vapour pressure of acetone at 20°C is 185 torr. When 1.2g of a non-volatile substance was dissolved in 100g of acetone at 20°C , its vapour pressure was 183 torr. The molar mass (gmol^{-1}) of the substance is :

[2015]

Options:

- A. 128
- B. 488
- C. 32
- D. 64

Answer: D

Solution:

Using relation,

$$\frac{P^\circ - P_s}{P^\circ} = \frac{w_2 M_1}{w_1 M_2}$$

where w_1, M_1 = mass in g and mol. mass of solvent

w_2, M_2 = mass in g and mol. mass of solute

$P^\circ = 185 \text{ torr}, P_s = 183 \text{ torr}$

$$\frac{185 - 183}{185} = \frac{1.2 \times 58}{100 \times M_2}$$

(Mol. mass of acetone = 58)

$M_2 = 64.68 \approx 64$

$\therefore$ Molar mass of substance = 64

Question 178

Determination of the molar mass of acetic acid in benzene using freezing point depression is affected by:

[Online April 11, 2015]

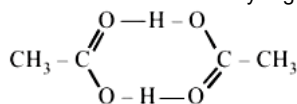
Options:

- A. partial ionization
- B. dissociation
- C. complex formation
- D. association

Answer: D

Solution:

Acetic acid contains carboxylic group $-\text{COOH}$ which can form H-bonding so acetic acid dimerises.



Question 179

Choose the correct statement with respect to the vapour pressure of a liquid among the following:
[Online April 19, 2014]

Options:

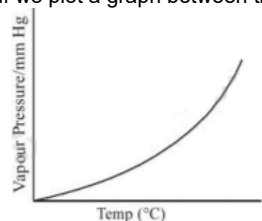
- A. Increases linearly with increasing temperature
- B. Increases non-linearly with increasing temperature
- C. Decreases linearly with increasing temperature

D. Decreases non-linearly with increasing temperature

Answer: B

Solution:

If we plot a graph between the vapour pressure and the temperature. We would get a curve that rises faster as T increases, giving a curved line.



Question180

For an ideal solution of two components A and B, which of the following is true?
[Online April 19,2014]

Options:

- A. $\Delta H_{\text{mixing}} < 0$ (zero)
- B. $\Delta H_{\text{mixing}} > 0$ (zero)
- C. A – B interaction is stronger than A – A and B – B interactions
- D. A – A, B – B and A – B interactions are identical.

Answer: D

Solution:

Solutions in which solute-solute and solvent-solvent interactions are almost similar to solute solvent interactions are known as ideal solution.

Question181

Consider separate solutions of 0.500M C₂H₅OH (aq), 0.100 M M g₃(PO₄)₂(aq), 0.250M K Br(aq) and 0.125M N a₃PO₄(aq) at 25°C. Which statement is true about these solutions, assuming all salts to be strong electrolytes?
[2014]

Options:

- A. They all have the same osmotic pressure.
- B. 0.100M M g₃(PO₄)₂(aq) has the highest osmotic pressure.
- C. 0.125M N a₃PO₄(aq) has the highest osmotic pressure.
- D. 0.500M C₂H₅OH (aq) has the highest osmotic pressure.

Answer: A

Solution:

$$\pi = iCRT \quad \pi_{\text{C}_2\text{H}_5\text{OH}} = 1 \times 0.500 \times R \times T = 0.5RT$$

$$\pi_{\text{M g}_3(\text{PO}_4)_2} = 5 \times 0.100 \times R \times T = 0.5RT$$

$$\pi_{\text{KBr}} = 2 \times 0.250 \times R \times T = 0.5RT$$

$$\pi_{\text{Na}_3\text{PO}_4} = 4 \times 0.125 \times RT = 0.5RT$$

Since the osmotic pressure of all the given solutions is equal. Hence all are isotonic solutions.

Question182

The observed osmotic pressure for a 0.10M solution of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$ at 25°C is 10.8atm. The expected and experimental (observed) values of van't Hoff factor (i) will be respectively:

($R = 0.082\text{LatmK}^{-1}\text{mol}^{-1}$)

[Online April 19, 2014]

Options:

A. 5 and 4.42

B. 4 and 4.00

C. 5 and 3.42

D. 3 and 5.42

Answer: A

Solution:

Given, $\pi_{\text{ob}} = 10.8\text{atm}$

$$\pi_{\text{nor}} = CRT = 0.10 \times 0.0821 \times 298 = 2.446$$

Now experimental value of (i)

$$= \frac{\text{Observed osmotic pressure}}{\text{Normal osmotic pressure}} = \frac{10.8}{2.446} = 4.42$$

Question183

The molarity of a solution obtained by mixing 750mL of 0.5(M)HCl with 250mL of 2(M)HCl will be: [2013]

Options:

A. 0.875M

B. 1.00M

C. 1.75M

D. 0.975M

Answer: A

Solution:

From molarity equation :

$$M_1V_1 + M_2V_2 = M \times V$$

$$M = \frac{M_1V_1 + M_2V_2}{V} \text{ where } V = \text{total volume}$$

$$= \frac{750 \times 0.5 + 250 \times 2}{1000}$$

$$= 0.875M$$

Question184

Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be :

[Online April 23, 2013]

Options:

A. 0.137

B. 0.237

C. 0.435

D. 0.205

Answer: B

Solution:

$$P_A = P_A^\circ \times X_A = \text{total pressure} \times y_A$$

$$P_B = P_B^\circ \times X_B = \text{total pressure} \times y_B$$

where x and y represents mole fraction in liquid and vapour phase respectively.

$$\frac{P_B^\circ X_B}{P_A^\circ X_A} = \frac{y_B}{y_A}; \frac{P_B^\circ (1 - X_A)}{P_A^\circ X_A} = \frac{1 - y_A}{y_A}$$

$$\text{on putting values } \frac{119(1 - 0.50)}{37 \times 0.50} = \frac{1 - y_A}{y_A}$$

$$\text{on solving } y_A = 0.237$$

Question185

12g of a nonvolatile solute dissolved in 108g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is :

[Online April9, 2013]

Options:

A. 80

B. 60

C. 20

D. 40

Answer: C

Solution:

$$\frac{P^\circ - P_s}{P^\circ} = \frac{W_B \times M_A}{M_B \times W_A}$$

$$0.1 = \frac{12 \times 18}{M_B \times 108}$$

$$m_B = \frac{12 \times 18}{0.1 \times 108} = 20$$

Question186

A molecule M associates in a given solvent according to the equation $M \rightleftharpoons (M)_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is:

[Online April 22, 2013]

Options:

- A. 3
- B. 5
- C. 2
- D. 4

Answer: C

Solution:

van't Hoff factor (i) and the degree of association are related as below:

$$i = 1 - \alpha \left(1 - \frac{1}{n} \right)$$

$$0.9 = 1 - 0.2 \left(1 - \frac{1}{n} \right)$$

On solving,

$$\left(1 - \frac{1}{n} \right) = \frac{1}{2}$$

$$\frac{1}{n} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\therefore n = 2$$

Question187

How many grams of methyl alcohol should be added to 10 litre tank of water to prevent its freezing at 268K ?(K_f for water is 1.86K kgmol⁻¹)

[Online April 25, 2013]

Options:

- A. 880.07g
- B. 899.04g
- C. 886.02g
- D. 868.06g

Answer: D

Solution:

$$\Delta T_f = K_f m$$

where m = molality

$$273 - 268 = 1.86 \times \frac{w}{M \times V}$$

$$5 = 1.86 \times \frac{w}{32 \times 10}$$

$$w = \frac{5 \times 32 \times 10}{1.86} = 860.2 \approx 868.06g$$

Question188

The density of a solution prepared by dissolving 120g of urea (mol. mass = 60u) in 1000g of water is 1.15g/mL. The molarity of this solution is :

[2012]

Options:

- A. 0.50M

B. 1.78M

C. 1.02M

D. 2.05M

Answer: D

Solution:

$$\begin{aligned}\text{Molarity} &= \frac{\text{Moles of solute}}{\text{Volume of solution (l)}} \\ \text{Mass of solution} &= 1000 + 120 = 1120 \\ d &= \frac{M}{V}; V = \frac{M}{d} = \frac{1120}{1.15} \text{mL} \\ &= \frac{120 \times 1.15}{60 \times 1120} \times 1000 = 2.05\text{M}\end{aligned}$$

Question189

K_f for water is 1.86K kgmol^{-1} . If your automobile radiator holds 1.0kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must you add to get the freezing point of the solution lowered to -2.8°C ? [2012]

Options:

A. 72g

B. 93g

C. 39g

D. 27g

Answer: B

Solution:

$$\begin{aligned}\Delta T_f &= i \times K_f \times m \\ \text{Given } \Delta T_f &= 2.8, K_f = 1.86\text{K kgmol}^{-1}, i = 1 \\ (\text{ethylene glycol is a non-electrolyte}) \\ \text{Wt. of solvent} &= 1\text{kg} \\ \text{Let of wt of solute} &= x \\ \text{Mol. wt of ethylene glycol} &= 62 \\ 2.8 &= 1 \times 1.86 \times \frac{x}{62 \times 1} \\ \text{or } x &= \frac{2.8 \times 62}{1.86} = 93\text{g}\end{aligned}$$

Question190

A solution containing 0.85g of ZnCl_2 in 125.0g of water freezes at -0.23°C . The apparent degree of dissociation of the salt is (K_f for water = 1.86K kgmol^{-1} , atomic mass:

$\text{Zn} = 65.3$ and $\text{Cl} = 35.5$)

[Online May 12, 2012]

Options:

A. 1.36%

B. 73.5%

C. 7.35%

D. 2.47%

Answer: B

Solution:

$$\text{Mol.wt. (M}_B) = \frac{K_f \times W_B \times 1000}{\Delta T_f \times W_A}$$
$$= \frac{1.86 \times 0.85 \times 1000}{0.23 \times 125} \approx 55\text{g}$$

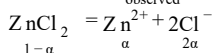
Where,

$$W_B = 0.85\text{g}$$

$$W_A = 125\text{g}$$

$$\Delta T_f = 0^\circ\text{C} - (-23^\circ\text{C}) = 23^\circ\text{C}$$

$$\text{Now, } i = \frac{M_{\text{normal}}}{M_{\text{observed}}} = \frac{136.3}{55} = 2.47$$



$$\text{Van't Hoff factor (i)} = \frac{1 - \alpha + \alpha + 2\alpha}{1} = 2.47$$

$$\therefore \alpha = 0.735 = 73.5\%$$

Question191

The freezing point of a 1.00m aqueous solution of H F is found to be -1.91°C . The freezing point constant of water, K_f is 1.86K kgmol^{-1} . The percentage dissociation of H F at this concentration is [Online May 26, 2012]

Options:

A. 30%

B. 10%

C. 5.2%

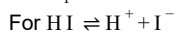
D. 2.7%

Answer: D

Solution:

$$\Delta T_f = i \times K_f \times m$$

$$i = \frac{\Delta T_f}{K_f \times m} = \frac{1.91}{1.86 \times 1} = 1.02$$



$$1 - \alpha + \alpha + \alpha = i = 1.027$$

$$1 + \alpha = 1.027$$

$$\alpha = 0.027 \text{ or } 2.7\%$$

Question192

Liquids A and B form an ideal solution. At 30°C , the total vapour pressure of a solution containing 1 mol of A and 2 mol of B is 250mmHg. The total vapour pressure becomes 300mmHg when 1 more mol of A is added to the first solution. The vapour pressures of pure A and B at the same temperature are [Online May 19, 2012]

Options:

A. 150, 450mmHg

B. 125, 150mmHg

C. 450, 150mmHg

D. 250, 300mmHg

Answer: C

Solution:

Let vapour pressure of A = P_A°

Vapour pressure of B = P_B°

In first solution,

$$\text{Mole fraction of A}(X_A) = \frac{1}{1+2} = \frac{1}{3}$$

$$\text{Mole fraction of B}(X_B) = \frac{2}{1+2} = \frac{2}{3}$$

According to Raoult's law,

$$\text{Total vapour pressure} = 250 = P_A^\circ X_A + P_B^\circ X_B$$

$$250 = \frac{1}{3}P_A^\circ + \frac{2}{3}P_B^\circ \dots (i)$$

In second solution

$$\text{Mole fraction of A}(X_A) = \frac{2}{2+2} = \frac{2}{4} = \frac{1}{2}$$

$$\text{Mole fraction of B}(X_B) = \frac{2}{4} = \frac{1}{2}$$

$$\therefore \text{Total vapour pressure} = 300 = P_A^\circ X_A + P_B^\circ X_B$$

$$300 = \frac{1}{2}P_A^\circ + \frac{1}{2}P_B^\circ \dots (ii)$$

Multiplying equation (i) by $\frac{1}{2}$ and equation (ii) by $\frac{1}{3}$

$$\frac{1}{6}P_A^\circ + \frac{2}{6}P_B^\circ = 125$$

$$\frac{1}{6}P_A^\circ + \frac{1}{6}P_B^\circ = 100$$

$$\frac{1}{6}P_B^\circ = 25$$

$P_B^\circ = 25 \times 6 = 150\text{mmHg}$ On substituting value of P_B° in equation (ii) we get

$$300 = P_A^\circ \times \frac{1}{2} + 150 \times \frac{1}{2}$$

$$P_A^\circ = 450\text{mmHg}$$

Question 193

A 5% solution of cane sugar (molar mass 342) is isotonic with 1% of a solution of an unknown solute. The molar mass of unknown solute in g/mol is : [2011RS]

Options:

A. 171.2

B. 68.4

C. 34.2

D. 136.2

Answer: B

Solution:

For isotonic solutions

$$\pi_1 = \pi_2$$

$$C_1 = C_2$$

$$\frac{5/342}{0.1} = \frac{1/M}{0.1}$$

$$\frac{5}{342} = \frac{1}{M}$$

$$\Rightarrow M = \frac{342}{5} = 68.4 \text{ g/mol}$$

Question 194

On mixing, heptane and octane form an ideal solution. At 373K, the vapour pressures of the two liquid components (heptane and octane) are 105kPa and 45kPa respectively. Vapour pressure of the solution obtained by mixing 25.0g of heptane and 35g of octane will be (molar mass of heptane = 100 g mol^{-1} and of octane = 114 g mol^{-1})
[2010]

Options:

- A. 72.0kPa
- B. 36.1kPa
- C. 96.2kPa
- D. 144.5kPa

Answer: A

Solution:

$$\begin{aligned}
 P_{\text{Total}} &= P_A^{\circ} X_A + P_B^{\circ} X_B \\
 &= P_{\text{Heptane}}^{\circ} X_{\text{Heptane}} + P_{\text{Octane}}^{\circ} X_{\text{Octane}} \\
 &= 105 \times \frac{25/100}{\frac{25}{100} + \frac{35}{114}} + 45 \times \frac{35/114}{\frac{25}{100} + \frac{35}{114}} \\
 &= 105 \times \frac{0.25}{0.25 + 0.3} + 45 \times \frac{0.3}{0.25 + 0.3} \\
 &= \frac{105 \times 0.25}{0.55} + \frac{45 \times 0.3}{0.55} = \frac{26.25 + 13.5}{0.55} \\
 &= 72 \text{ kPa}
 \end{aligned}$$

Question 195

If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1kg of water, is ($K_f = 1.86 \text{ K kg mol}^{-1}$)
[2010]

Options:

- A. 0.372K
- B. 0.0558K
- C. 0.0744K
- D. 0.0186K

Answer: B

Solution:

Sodium sulphate dissociates as
 $\text{Na}_2\text{SO}_4(\text{s}) \longrightarrow 2\text{Na}^+ + \text{SO}_4^{2-}$
 Hence van't Hoff factor, $i = 3$

$$\begin{aligned}\text{Now } \Delta T_f &= iK_f \cdot m \\ &= 3 \times 1.86 \times 0.01 = 0.0558\text{K}\end{aligned}$$

Question 196

Two liquids X and Y form an ideal solution. At 300K, vapour pressure of the solution containing 1mol of X and 3mol of Y is 550mmHg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10mmHg. Vapour pressure (in mmHg) of X and Y in their pure states will be, respectively:
[2009]

Options:

- A. 300 and 400
- B. 400 and 600
- C. 500 and 600
- D. 200 and 300

Answer: B

Solution:

$$\begin{aligned}P_{\text{total}} &= P_x^\circ X_x + P_y^\circ X_y \\ 550 &= P_x^\circ \times \frac{1}{4} + P_y^\circ \times \frac{3}{4} \\ P_x^\circ + 3P_y^\circ &= 550 \times 4 \dots (i) \\ \text{In second case,} \\ P_{\text{total}} &= P_x^\circ \times \frac{1}{5} + P_y^\circ \times \frac{4}{5} \\ P_x^\circ + 4P_y^\circ &= 560 \times 5 \dots (ii) \\ \text{Subtract (i) from (ii)} \\ \therefore P_y^\circ &= 560 \times 5 - 550 \times 4 = 600 \\ \therefore P_x^\circ &= 400\end{aligned}$$

Question 197

A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution?
[2009]

Options:

- A. The solution is non-ideal, showing -ve deviation from Raoult's Law.
- B. The solution is non-ideal, showing + ve deviation from Raoult's Law.
- C. n-Heptane shows + ve deviation while ethanol shows - ve deviation from Raoult's Law.
- D. The solution formed is an ideal solution.

Answer: B

Solution:

For this solution intermolecular interactions between n-heptane and ethanol are weaker than n-heptane-n-heptane & ethanol-ethanol interactions, hence the solution of n-heptane and ethanol is non-ideal and shows positive deviation from Raoult's law.

Question 198

At 80°C, the vapour pressure of pure liquid 'A' is 520mm Hg and that of pure liquid 'B' is 1000mmHg. If a mixture solution of 'A' and 'B' boils at 80°C and at 1 atm pressure, the amount of 'A' in the mixture is (1atm = 760mmHg) [2008]

Options:

- A. 52 mol percent
- B. 34mol percent
- C. 48 mol percent
- D. 50mol percent

Answer: D

Solution:

At 1 atmospheric pressure the boiling point of mixture is 80°C.
 At boiling point the vapour pressure of mixture, $P_T = 1 \text{ atmosphere} = 760\text{mmHg}$.
 Using the relation,
 $P_T = P_A^\circ X_A + P_B^\circ X_B$, we get
 $P_B^\circ = 1000\text{mmHg}$, $X_A + X_B = 1$ }
 or $760 = 520X_A + 1000 - 1000X_A$
 or $480X_A = 240$
 or $X_A = \frac{240}{480} = \frac{1}{2}$ or 50 mol. percent

Question199

The vapour pressure of water at 20°C is 17.5mmHg. If 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2g of water at 20°C, the vapour pressure of the resulting solution will be [2008]

Options:

- A. 17.325mmHg
- B. 15.750mmHg
- C. 16.500mmHg
- D. 17.500mmHg

Answer: A

Solution:

Note: On addition of glucose to water, vapour pressure of water will decrease. The vapour pressure of a solution of glucose in water can be calculated using the relation

$$\frac{P^\circ - P_s}{P^\circ} = \frac{\text{Moles of glucose in solution}}{\text{Moles of water in solution}}$$

$$\text{or } \frac{17.5 - P_s}{17.5} = \frac{18/180}{178.2/18} \quad [\because P^\circ = 17.5]$$

$$\text{or } 17.5 - P_s = \frac{0.1 \times 17.5}{9.9}$$

$$\text{or } P_s = 17.325\text{mmHg.}$$

Question200

A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290mm at 300K . The vapour pressure of propyl alcohol is 200mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be [2007]

Options:

- A. 360
- B. 350
- C. 300
- D. 700

Answer: B

Solution:

$$\begin{aligned}
 P_A^\circ &= ?, \text{ Given } P_B^\circ = 200\text{mm}, X_A = 0.6, \\
 X_B &= 1 - 0.6 = 0.4, P = 290 \\
 P &= P_A + P_B = P_A^\circ X_A + P_B^\circ X_B \\
 290 &= P_A^\circ \times 0.6 + 200 \times 0.4 \\
 \therefore P_A^\circ &= 350\text{mm}
 \end{aligned}$$

Question201

Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is [2007]

Options:

- A. 1/2
- B. 2/3
- C. $\frac{1}{3} \times \frac{273}{298}$
- D. 1/3

Answer: D

Solution:

$$\begin{aligned}
 &\text{Let the mass of methane and oxygen} = m\text{g.} \\
 &\text{Mole fraction of } O_2 \\
 &= \frac{\text{Moles of } O_2}{\text{Moles of } O_2 + \text{Moles of } CH_4} \\
 &= \frac{m/32}{m/32 + m/16} = \frac{m/32}{3m/32} = \frac{1}{3} \\
 &\text{Partial pressure of } O_2 = \text{Total pressure} \times \text{mole fraction of } O_2, \\
 &P_{O_2} = P \times \frac{1}{3} = \frac{1}{3}P
 \end{aligned}$$

Question202

A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60gmol^{-1}) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0gcm^{-3} , molar mass

**of the substance will be
[2007]**

Options:

A. 210.0 g mol^{-1}

B. 90.0 g mol^{-1}

C. 115.0 g mol^{-1}

D. 105.0 g mol^{-1} .

Answer: A

Solution:

Osmotic pressure (π) of isotonic solutions are equal. For solution of unknown substance

($\pi = CRT$)

$$C_1 = \frac{5.25/M}{V}$$

For solution of urea,

$$C_2(\text{concentration}) = \frac{1.5/60}{V}$$

Given, $\pi_1 = \pi_2$

$\therefore \pi = CRT$

$\therefore C_1 RT = C_2 RT$

or $C_1 = C_2$

$$\text{or } \frac{5.25/M}{V} = \frac{1.5/60}{V}$$

$$\therefore M = 210 \text{ g/mol}$$

Question 203

**Among the following mixtures, dipole-dipole as the major interaction, is present in
[2006]**

Options:

A. KCl and water

B. benzene and carbon tetrachloride

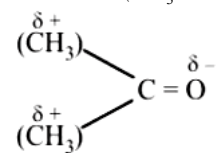
C. benzene and ethanol

D. acetonitrile and acetone

Answer: D

Solution:

Acetonitrile ($\overset{\delta+}{\text{CH}_3} - \text{C} \equiv \text{N}^{\delta-}$) and acetone



both are polar molecules, hence dipole-dipole interaction exists between them. Between KCl and water, ion-dipole interaction is found and in benzene-ethanol and benzene- carbon tetrachloride dispersion force is present.

Question 204

Benzene and toluene form nearly ideal solution. At 20°C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20°C for a solution containing 78g of benzene and 46g of toluene in torr is [2005]

Options:

- A. 53.5
- B. 37.5
- C. 25
- D. 50

Answer: D

Solution:

Vapour pressure of benzene = 75 torr

Vapour pressure of toluene = 22 torr

mass of benzene in = 78g

Moles of benzene = $\frac{78}{78} = 1\text{mol}$

∴ (Mol.wt of benzene = 78)

Mass of toluene in solution = 46g

Hence moles of toluene = $\frac{46}{92} = 0.5\text{mol}$

Now partial pressure of benzene

$$= P_b^0 \cdot X_b = 75 \times \frac{1}{1+0.5} = 75 \times \frac{1}{1.5}$$

$$= 75 \times \frac{2}{3} = 50$$

Question205

Equimolar solutions in the same solvent have [2005]

Options:

- A. Different boiling and different freezing points
- B. Same boiling and same freezing points
- C. Same freezing point but different boiling points
- D. Same boiling point but different freezing points

Answer: B

Solution:

Equimolar solutions of normal solutes in the same solvent will have the same b. pts and same f . pts.

Question206

For which of the following parameters, the structural isomers C₂H₅OH and CH₃OCH₃ would be expected to have the same values?(Assume ideal behaviour). [2004]

Options:

- A. Boiling points

- B. Vapour pressure at the same temperature
- C. Heat of vaporization
- D. Gaseous densities at the same temperature and pressure

Answer: D

Solution:

Gaseous densities of ethanol and dimethyl ether would be same at same temperature and pressure. The heat of vaporisation, V.P. and b.pts will differ due to H bonding in ethanol.

Question207

**Which of the following liquid pairs shows a positive deviation from Raoult's law?
[2004]**

Options:

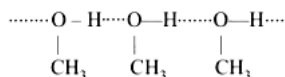
- A. Water - nitric acid
- B. Benzene-methanol
- C. Water - hydrochloric acid
- D. Acetone-chloroform

Answer: B

Solution:

Note: Positive deviations are shown by such solutions in which solvent-solvent and solute-solute interactions are stronger than the solvent-solute interactions. In such solutions, the interactions among molecules becomes weaker. Therefore, their escaping tendency increases which results in the increase in their partial vapour pressure.

In a solution of methanol there exists intermolecular H-bonding.



In this solution benzene molecules come between methanol molecules which weaken intermolecular forces. This results in increase in vapour pressure.

Question208

**Which one of the following aqueous solutions will exhibit highest boiling point?
[2004]**

Options:

- A. 0.015M urea
- B. 0.01M KNO₃
- C. 0.01M Na₂SO₄
- D. 0.015M glucose

Answer: C

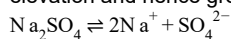
Solution:

$$\because \Delta T_b = T_b - T^\circ$$

Where T_b = b. pt of solution

$$T_b^\circ = \text{b.pt of solvent or } T_b = T_b^\circ + \Delta T_b$$

Note: Elevation in boiling point is a colligative property, which depends upon the no. of particles. Thus greater the number of particles, greater is its elevation and hence greater will be its boiling point.



Since Na_2SO_4 has maximum number of particles (3), hence has maximum boiling point.

Question209

Which one of the following statements is FALSE?

[2004]

Options:

- A. The correct order of osmotic pressure for 0.01M aqueous solution of each compound is $\text{BaCl}_2 > \text{KCl} > \text{CH}_3\text{COOH} > \text{sucrose}$
- B. The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$, where M is the molarity of the solution
- C. Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction
- D. Two sucrose solutions of same molality prepared in different solvents will have same freezing point depression

Answer: D

Solution:

$$\Delta T_f = i \times K_f \times m.$$

Since K_f has different values for different solvents, hence even if m is same, ΔT_f will be different.

Question210

If liquids A and B form an ideal solution

[2003]

Options:

- A. the entropy of mixing is zero
- B. the free energy of mixing is zero
- C. the free energy as well as the entropy of mixing is zero
- D. the enthalpy of mixing is zero

Answer: D

Solution:

When A and B form an ideal solution, $\Delta H_{\text{mix}} = 0$

Question211

A pressure cooker reduces cooking time for food because

[2003]

Options:

- A. boiling point of water involved in cooking is increased
- B. the higher pressure inside the cooker crushes the food material
- C. cooking involves chemical changes helped by a rise in temperature

D. heat is more evenly distributed in the cooking space

Answer: A

Solution:

Note: On increasing pressure, the temperature is also increased. Thus in pressure cooker due to increase in pressure the b.p. of water increases.

Question212

In a 0.2 molal aqueous solution of a weak acid H X , the degree of ionization is 0.3. Taking K_f for water as 1.85, the freezing point of the solution will be nearest to [2003]

Options:

A. -0.360°C

B. -0.260°C

C. $+0.480^{\circ}\text{C}$

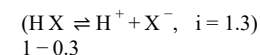
D. -0.480°C

Answer: D

Solution:

$$\Delta T_f = i \times K_f \times m;$$

$$\Delta T_f = 1.85 \times 0.2 \times 1.3 = 0.480^{\circ}\text{C}$$



$$\Delta T_f = T_f^{\circ} - T_f$$

$$\therefore T_f = 0 - 0.480^{\circ}\text{C} = -0.480^{\circ}\text{C}$$

Question213

In mixture A and B components show -ve deviation as [2002]

Options:

A. $\Delta V_{\text{mix}} > 0$ [2002]

B. $\Delta H_{\text{mix}} < 0$

C. A – B interaction is weaker than A – A and B – B interaction

D. A – B interaction is stronger than A – A and B – B interaction.

Answer: D

Solution:

In solution containing A and B component showing negative deviation A-A and B-B interactions are weaker than that of A-B interactions. For such solutions,
 $\Delta H = -\text{ve}$ and $\Delta V = -\text{ve}$

Question 214

Freezing point of an aqueous solution is $(-0.186)^{\circ}\text{C}$. Elevation of boiling point of the same solution is $K_b = 0.512^{\circ}\text{C}$, $K_f = 1.86^{\circ}\text{C}$, find the increase in boiling point.
[2002]

Options:

- A. 0.186°C
- B. 0.0512°C
- C. 0.092°C
- D. 0.2372°C .

Answer: B

Solution:

$$\Delta T_b = K_b \frac{W_B}{M_B \times W_A} \times 1000$$

$$\Delta T_f = K_f \frac{W_B}{M_B \times W_A} \times 1000$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f}, \quad \frac{\Delta T_b}{0.186} = \frac{0.512}{1.86}$$
$$= 0.0512^{\circ}\text{C}.$$
